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Learning Objectives of This Lecture

1. Understand the relationship between **z-transform** and **Laplace transform**;
2. Be familiar with the z-transforms and their Regions of Convergence (ROC) of **common sequences**;
3. Be able to analyze the **ROC** of various types of sequences;
4. Skillfully master the **method of partial fractions** to find the **inverse z-transform**

7.1 Definition of z-Transform

The z-transform is introduced with the help of the Laplace transform of sampled signals.

For a continuous causal signal $x(t)$ subjected to uniform impulse sampling, the expression of the sampled signal $x_s(t)$ is

$$x_s(t) = x(t)\delta_T(t) = \sum_{n=0}^{\infty} x(nT)\delta(t - nT)$$

where T is the sampling interval.

Unilateral Laplace Transform

$$X_s(s) = \int_0^{\infty} x_s(t)e^{-st}dt = \sum_{n=0}^{\infty} x(nT) \int_0^{\infty} \delta(t - nT)e^{-st}dt = \sum_{n=0}^{\infty} x(nT)e^{-snT}$$

Let $z = e^{sT}$, where **z is a complex variable**. The relationship between the Laplace transform of the sampled signal and the z-transform of the sequence is as follows:

$$X(z) \Big|_{z=e^{sT}} = X_s(s)$$

7.1 Definition of z-Transform

$$z = e^{sT}, \quad X(z) = \sum_{n=0}^{\infty} x(nT)z^{-n}$$

Usually let $T=1, z = e^s$

$$X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$$

Unilateral z-Transform

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

Bilateral z-Transform

The z-transform of a sequence is **a power series of the complex variable z^{-1}** , and **its coefficients are $x(n)$** .

The unilateral and bilateral z-transforms of a causal sequence are equivalent.

We focus on the unilateral, while also considering the bilateral.

Why is **z-transform** introduced?

- The z-transform is a powerful tool for **solving difference equations** (converting difference equations into algebraic equations), similar to the Laplace transform in continuous-time systems.
- The z-transform is also used in **the analysis and design of digital filters**, as well as various types of digital signal processing problems.

Generation and Development of z-Transform:

- Preliminary understanding in the 18th century;
- Further development and applications in the 1950s–1960s.

The z-transform of a sequence is meaningful under any conditions.

The statement is ()

☐ A True

☒ B False

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7.2 z-Transform of Typical Sequences

1. z-Transform of the Unit Sample Sequence

$$ZT[\delta(n)] = \sum_{n=0}^{\infty} \delta(n)z^{-n} = 1 \quad (0 \leq |z| \leq \infty, \text{ Convergent on the entire plane})$$

**Unilateral (Bilateral)
z-Transform**

$$ZT[\delta(n-1)] = \sum_{n=0}^{\infty} \delta(n-1)z^{-n} = z^{-1} \quad (|z| > 0)$$

**Unilateral (Bilateral)
z-Transform**

$$ZT[\delta(n+1)] = \sum_{n=-\infty}^{-1} \delta(n+1)z^{-n} + \sum_{n=0}^{\infty} \delta(n+1)z^{-n} = z^1 + 0 = z \quad (|z| < \infty)$$

**Bilateral
z-Transform**

2. z-Transform of the Unit Step Sequence

$$ZT[u(n)] = \sum_{n=0}^{\infty} u(n)z^{-n} = \sum_{n=0}^{\infty} z^{-n} = \frac{1}{1 - z^{-1}} = \frac{z}{z - 1} \quad (|z^{-1}| < 1 \rightarrow |z| > 1)$$

Unilateral (Bilateral) z-Transform

The z-transform of the sequence $a^n u(n)$ ($|a| \neq 1$) is ()

A $\frac{z}{a-z} (|z| < |a|)$

B $\frac{z}{a-z} (|z| > |a|)$

C $\frac{z}{z-a} (|z| < |a|)$

D $\frac{z}{z-a} (|z| > |a|)$

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7.2 z-Transform of Typical Sequences

3. z-Transform of the Unilateral Exponential Sequence

$$ZT[a^n u(n)] = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1 - az^{-1}} = \frac{z}{z - a} \quad (|az^{-1}| < 1 \rightarrow |z| > |a|)$$

4. z-Transform of the Ramp Sequence

$$ZT[nu(n)] = \sum_{n=0}^{\infty} nu(n)z^{-n} = \sum_{n=0}^{\infty} nz^{-n}$$

From $\sum_{n=0}^{\infty} (z^{-1})^n = \frac{1}{1 - z^{-1}} \quad (|z| > 1)$ (1)

Differentiate both sides of (1) with respect to z^{-1} , $\sum_{n=0}^{\infty} n(z^{-1})^{n-1} = \frac{1}{(1 - z^{-1})^2}$ (2)

Multiply both sides of (2) by z^{-1} , $ZT[nu(n)] = \sum_{n=0}^{\infty} nz^{-n} = \frac{z}{(z - 1)^2} \quad (|z| > 1)$

5. z-Transform of the Unilateral Complex Exponential Sequence

$$ZT[e^{j\omega_0 n}u(n)] = \frac{z}{z - e^{j\omega_0}} \quad |z| > |e^{j\omega_0}| = 1$$

$$ZT[e^{-j\omega_0 n}u(n)] = \frac{z}{z - e^{-j\omega_0}} \quad |z| > |e^{-j\omega_0}| = 1$$

$$ZT[\beta^n e^{j\omega_0 n}u(n)] = \frac{z}{z - \beta e^{j\omega_0}} \quad |z| > |\beta e^{j\omega_0}| = |\beta|$$

$$ZT[\beta^n e^{-j\omega_0 n}u(n)] = \frac{z}{z - \beta e^{-j\omega_0}} \quad |z| > |\beta e^{-j\omega_0}| = |\beta|$$

7.2 z-Transform of Typical Sequences

6. z-Transform of the Unilateral Sinusoidal Sequence

$$ZT[\cos(\omega_0 n)u(n)] = ZT[(e^{j\omega_0 n} + e^{-j\omega_0 n}) / 2 \cdot u(n)]$$

$$= \left(\frac{z}{z - e^{j\omega_0}} + \frac{z}{z - e^{-j\omega_0}} \right) / 2$$

$$= \frac{z(z - \cos \omega_0)}{z^2 - 2z \cos \omega_0 + 1} \quad (|z| > 1)$$

$$ZT[\sin(\omega_0 n)u(n)] = ZT[(e^{j\omega_0 n} - e^{-j\omega_0 n}) / (2j) \cdot u(n)]$$

$$= \left(\frac{z}{z - e^{j\omega_0}} - \frac{z}{z - e^{-j\omega_0}} \right) / 2j$$

$$= \frac{z \sin \omega_0}{z^2 - 2z \cos \omega_0 + 1} \quad (|z| > 1)$$

7.2 z-Transform of Typical Sequences

7. z-Transform of the Unilateral Exponential Sinusoidal Sequence

$$\begin{aligned} ZT[\beta^n \cos(\omega_0 n)u(n)] &= ZT[\beta^n (e^{j\omega_0 n} + e^{-j\omega_0 n}) / 2] \\ &= \left(\frac{z}{z - \beta e^{j\omega_0}} + \frac{z}{z - \beta e^{-j\omega_0}} \right) / 2 \\ &= \frac{z(z - \beta \cos \omega_0)}{z^2 - 2z\beta \cos \omega_0 + \beta^2} \quad (|z| > |\beta|) \end{aligned}$$

$$\begin{aligned} ZT[\beta^n \sin(\omega_0 n)u(n)] &= ZT[\beta^n (e^{j\omega_0 n} - e^{-j\omega_0 n}) / 2j] \\ &= \left(\frac{z}{z - \beta e^{j\omega_0}} - \frac{z}{z - \beta e^{-j\omega_0}} \right) / 2j \\ &= \frac{z\beta \sin \omega_0}{z^2 - 2z\beta \cos \omega_0 + \beta^2} \quad (|z| > |\beta|) \end{aligned}$$

7.3 Region of Convergence of z-Transform

$$X(z) = ZT[x(n)] = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

If $x(n)$ is bounded, **the set of all z-values for which the above series converges** is called the **Region of Convergence (ROC)**.

The **necessary and sufficient condition** for the series to converge is satisfying **the absolute summability condition**:

$$\sum_{n=-\infty}^{\infty} |x(n)z^{-n}| < \infty$$

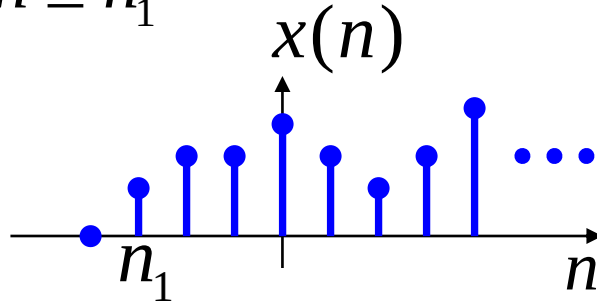
The left side forms a positive-term series, and common determination methods for the convergence of positive-term series can be used:

Ratio Test: $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$

Root Test: $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1$

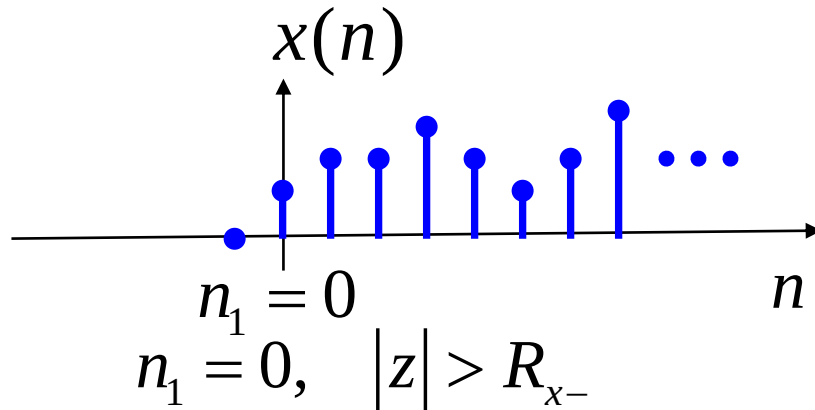
7.3 Region of Convergence of z-Transform

1. Right-Sided Sequence: A sequence that has non-zero finite values only in the interval $n \geq n_1$



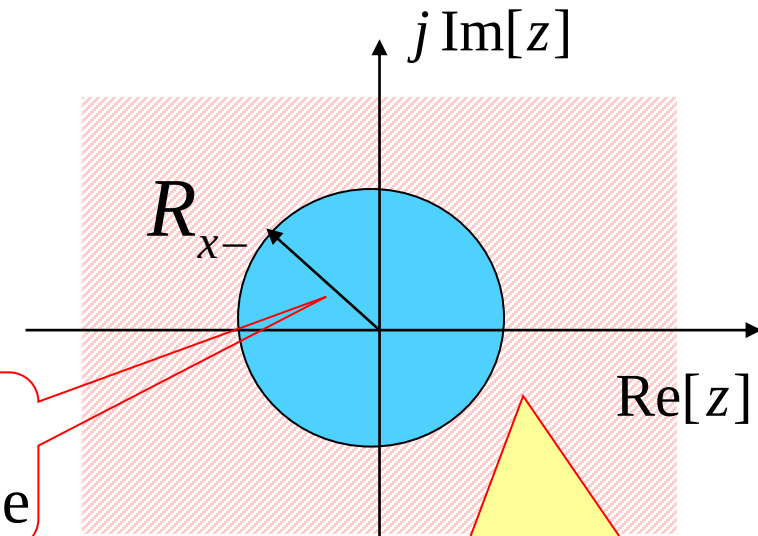
$$X(z) = \sum_{n=n_1}^{\infty} x(n)z^{-n} \quad n_1 \leq n \leq \infty$$

$n_1 < 0$, $R_{x-} < |z| < \infty$ (The ROC does not include $|z| = \infty$, because ∞^{-n_1} is infinite)



$$n_1 = 0, \quad |z| > R_{x-}$$

Radius of Convergence

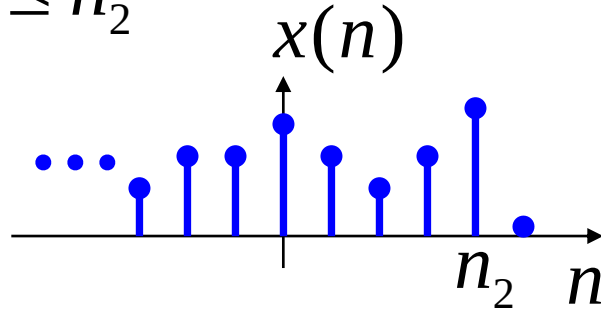


The ROC is outside the circle. If $n_1 < 0$, $|z| = \infty$ is not included.

A **causal sequence** is a special case of a right-sided sequence, and its ROC is $|z| > R_{x-}$

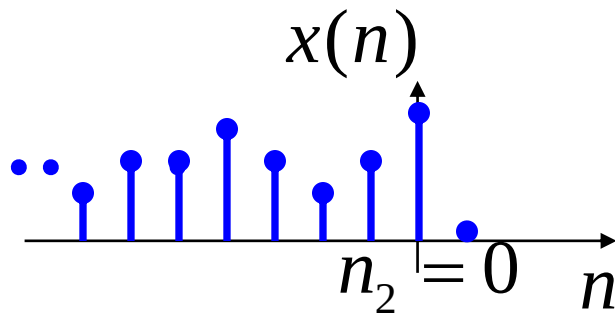
7.3 Region of Convergence of z-Transform

2. Left-Sided Sequence: A sequence $x(n)$ that has non-zero finite values only in the interval $n \leq n_2$



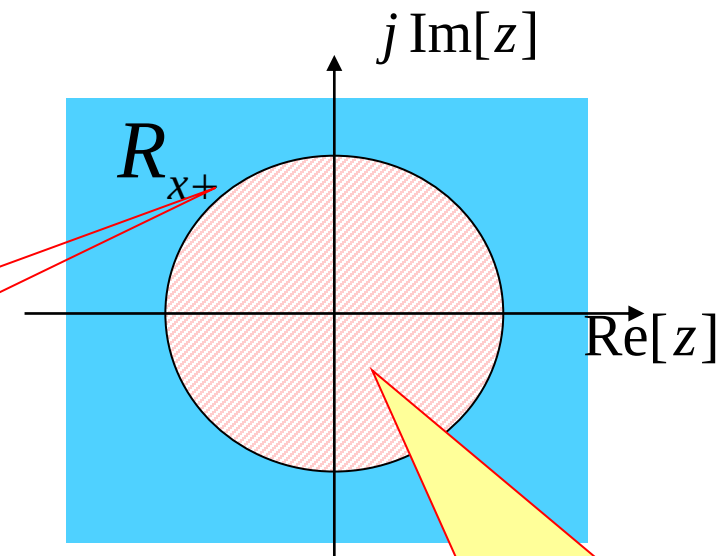
$$X(z) = \sum_{n=-\infty}^{n_2} x(n)z^{-n} \quad -\infty \leq n \leq n_2$$

$n_2 > 0$, $0 < |z| < R_{x+}$ (The ROC does not include $z = 0$, because the negative power of 0 is meaningless)



$$n_2 = 0, \quad |z| < R_{x+}$$

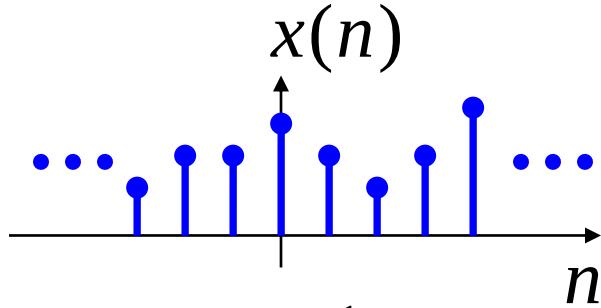
Radius of Convergence



The ROC is inside the circle. If $n_2 > 0$, $z=0$ is not included.

7.3 Region of Convergence of z-Transform

3. Double-Sided Sequence: A sequence that has non-zero finite values in the interval $-\infty \leq n \leq \infty$



$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} \quad -\infty \leq n \leq \infty$$

$$X(z) = \sum_{n=-\infty}^{-1} x(n)z^{-n} + \sum_{n=0}^{\infty} x(n)z^{-n}$$

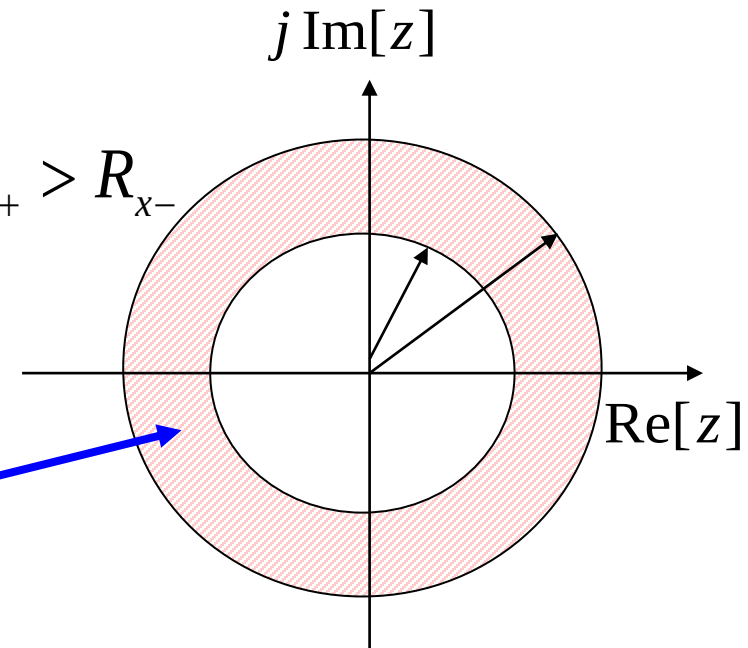
Converges inside
the circle
 $|z| < R_{x+}$

Converges outside
the circle
 $|z| > R_{x-}$

$$R_{x+} > R_{x-}$$

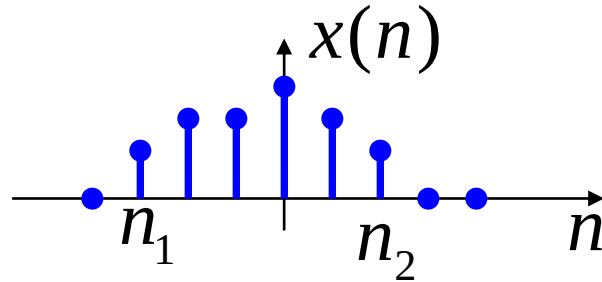
$R_{x+} > R_{x-}$ There is an annular ROC

$R_{x+} < R_{x-}$ No ROC



7.3 Region of Convergence of z-Transform

4. Finite-Length Sequence: A sequence that has non-zero finite values in the finite interval $n_1 \leq n \leq n_2$



$$X(z) = \sum_{n=n_1}^{n_2} x(n)z^{-n} \quad n_1 \leq n \leq n_2$$

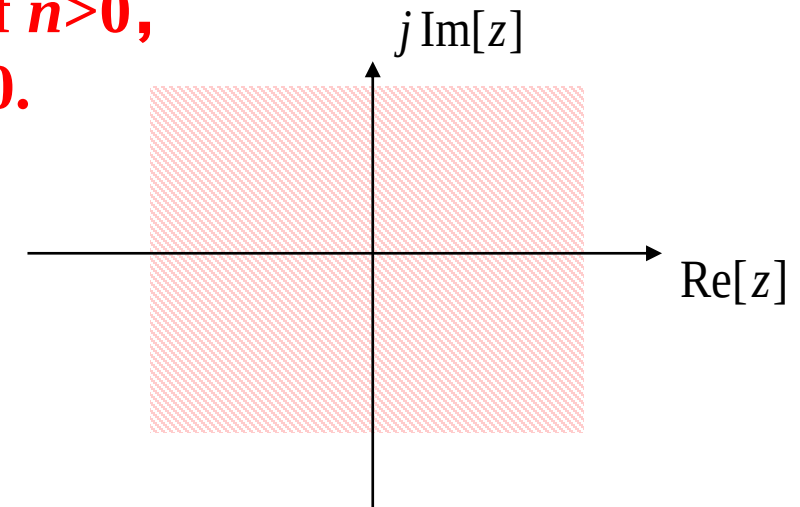
Divided into three cases:

$$n_1 < 0, n_2 > 0 \quad 0 < |z| < \infty$$

$$n_1 \geq 0, n_2 > 0 \quad 0 < |z| \leq \infty$$

$$n_1 < 0, n_2 \leq 0 \quad 0 \leq |z| < \infty$$

**0^{-n} is meaningless if $n > 0$,
 ∞^{-n} is infinite if $n < 0$.**



The ROC is at least the entire z-plane but 0 and ∞ (If pole-zero cancellation occurs, the ROC expands to the entire z-plane).

Which of the following statements are correct? ()

- ☒ A The unilateral z-transform's transform pair has a one-to-one correspondence with the sequence.
- ☒ B The unilateral z-transform's has a unique ROC.
- ☐ C The bilateral z-transform has a one-to-one correspondence with the sequence.
- ☒ D The bilateral z-transform may correspond to different sequences.

提交

7.3 Region of Convergence of z-Transform

For the unilateral z-transform, there is a one-to-one correspondence between the sequence and the transform expression, and it has a unique ROC.

For the bilateral z-transform, different sequences may map to the same transform expression under different ROCs.

Example: $x_1(n) = a^n u(n)$

$$X_1(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a} \quad |z| > |a|$$

$$x_2(n) = -a^n u(-n-1)$$

$$\begin{aligned} X_2(z) &= -\sum_{n=-\infty}^{-1} a^n z^{-n} = 1 - \sum_{n=0}^{\infty} (a^{-1}z)^n \\ &= 1 - \frac{1}{1 - a^{-1}z} = \frac{z}{z - a} \quad |z| < |a| \end{aligned}$$

For the z-transform of a given sequence, **its ROC must be specified.**

7.3 Region of Convergence of z-Transform

Example 8-1: Find the unilateral and bilateral z-transforms of the sequence $x(n)=a^n u(n)-b^n u(-n-1)$ and determine its ROC (where $b>a>0$).

Solution: This is a two-sided sequence.

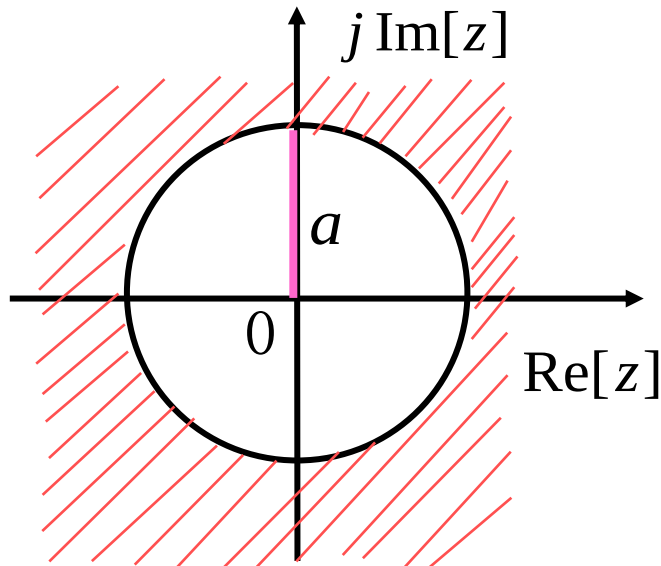
Unilateral z-transform:
$$X(z) = \sum_{n=0}^{\infty} x(n)z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n$$

If $|z| > a$,
$$X(z) = \frac{1}{1-az^{-1}} = \frac{z}{z-a}$$

Bilateral z-transform:
$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} [a^n u(n) - b^n u(-n-1)]z^{-n} \\ &= \sum_{n=0}^{\infty} a^n z^{-n} - \sum_{n=-\infty}^{-1} b^n z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n} + 1 - \sum_{n=0}^{\infty} b^{-n} z^n \end{aligned}$$

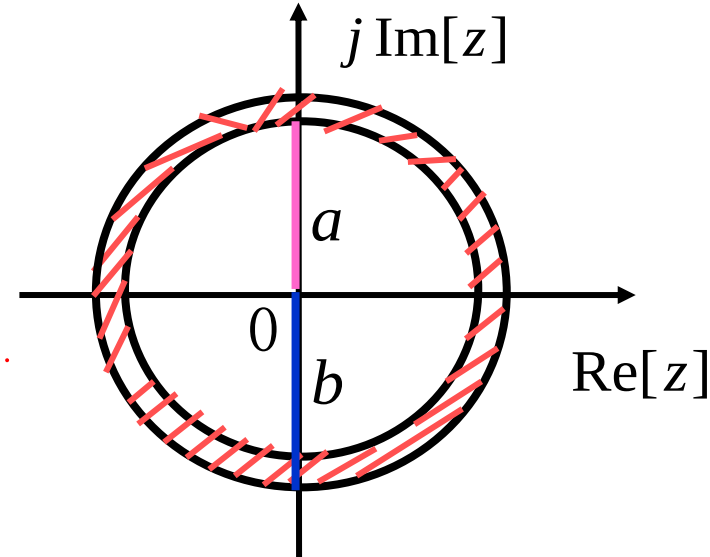
If $a < |z| < b$,
$$X(z) = \frac{1}{1-az^{-1}} + 1 + \frac{b}{1-bz^{-1}} = \frac{z}{z-a} + \frac{z}{z-b}$$

7.3 Region of Convergence of z-Transform



ROC of the sequence's
unilateral z-transform

$$X(z) = \frac{z}{z-a} \quad (|z| > a)$$



ROC of the sequence's
bilateral z-transform

$$X(z) = \frac{z}{z-a} + \frac{z}{z-b} \quad (a < |z| < b)$$

The ROC typically does not contain any poles (unless zero-pole cancellation occurs), and it is usually bounded by poles. The ROC of a right-sided sequence extends outward from the outermost (largest-magnitude) pole of $X(z)$ to $z \rightarrow \infty$ (may include ∞); The ROC of a left-sided sequence extends inward from the innermost (smallest-magnitude) pole of $X(z)$ to $z \rightarrow 0$ (may include 0).

7.4 Inverse z-Transform

$$X(z) = ZT[x(n)] = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

Partial Fraction Expansion Method:
$$X(z) = \frac{b_0 + b_1z + \dots + b_{r-1}z^{r-1} + b_rz^r}{a_0 + a_1z + \dots + a_{k-1}z^{k-1} + a_kz^k}$$

1. $X(z)$ has only first-order poles

$$X(z) = A_0 + \sum_{m=1}^M \frac{A_m z}{z - z_m} = A_0 + \frac{A_1 z}{z - z_1} + \dots + \frac{A_M z}{z - z_M} \quad A_0 = X(z) \Big|_{z=0} = \frac{b_0}{a_0}$$

Let z_m be a first-order pole of $X(z)$ $A_m = \left[(z - z_m) \frac{X(z)}{z} \right]_{z=z_m}$ $X(z) = A_0 + \sum_{m=1}^M \frac{z A_m}{z - z_m}$

or $X(z) = A_0 + \sum_{m=1}^M \frac{A_m}{1 - z_m z^{-1}} = A_0 + \frac{A_1}{1 - z_1 z^{-1}} + \dots + \frac{A_M}{1 - z_M z^{-1}} \quad A_m = \left[(1 - z_m z^{-1}) X(z) \right]_{z=z_m}$

Inverse Transform: $x(n) = A_0 \delta(n) + \sum_{m=1}^M A_m z_m^n = A_0 \delta(n) + A_1 z_1^n + \dots + A_M z_M^n$

The inverse transform $x(n)$ of $X(z) = \frac{z^2}{z^2 - 1.5z + 0.5}$ ($|z| > 1$) ()

- ☐ A $(1-0.5^n)u(n)$
- ☒ B $(2-0.5^n)u(n)$
- ☐ C $2(1-0.5^n)u(n)$
- ☐ D $2(1+0.5^n)u(n)$

提交

7.4 Inverse z-Transform

Example 8-2: Find the inverse transform $x(n)$ of $X(z) = \frac{z^2}{z^2 - 1.5z + 0.5}$ ($|z| > 1$).

Solution: This is a **right-sided sequence**.

Method 1: $X(z) = \frac{z^2}{(z-1)(z-0.5)}$ $\frac{X(z)}{z} = \frac{A_1}{z-0.5} + \frac{A_2}{z-1}$

$$\text{where } A_1 = \left[\frac{X(z)}{z} (z-0.5) \right]_{z=0.5} = -1, \quad A_2 = \left[\frac{X(z)}{z} (z-1) \right]_{z=1} = 2$$

$$\text{then } X(z) = \frac{2z}{z-1} - \frac{z}{z-0.5}$$

$$x(n) = (2 - 0.5^n)u(n)$$

Method 2:

$$X(z) = \frac{z^2}{(z-1)(z-0.5)} = \frac{1}{(1-z^{-1})(1-0.5z^{-1})}$$

$$X(z) = \frac{A_1}{1-0.5z^{-1}} + \frac{A_2}{1-z^{-1}}$$

$$\text{where } A_1 = \left[X(z)(1-0.5z^{-1}) \right]_{z=0.5} = -1, \quad A_2 = \left[X(z)(1-z^{-1}) \right]_{z=1} = 2$$

$$\text{then } X(z) = \frac{2}{1-z^{-1}} - \frac{1}{1-0.5z^{-1}}$$

$$x(n) = (2-0.5^n)u(n)$$

2. In addition to M first-order poles, $X(z)$ also has an N -th order pole at $z=z_i$

$$X(z) = A_0 + \sum_{m=1}^M \frac{A_m z}{z - z_m} + \sum_{j=1}^N \frac{B_j z}{(z - z_i)^j} \quad (|z| > \max\{z_1, \dots, z_M, z_i\})$$

$$\text{Let } X_1(z) = (z - z_i)^N \frac{X(z)}{z}$$

$$B_N = X_1(z) \Big|_{z=z_i} \quad B_j = \frac{1}{(N-j)!} \left[\frac{d^{N-j}}{dz^{N-j}} X_1(z) \right]_{z=z_i} \quad (j = 1, \dots, N-1)$$

$$\text{If it is a second - order pole } N = 2: \quad X_1(z) = (z - z_i)^2 \frac{X(z)}{z}$$

$$B_2 = X_1(z) \Big|_{z=z_i} \quad B_1 = \left[\frac{d}{dz} X_1(z) \right]_{z=z_i}$$

Inverse transform: $x(n) = A_0 \delta(n) + (A_1 z_1^n + \dots + A_M z_M^n + B_1 z_i^n + B_2 n z_i^{n-1}) u(n)$

7.4 Inverse z-Transform

Example 8-3: Find the inverse transform of $X(z) = \frac{z^2}{z^2 - 4z + 4} \quad (|z| > 2)$.

Solution: $x(n)$ is a **right-sided sequence**.

$$X(z) = \frac{z^2}{(z-2)^2} \quad X_1(z) = \frac{X(z)}{z} = \frac{B_1}{z-2} + \frac{B_2}{(z-2)^2}$$

where

$$B_2 = \left[(z-2)^2 X_1(z) \right]_{z=2} = [z]_{z=2} = 2, \quad B_1 = \left\{ \frac{d}{dz} \left[(z-2)^2 X_1(z) \right] \right\}_{z=2} = 1$$

$$\text{then } X(z) = \frac{z}{z-2} + \frac{2z}{(z-2)^2}$$

$$x(n) = 2^n u(n) + n2^n u(n) = (n+1)2^n u(n)$$

7.4 Inverse z-Transform

Generally, $X(z)$ can be written as:

$$X(z) = \frac{b_0 + b_1 z + \cdots + b_{r-1} z^{r-1} + b_r z^r}{a_0 + a_1 z + \cdots + a_{k-1} z^{k-1} + a_k z^k}$$

When using the **partial fraction expansion** method to find the inverse transform, you need to master the inverse transforms of basic forms.

Note: The inverse transform is related to the ROC.

$$\frac{z}{z - z_m} (|z| > |z_m|) \Leftrightarrow z_m^n u(n)$$

$$\frac{z}{z - z_m} (|z| < |z_m|) \Leftrightarrow -z_m^n u(-n-1)$$

Discussion: Only **proper fractions** can be expanded by **partial fractions**, but the expanded form needs to be **multiplied by z** to have the basic form of the inverse z-transform mentioned above.

To perform partial fraction expansion on $\frac{X(z)}{z}$, $\frac{X(z)}{z}$ is required to be a proper fraction — which means **$k \geq r$ is needed to ensure $X(z)$ converges at $z=\infty$. The ROC of the z-transform of a causal sequence is $|z| > R_x$, and $k \geq r$ is a necessary and sufficient condition for convergence.**

7.4 Inverse z-Transform

Example 8-4: $X(z) = \frac{-5z}{3z^2 - 7z + 2} \quad \left(\frac{1}{3} < |z| < 2\right)$, find $x(n)$.

Two-Sided Sequence

Solution:
$$\frac{X(z)}{z} = \frac{-\frac{5}{3}}{z^2 - \frac{7}{3}z + \frac{2}{3}} = \frac{-\frac{5}{3}}{\left(z - \frac{1}{3}\right)(z - 2)} = \frac{1}{z - \frac{1}{3}} - \frac{1}{z - 2}$$

$$X(z) = \frac{z}{z - \frac{1}{3}} - \frac{z}{z - 2}$$

Right-Sided Sequence

Left-Sided Sequence

$$x(n) = \left(\frac{1}{3}\right)^n u(n) + 2^n u(-n-1)$$

The same z-transform expression with different ROCs correspond to different sequences.

$$X(z) = \frac{z}{z - \frac{1}{3}} - \frac{z}{z - 2}$$

$$\text{if } |z| < \frac{1}{3} \quad x(n) = -\left(\frac{1}{3}\right)^n u(-n-1) + 2^n u(-n-1)$$

$$\text{if } |z| > 2 \quad x(n) = \left(\frac{1}{3}\right)^n u(n) - 2^n u(n)$$

1. Repeat all the examples in the lecture notes.
2. 7.17(e) and 7.24 (a) in the textbook.

Review of Last Lecture

7.1 Definition of z-Transform

7.2 z-Transform of Typical Sequences

7.3 Region of Convergence of z-Transform

7.4 Inverse z-Transform

7.1 Definition of z-Transform

For a continuous causal signal $x(t)$ sampled at interval T , the sampled signal $x_s(t)$ is

$$x_s(t) = x(t)\delta_T(t) = \sum_{n=0}^{\infty} x(nT)\delta(t - nT)$$

Unilateral Laplace Transform

$$X_s(s) = \int_0^{\infty} x_s(t)e^{-st}dt = \sum_{n=0}^{\infty} x(nT) \int_0^{\infty} \delta(t - nT)e^{-st}dt = \sum_{n=0}^{\infty} x(nT)e^{-snT}$$

Let $z = e^{sT}$, where **z is a complex variable**. The relationship between the Laplace transform of the sampled signal and the z-transform of the sequence is as follows:

$$X(z) \Big|_{z=e^{sT}} = X_s(s)$$

Usually let $T=1$, $z = e^s$

**Unilateral
z-Transform**

$$X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$$

**Bilateral
z-Transform**

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

The z-transform of a sequence is **a power series of the complex variable z^{-1}** , and **its coefficients are $x(n)$** .

Which of the following statements are correct? ()

A

The z-transform is helpful to solve difference equations.

B

For a causal sequence, the unilateral z-transform is equivalent to the bilateral z-transform.

C

The unilateral z-transform of an anti-causal sequence is equivalent to the bilateral z-transform.

D

The unilateral z-transform of an anti-causal sequence is zero.

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7.2 z-Transform of Typical Sequences

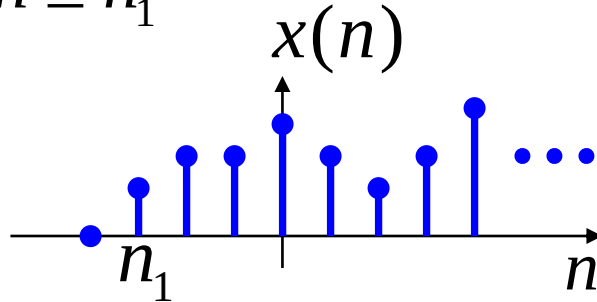
Sequence	Z-Transform	Type of Z-Transform
$\delta(n)$	1 (Converges over the entire z-plane)	Unilateral/Bilateral Z-Transform
$\delta(n-1)$	$z^{-1} \quad (z > 0)$	Unilateral/Bilateral Z-Transform
$\delta(n+1)$	0	Unilateral Z-Transform
	$z \quad (z < \infty)$	Bilateral Z-Transform
$u(n)$	$\frac{z}{z-1} \quad (z > 1)$	Unilateral/Bilateral Z-Transform
$a^n u(n)$	$\frac{z}{z-a} \quad (z > a)$	Unilateral/Bilateral Z-Transform
$nu(n)$	$\frac{z}{(z-1)^2} \quad (z > 1)$	Unilateral/Bilateral Z-Transform

7.2 z-Transform of Typical Sequences

Sequence	Z-Transform	Type of Z-Transform
$e^{-j\omega_0 n} u(n)$	$\frac{z}{z - e^{-j\omega_0}} \quad (z > 1)$	Unilateral/Bilateral Z-Transform
$\beta e^{-j\omega_0 n} u(n)$	$\frac{z}{z - \beta e^{-j\omega_0}} \quad (z > \beta)$	Unilateral/Bilateral Z-Transform
$\cos(\omega_0 n) u(n)$	$\frac{z(z - \cos \omega_0)}{z^2 - 2z \cos \omega_0 + 1} \quad (z > 1)$	Unilateral/Bilateral Z-Transform
$\sin(\omega_0 n) u(n)$	$\frac{z \sin \omega_0}{z^2 - 2z \cos \omega_0 + 1} \quad (z > 1)$	Unilateral/Bilateral Z-Transform
$\beta^n \cos(\omega_0 n) u(n)$	$\frac{z(z - \beta \cos \omega_0)}{z^2 - 2z \beta \cos \omega_0 + \beta^2} \quad (z > \beta)$	Unilateral/Bilateral Z-Transform
$\beta^n \sin(\omega_0 n) u(n)$	$\frac{z \beta \sin \omega_0}{z^2 - 2z \beta \cos \omega_0 + \beta^2} \quad (z > \beta)$	Unilateral/Bilateral Z-Transform

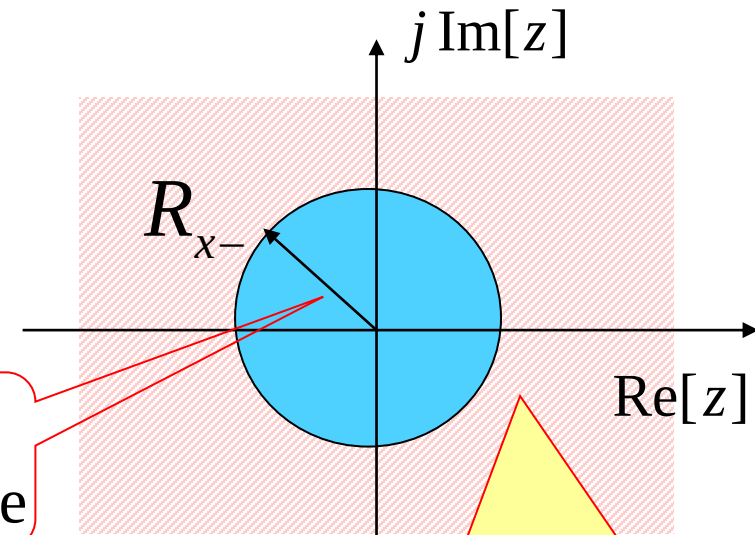
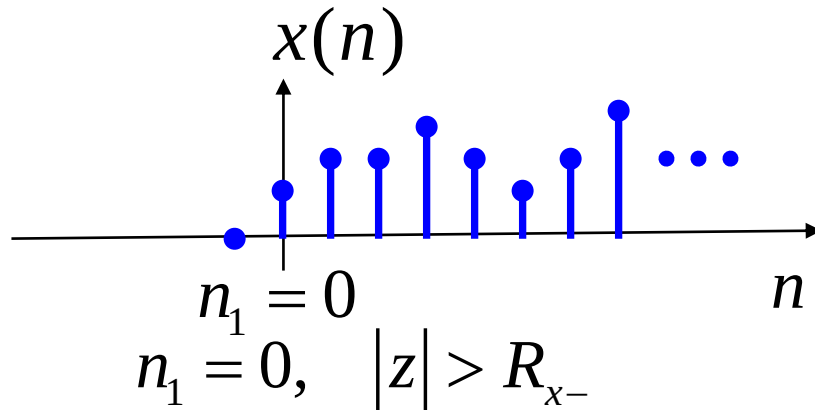
7.3 Region of Convergence of z-Transform

1. Right-Sided Sequence: A sequence that has non-zero finite values only in the interval $n \geq n_1$



$$X(z) = \sum_{n=n_1}^{\infty} x(n)z^{-n} \quad n_1 \leq n \leq \infty$$

$n_1 < 0$, $R_{x-} < |z| < \infty$ (The ROC does not include $|z| = \infty$, because ∞^{-n_1} is infinite)

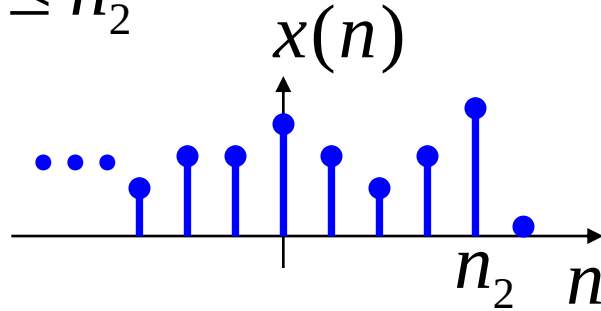


The ROC is outside the circle. If $n_1 < 0$, $|z| = \infty$ is not included.

A **causal sequence** is a special case of a right-sided sequence, and its ROC is $|z| > R_{x-}$

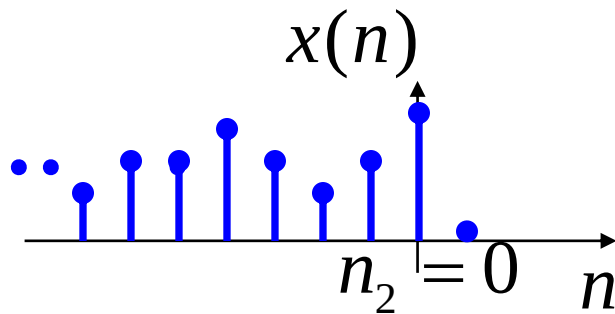
7.3 Region of Convergence of z-Transform

2. Left-Sided Sequence: A sequence $x(n]$ that has non-zero finite values only in the interval $n \leq n_2$



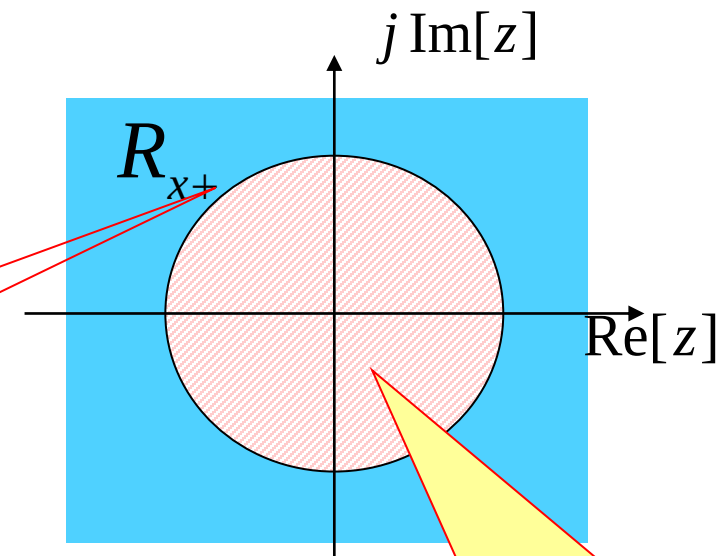
$$X(z) = \sum_{n=-\infty}^{n_2} x(n)z^{-n} \quad -\infty \leq n \leq n_2$$

$n_2 > 0$, $0 < |z| < R_{x+}$ (The ROC does not include $z = 0$, because the negative power of 0 is meaningless)



$$n_2 = 0, \quad |z| < R_{x+}$$

Radius of Convergence

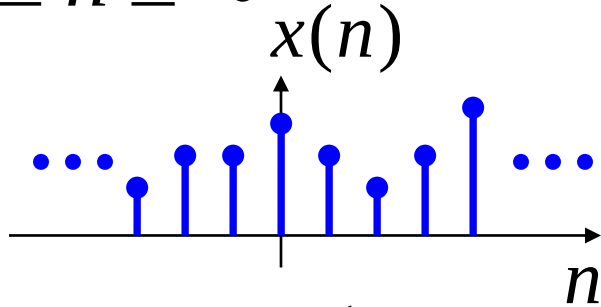


The ROC is inside the circle. If $n_2 > 0$, $z=0$ is not included.

7.3 Region of Convergence of z-Transform

3. Double-Sided Sequence: A sequence that has non-zero finite values in the interval

$$-\infty \leq n \leq \infty$$



$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} \quad -\infty \leq n \leq \infty$$

$$X(z) = \sum_{n=-\infty}^{-1} x(n)z^{-n} + \sum_{n=0}^{\infty} x(n)z^{-n}$$

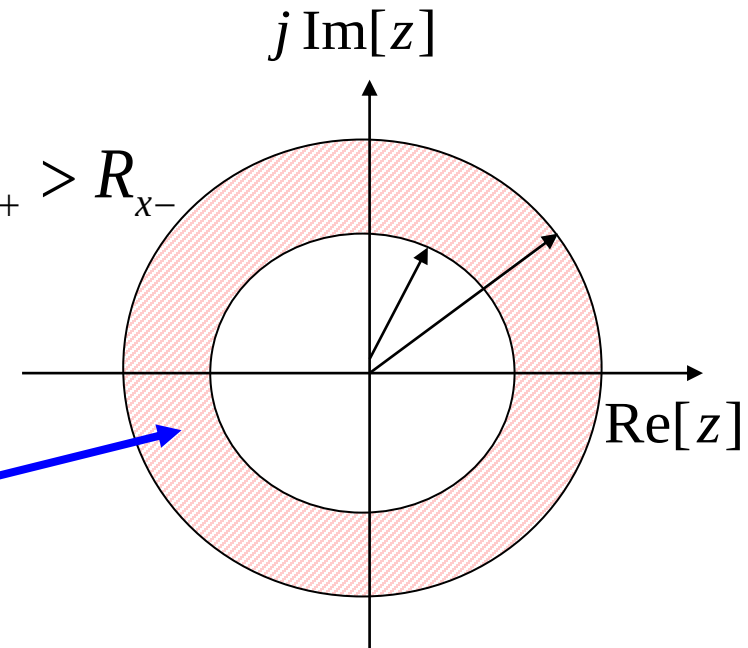
Converges inside
the circle
 $|z| < R_{x+}$

Converges outside
the circle
 $|z| > R_{x-}$

$$R_{x+} > R_{x-}$$

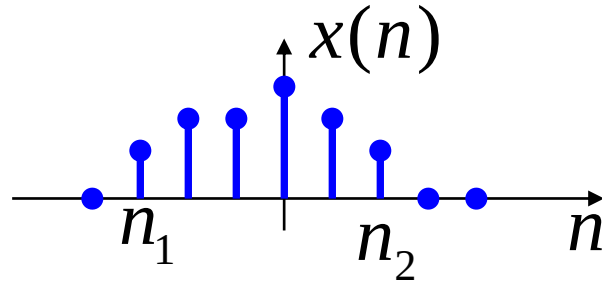
$R_{x+} > R_{x-}$ There is an annular ROC

$R_{x+} < R_{x-}$ No ROC



7.3 Region of Convergence of z-Transform

4. Finite-Length Sequence: A sequence that has non-zero finite values in the finite interval $n_1 \leq n \leq n_2$



$$X(z) = \sum_{n=n_1}^{n_2} x(n)z^{-n} \quad n_1 \leq n \leq n_2$$

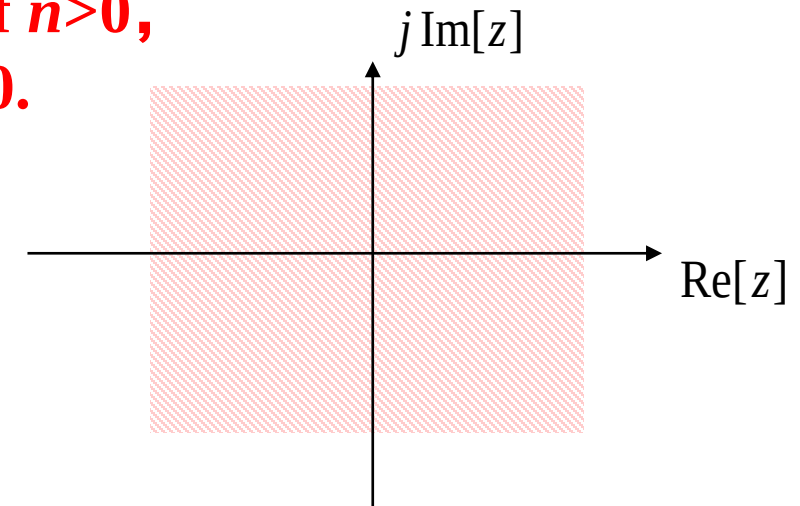
Divided into three cases:

$$n_1 < 0, n_2 > 0 \quad 0 < |z| < \infty$$

$$n_1 \geq 0, n_2 > 0 \quad 0 < |z| \leq \infty$$

$$n_1 < 0, n_2 \leq 0 \quad 0 \leq |z| < \infty$$

**0^{-n} is meaningless if $n > 0$,
 ∞^{-n} is infinite if $n < 0$.**



The ROC is at least the entire z-plane but 0 and ∞ (If pole-zero cancellation occurs, the ROC expands to the entire z-plane).

Which of the following statements are correct? ()

A

Using the partial fraction method to find the inverse transform of $X(z)$ requires that $X(z)$ is a proper fraction.

B

Sequence shifting does not affect the zeros and poles of the z-transform (except at 0 and ∞)

C

The condition for applying the final value theorem is that all poles must lie inside the unit circle.

D

When pole-zero cancellation occurs, the ROC of $X(z)$ may include poles.

提交

7.4 Inverse z-Transform

$$X(z) = ZT[x(n)] = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

Partial Fraction Expansion Method:

$$X(z) = \frac{b_0 + b_1z + \cdots + b_{r-1}z^{r-1} + b_rz^r}{a_0 + a_1z + \cdots + a_{k-1}z^{k-1} + a_kz^k}$$

1. $X(z)$ has only first-order poles

$$X(z) = A_0 + \sum_{m=1}^M \frac{A_m z}{z - z_m} = A_0 + \frac{A_1 z}{z - z_1} + \cdots + \frac{A_M z}{z - z_M} \quad A_0 = X(z) \Big|_{z=0} = \frac{b_0}{a_0}$$

Let z_m be a first-order pole of $X(z)$ $A_m = \left[(z - z_m) \frac{X(z)}{z} \right]_{z=z_m}$ $X(z) = A_0 + \sum_{m=1}^M \frac{z A_m}{z - z_m}$

or $X(z) = A_0 + \sum_{m=1}^M \frac{A_m}{1 - z_m z^{-1}} = A_0 + \frac{A_1}{1 - z_1 z^{-1}} + \cdots + \frac{A_M}{1 - z_M z^{-1}} \quad A_m = \left[(1 - z_m z^{-1}) X(z) \right]_{z=z_m}$

Inverse Transform: $x(n) = A_0 \delta(n) + \sum_{m=1}^M A_m z_m^n = A_0 \delta(n) + A_1 z_1^n + \cdots + A_M z_M^n$

2. In addition to M first-order poles, $X(z)$ also has an N -th order pole at $z=z_i$

$$X(z) = A_0 + \sum_{m=1}^M \frac{A_m z}{z - z_m} + \sum_{j=1}^N \frac{B_j z}{(z - z_i)^j} \quad (|z| > \max\{z_1, \dots, z_M, z_i\})$$

$$\text{Let } X_1(z) = (z - z_i)^N \frac{X(z)}{z}$$

$$B_N = X_1(z) \Big|_{z=z_i} \quad B_j = \frac{1}{(N-j)!} \left[\frac{d^{N-j}}{dz^{N-j}} X_1(z) \right]_{z=z_i} \quad (j = 1, \dots, N-1)$$

$$\text{If it is a second - order pole } N = 2: \quad X_1(z) = (z - z_i)^2 \frac{X(z)}{z}$$

$$B_2 = X_1(z) \Big|_{z=z_i} \quad B_1 = \left[\frac{d}{dz} X_1(z) \right]_{z=z_i}$$

Inverse transform: $x(n) = A_0 \delta(n) + (A_1 z_1^n + \dots + A_M z_M^n + B_1 z_i^n + B_2 n z_i^{n-1}) u(n)$

7.4 Inverse z-Transform

Generally, $X(z)$ can be written as:

$$X(z) = \frac{b_0 + b_1 z + \cdots + b_{r-1} z^{r-1} + b_r z^r}{a_0 + a_1 z + \cdots + a_{k-1} z^{k-1} + a_k z^k}$$

When using the **partial fraction expansion** method to find the inverse transform, you need to master the inverse transforms of basic forms.

Note: The inverse transform is related to the ROC.

$$\frac{z}{z - z_m} (|z| > |z_m|) \Leftrightarrow z_m^n u(n)$$

$$\frac{z}{z - z_m} (|z| < |z_m|) \Leftrightarrow -z_m^n u(-n-1)$$

Discussion: Only **proper fractions** can be expanded by **partial fractions**, but the expanded form needs to be **multiplied by z** to have the basic form of the inverse z-transform mentioned above.

To perform partial fraction expansion on $\frac{X(z)}{z}$, $\frac{X(z)}{z}$ is required to be a proper fraction — which means **$k \geq r$ is needed to ensure $X(z)$ converges at $z=\infty$. The ROC of the z-transform of a causal sequence is $|z| > R_x$, and $k \geq r$ is a necessary and sufficient condition for convergence.**

Contents of This Lecture

7.5 Basic Properties of the z-Transform

7.6 Relationship Between the z-Transform
and the Laplace Transform

Learning Objectives of This Lecture

1. Master the **basic properties of the z-transform** proficiently;
2. Be able to analyze the impact on the **Region of Convergence** (ROC) when applying the basic properties of the z-transform;
3. Be familiar with the mapping relationship between the **s-plane** and the **z-plane**.

7.5 Basic Properties of the z-Transform

7.5.1 Linearity

$$\text{if } ZT[x(n)] = X(z) \quad (R_{x1} < |z| < R_{x2}),$$

$$ZT[y(n)] = Y(z) \quad (R_{y1} < |z| < R_{y2})$$

$$\text{then } ZT[ax(n) + by(n)] = aX(z) + bY(z)$$

$$\max(R_{x1}, R_{y1}) < |z| < \min(R_{x2}, R_{y2})$$

the ROC is the overlapping part

Note: If **some zeros and poles cancel each other out** in the linear combination,
the ROC may expand.

7.5.2 Shift Property

1. Bilateral z-transform of a Shifted Bilateral Sequence

$$ZT[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$ZT[x(n-m)] = z^{-m} X(z)$$

$$ZT[x(n+m)] = z^m X(z)$$

m is any positive integer.

If $x(n)$ is a double-sided sequence and its bilateral z-transform is $X(z)$, shifting the sequence $x(n)$ will change the ROC of $X(z)$. This statement is ().

☐ A True

☒ B False

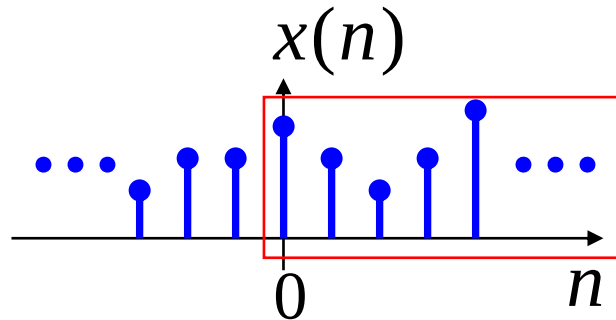
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7.5 Basic Properties of the z-Transform

2. Unilateral z-transform of a Left-Shifted Bilateral Sequence

$$X(z) = \sum_{n=0}^{\infty} x(n)u(n)z^{-n}$$

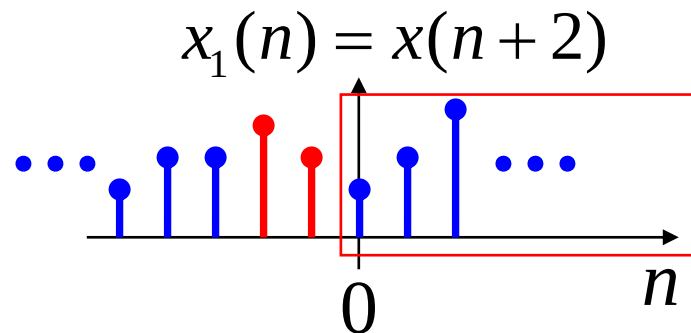
$$ZT[x(n+m)u(n)] = \sum_{n=0}^{\infty} x(n+m)z^{-n}$$



$$= z^m \sum_{n=0}^{\infty} x(n+m)z^{-(n+m)} = z^m \sum_{k=m}^{\infty} x(k)z^{-k}$$

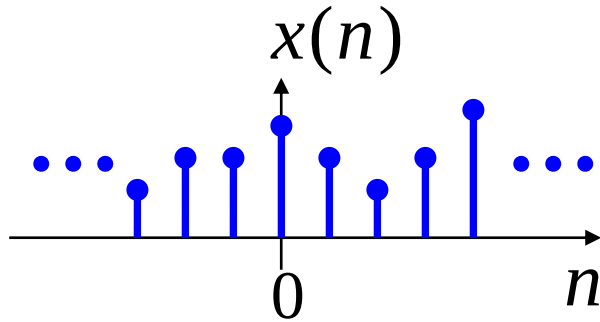
$$= z^m \left[\sum_{k=0}^{\infty} x(k)z^{-k} - \sum_{k=0}^{m-1} x(k)z^{-k} \right]$$

$$= z^m \left[X(z) - \sum_{k=0}^{m-1} x(k)z^{-k} \right]$$



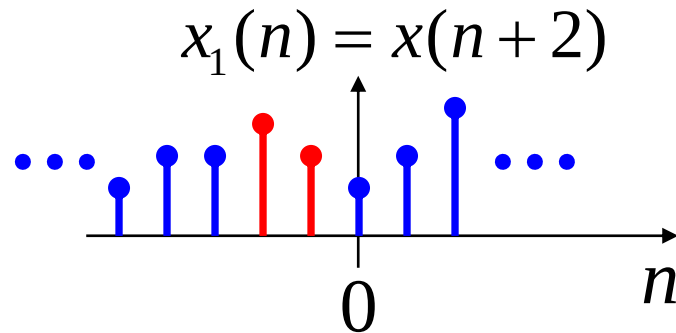
**reduces the contribution
of some points**

7.5 Basic Properties of the z-Transform



$$X(z) = x(0) + x(1)z^{-1} + x(2)z^{-2} + \dots$$

$$z^2 X(z) = x(0)z^2 + x(1)z + x(2) + x(3)z^{-1} + \dots$$



$$X_1(z) = x_1(0) + x_1(1)z^{-1} + x_1(2)z^{-2} + \dots$$

$$= x(2) + x(3)z^{-1} + x(4)z^{-2} + \dots$$

$$= z^2 X(z) - x(0)z^2 - x(1)z$$

$$= z^2 \left[X(z) - x(0) - x(1)z^{-1} \right]$$

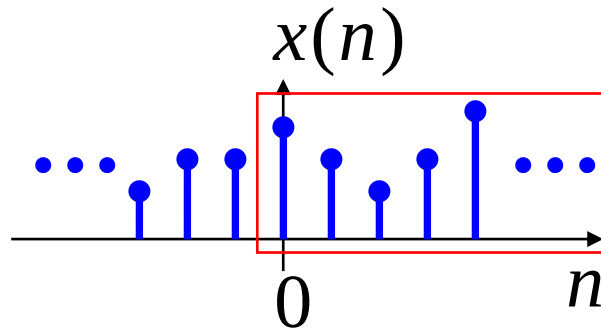
7.5 Basic Properties of the z-Transform

3. Unilateral z-Transform of a Right-Shifted Bilateral Sequence

$$X(z) = \sum_{n=0}^{\infty} x(n)u(n)z^{-n}$$

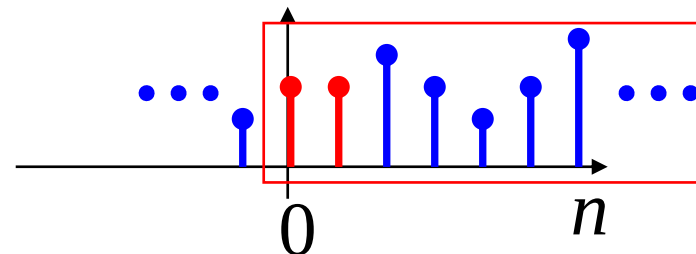
$$ZT[x(n-m)u(n)] = z^{-m} \left[X(z) + \sum_{k=-m}^{-1} x(k)z^{-k} \right]$$

adds the contribution of some points



$$X(z) = x(0) + x(1)z^{-1} + x(2)z^{-2} + \dots$$

$$x_2(n) = x(n-2)$$



$$X_2(z) = x_2(0) + x_2(1)z^{-1} + x_2(2)z^{-2} + \dots$$

$$= x(-2) + x(-1)z^{-1} + x(0)z^{-2} + \dots$$

$$= z^{-2}X(z) + x(-2) + x(-1)z^{-1}$$

$$= z^{-2} \left[X(z) + x(-2)z^2 + x(-1)z \right]$$

4. Unilateral z-Transform of a Shifted Causal Sequence $x(n)$

$$\because \sum_{k=-m}^{-1} x(k)z^{-k} = 0$$

$$ZT[x(n-m)u(n)] = z^{-m} X(z)$$

$$ZT[x(n+m)u(n)] = z^m \left[X(z) - \sum_{k=0}^{m-1} x(k)z^{-k} \right]$$

Sequence shifting only changes the zero-pole situation of the z-transform at **$z=0$ and ∞** .

Given the sequence $x(n) = \left(\frac{1}{3}\right)^n [u(n) - u(n - 8)]$, its ROC is ()

- ☐ A $0 < |z| < \frac{1}{3}$
- ☐ B $|z| > \frac{1}{3}$
- ☒ C $|z| > 0$
- ☐ D The entire z -plane

提交

7.5 Basic Properties of the z-Transform

Example 7-5: Find the z-transform and ROC of $x(n) = \left(\frac{1}{3}\right)^n [u(n) - u(n-8)]$

Solution: The sequence is a **finite-length sequence**.

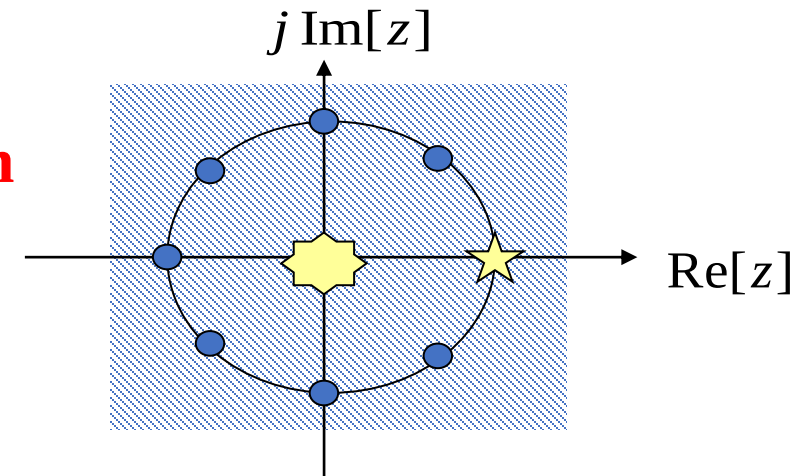
$$X(z) = \sum_{n=0}^7 \left(\frac{1}{3}\right)^n z^{-n} = \sum_{n=0}^7 \left(\frac{1}{3} z^{-1}\right)^n = \frac{1 - \left(\frac{1}{3} z^{-1}\right)^8}{1 - \frac{1}{3} z^{-1}} = \frac{z^8 - \left(\frac{1}{3}\right)^8}{z^7 \left(z - \frac{1}{3}\right)}$$

$$z^8 = \left(\frac{1}{3}\right)^8 \quad z = \frac{1}{3} e^{j\frac{2k\pi}{8}} \quad (k = 0, \dots, 7) \longrightarrow \text{Eight zeros}$$

$$z = 0 \longrightarrow \text{Seventh-order pole}$$

$$z = 1/3 \longrightarrow \text{First-order pole, which cancels with one zero (expanding the ROC beyond } |z| > 1/3)$$

$$\therefore |z| > 0$$



7.5 Basic Properties of the z-Transform

7.5.3 z-Domain Differentiation (Time-Domain Linear Weighting)

$$\text{if } ZT[x(n)] = X(z)$$

$$ZT[nx(n)] = -z \frac{d}{dz} X(z)$$

Multiplying the time-domain sequence by n is equivalent to differentiating in the z-domain and multiplying by $(-z)$.

$$ZT[n^2 x(n)] = -z \frac{d}{dz} \left[-z \frac{dX(z)}{dz} \right] = \left[-z \frac{d}{dz} \right]^2 X(z)$$

$$ZT[n^m x(n)] = \left[-z \frac{d}{dz} \right]^m X(z)$$

7.5 Basic Properties of the z-Transform

Example 7–6: Find the z-transforms of the ramp sequence $nu(n)$ and the sequence $n^2u(n)$.

Solution: $u(n) \xrightarrow{z} \frac{z}{z-1} \quad (|z| > 1)$

$$nu(n) \xrightarrow{z} -z \frac{d}{dz} \left(\frac{z}{z-1} \right) = \frac{z}{(z-1)^2} \quad (|z| > 1)$$

$$\begin{aligned} n^2u(n) \xrightarrow{z} \left[-z \frac{d}{dz} \right]^2 \left(\frac{z}{z-1} \right) &= -z \frac{d}{dz} \left[\frac{z}{(z-1)^2} \right] \\ &= \frac{z(z+1)}{(z-1)^3} \quad (|z| > 1) \end{aligned}$$

7.5 Basic Properties of the z-Transform

Common Sequence z-Transforms Obtained Using the z-Domain Differentiation Property

Z-Transform	Causal Sequence	Z-Transform	Causal Sequence
$\frac{z}{z-1} \quad (z > 1)$	$u(n)$	$\frac{z}{(z-1)^2} \quad (z > 1)$	$nu(n)$
$\frac{z}{z-a} \quad (z > a)$	$a^n u(n)$	$\frac{az}{(z-a)^2} \quad (z > a)$	$na^n u(n)$
$\frac{z^2}{(z-a)^2} \quad (z > a)$	$(n+1)a^n u(n)$	$\frac{a^2 z}{(z-a)^3} \quad (z > a)$	$\frac{n(n-1)}{2!} a^n u(n)$
$\frac{z^3}{(z-a)^3} \quad (z > a)$	$\frac{(n+1)(n+2)}{2!} a^n u(n)$		

Right-shift the sequence by 1 multiply by coefficient a
Right-shift the sequence by 2 multiply by coefficient a^2

Note: Right – shift $(n+1)a^n u(n)$ by 1, multiply by a , resulting in $na^n u(n-1) = 0 \cdot a^0 \delta(n) + 1 \cdot a \delta(n-1) + 2a^2 \delta(n-2) + \dots = na^n u(n)$

If the ROC of the z-transform is $|z| < |a|$, the inverse transform is an anticausal sequence: add a **negative sign** before the sequences in the above table, and replace $u(n)$ with $u(-n-1)$.

7.5 Basic Properties of the z-Transform

7.5.4 z-Domain Scaling (Time-Domain Exponential Weighting)

$$ZT[x(n)] = X(z) \quad (R_{x1} < |z| < R_{x2})$$

$$ZT[a^n x(n)] = X\left(\frac{z}{a}\right) \quad (R_{x1} < \left|\frac{z}{a}\right| < R_{x2} \Rightarrow |a|R_{x1} < |z| < |a|R_{x2}) \quad |a| > 1$$

$$ZT[a^{-n} x(n)] = X(az) \quad (R_{x1} < |az| < R_{x2} \Rightarrow \frac{R_{x1}}{|a|} < |z| < \frac{R_{x2}}{|a|}) \quad |a| > 1$$

$$ZT[(-1)^n x(n)] = X(-z) \quad (R_{x1} < |z| < R_{x2})$$

$$ZT[(-1)^n u(n)] = z/(z+1) \quad (|z| > 1)$$

Example 7-7: Find the z – transform of the sequence $\beta^n \cos(n\omega_0)u(n)$.

Solution: $\cos(n\omega_0)u(n) \xrightarrow{z} \frac{z(z - \cos \omega_0)}{z^2 - 2z \cos \omega_0 + 1}$

then

$$\beta^n \cos(\omega_0 n)u(n) \xrightarrow{z} \frac{\frac{z}{\beta} \left(\frac{z}{\beta} - \cos \omega_0 \right)}{\left(\frac{z}{\beta} \right)^2 - 2 \frac{z}{\beta} \cos \omega_0 + 1}$$
$$\rightarrow \frac{z(z - \beta \cos \omega_0)}{z^2 - 2z\beta \cos \omega_0 + \beta^2}$$

$$\left| \frac{z}{\beta} \right| > 1 \quad |z| > |\beta|$$

7.5 Basic Properties of the z-Transform

7.5.5 Initial Value Theorem

If $x(n)$ is a causal sequence, and $ZT[x(n)] = X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$

then $x(0) = \lim_{z \rightarrow \infty} X(z)$ (the signal jump at $n=0$ corresponds to high-frequency components)

7.5.6 Final Value Theorem

If $x(n)$ is a causal sequence, and $ZT[x(n)] = X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$

then $\lim_{n \rightarrow \infty} x(n) = \lim_{z \rightarrow 1} [(z-1)X(z)]$ ($n \rightarrow \infty$ corresponds to the DC component, i.e., $s=0$, $z=e^s=1$)

It requires that $x(n)$ converges as $n \rightarrow \infty$, i.e., the poles of $X(z)$ must lie **inside the unit circle (if on the unit circle, it should be a first-order pole and can only be at $z=1$)**.

Note the difference from the system stability condition: the condition **for the stability of a causal system** is that the poles of the system function must **lie inside the unit circle**.

7.5 Basic Properties of the z-Transform

The condition for using **the Final Value Theorem** is that the ROC includes **the unit circle or has a first-order pole at $z=1$.**

$x(n)$	Final Value ($n \rightarrow \infty$)	$X(z)$	ROC
$(2)^n$	No final value	$\frac{z}{z-2}$	$ z > 2$
$(1)^n$	Final value = 1	$\frac{z}{z-1}$	$ z > 1$
$(-1)^n$	No final value	$\frac{z}{z+1}$	$ z > 1$
$(0.5)^n$	Final value = 0	$\frac{z}{z-0.5}$	$ z > 0.5$

7.5 Basic Properties of the z-Transform

Example 7-8: Given the z-transform of a causal sequence $X(z) = \frac{1}{1-az^{-1}}$, $|z| > |a|$ (where a is a real number), find the initial value $x(0)$ and $x(\infty)$ of the sequence using the Initial Value Theorem and Final Value Theorem respectively, and derive $x(1)$.

Solution: From the Initial Value Theorem and Final Value Theorem:

$$x(0) = \lim_{z \rightarrow \infty} X(z) = \lim_{z \rightarrow \infty} \frac{1}{1-az^{-1}} = 1$$

$$x(\infty) = \lim_{z \rightarrow 1} (z-1)X(z) = \lim_{z \rightarrow 1} \frac{z-1}{1-az^{-1}} = \lim_{z \rightarrow 1} \frac{z(z-1)}{z-a} = \begin{cases} 1, & a = 1 \\ 0, & |a| < 1 \end{cases}$$

The final value does not exist when $|a| > 1$ or $a = -1$.

Since $x(1) = x(n+1)|_{n=0}$ and $x(n+1) \leftrightarrow z[X(z) - x(0)]$ (shift property of the unilateral z-transform)

$$\text{so } x(1) = \lim_{z \rightarrow \infty} z[X(z) - x(0)] = \lim_{z \rightarrow \infty} \left(\frac{z}{1-az^{-1}} - z \right) = \lim_{z \rightarrow \infty} \left(\frac{a}{1-az^{-1}} \right) = a$$

7.5 Basic Properties of the z-Transform

7.5.7 Time-Domain Convolution Theorem

Given two sequences $x(n)$ and $h(n)$, their z-transforms are:

$$ZT[x(n)] = X(z) \quad (R_{x1} < |z| < R_{x2})$$

$$ZT[h(n)] = H(z) \quad (R_{h1} < |z| < R_{h2})$$

Then the z-transform of the time-domain convolution of the two sequences is equivalent to the product of their z-transforms in the z-domain:

$$ZT[x(n) * h(n)] = X(z)H(z)$$
$$\max(R_{x1}, R_{h1}) < |z| < \min(R_{x2}, R_{h2})$$

The ROC is the overlapping part.

Note: If some zeros and poles cancel each other out, the ROC may expand.

7.5 Basic Properties of the z-Transform



$$\begin{aligned}\text{Proof: } ZT[x(n) * h(n)] &= ZT\left[\sum_{m=-\infty}^{\infty} x(m)h(n-m)\right] \\ &= \sum_{m=-\infty}^{\infty} x(m)ZT[h(n-m)] \\ &= \sum_{m=-\infty}^{\infty} x(m)z^{-m}H(z) \\ &= X(z)H(z)\end{aligned}$$

$$\begin{aligned}\text{Example: } ZT\left[\sum_{k=0}^n x(k)\right] &= ZT\{[x(n)u(n)] * u(n)\} \\ &= \frac{z}{z-1} X(z)\end{aligned}$$

Given that the excitation of a discrete-time system is $x(n) = u(n)$, and the unit sample response is $h(n) = a^n u(n) - a^{n-1} u(n-1)$ ($|a| < 1$), the ROC of the z-transform of the zero-state response is ()

- ☐ A $|a| < |z| < 1$
- ☐ B $|z| < |a|$
- ☒ C $|z| > |a|$
- ☐ D $|z| > 1$

提交

7.5 Basic Properties of the z-Transform

Example 7-9: Given that the excitation of a discrete-time system is $x(n) = u(n)$, and the unit sample response is $h(n) = a^n u(n) - a^{n-1} u(n-1)$, find the zero-state response of the system.

Solution: $X(z) = \frac{z}{z-1}, \quad |z| > 1$

$$H(z) = \frac{z}{z-a} - z^{-1} \frac{z}{z-a} = \frac{z-1}{z-a}, \quad |z| > |a|$$

$$Y_{zs}(z) = X(z)H(z) = \frac{z}{\textcolor{red}{z-1}} \cdot \frac{\textcolor{red}{z-1}}{z-a} = \frac{z}{z-a}, \quad |z| > |a|$$

The pole of $X(z)$ cancels with the zero of $H(z)$;
if $|a| < 1$, the ROC of $X(z)H(z)$ will expand.

the zero – state response is: $y_{zs}(n) = a^n u(n)$

Mapping Relationship Between the z-Plane and s-Plane Translation

$$z = e^{sT} \quad T \text{ is the time interval of the sequence}$$

Express s in rectangular coordinate form and **z in polar coordinate form:**

$$s = \sigma + j\omega, \quad z = re^{j\theta}$$

$$z = e^{(\sigma + j\omega)T} = e^{\sigma T} e^{j\omega T} = re^{j\theta}$$

$$\left\{ \begin{array}{l} r = e^{\sigma T} = e^{\frac{2\pi\sigma}{\omega_s}} \\ \theta = \omega T = 2\pi \frac{\omega}{\omega_s} \end{array} \right.$$

$$\omega_s = \frac{2\pi}{T} \text{ is the repetition frequency}$$

7.6 Relationship Between the z-Transform and the Laplace Transform

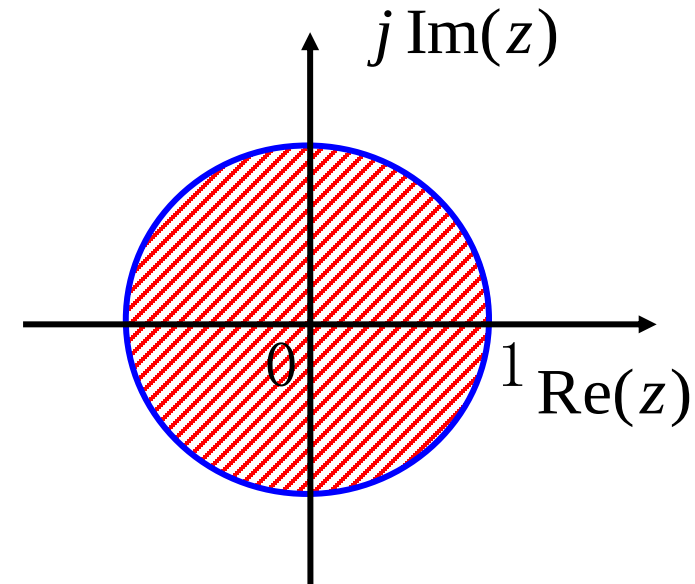
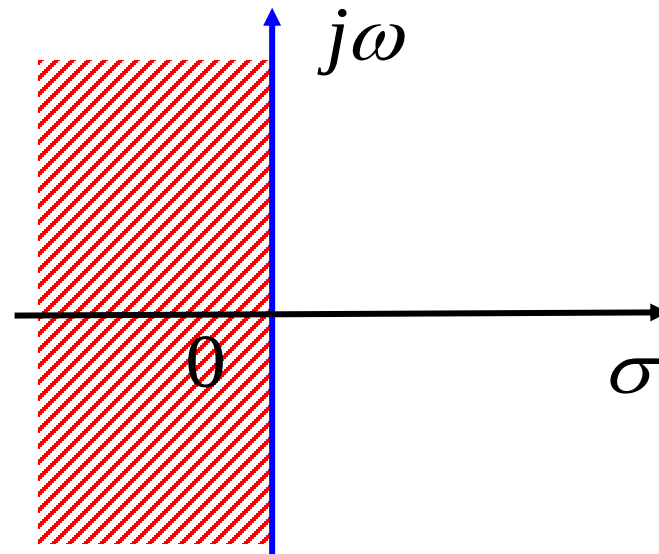


1) The imaginary axis on the s-plane $\xrightarrow{\text{mapping}}$ the unit circle ($r = 1$) in the z-plane

$$\sigma = 0 \quad s = j\omega \quad r = |z| = e^{\sigma T} = 1$$

2) The left half-plane on the s-plane $\xrightarrow{\text{mapping}}$ the interior of the unit circle ($r < 1$) on the z-plane

$$\sigma < 0, s = \sigma + j\omega \quad r = |z| = e^{\sigma T} < 1$$

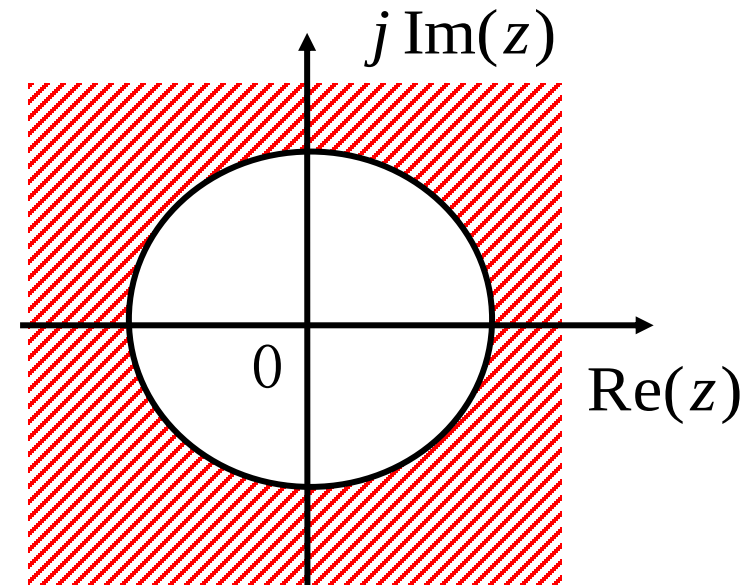
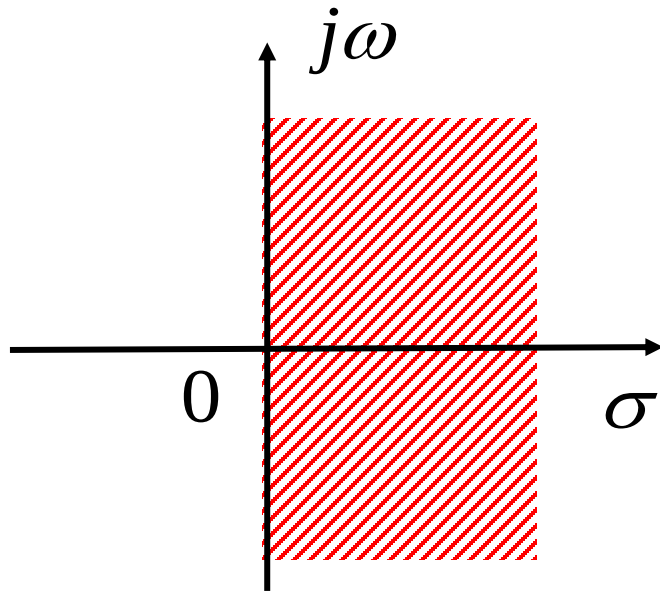


7.6 Relationship Between the z-Transform and the Laplace Transform



3) The right half-plane on the s-plane $\xrightarrow{\text{mapping}}$ the exterior of the unit circle ($r > 1$) on the z-plane

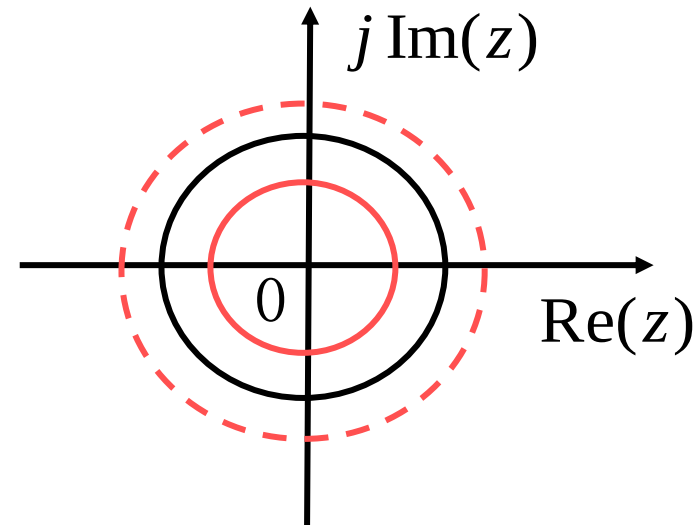
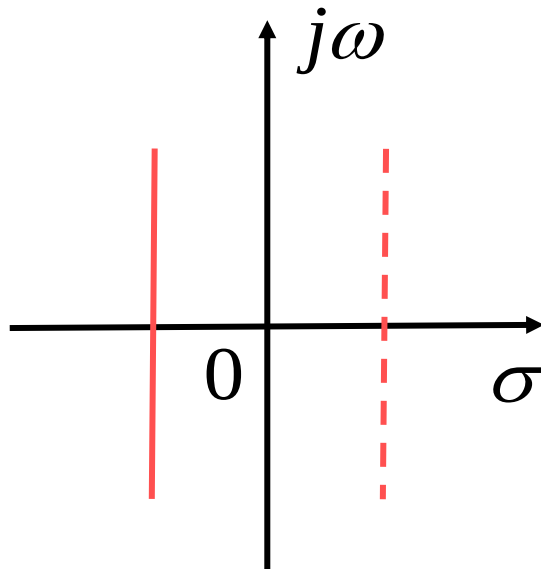
$$\sigma > 0, s = \sigma + j\omega \quad r = |z| = e^{\sigma T} > 1$$



7.6 Relationship Between the z-Transform and the Laplace Transform



4) Lines parallel to the imaginary axis on the s-plane $\xrightarrow{\text{mapping}}$ circles on the z-plane $\begin{pmatrix} \sigma > 0, r > 1 \\ \sigma < 0, r < 1 \end{pmatrix}$



7.6 Relationship Between the z-Transform and the Laplace Transform

5) The real axis ($\omega=0$) on the s-plane $\xrightarrow{\text{mapping}}$ the positive real axis ($\theta=0$) on the z-plane

Lines parallel to the real axis (ω is a constant) $\xrightarrow{\text{mapping}}$

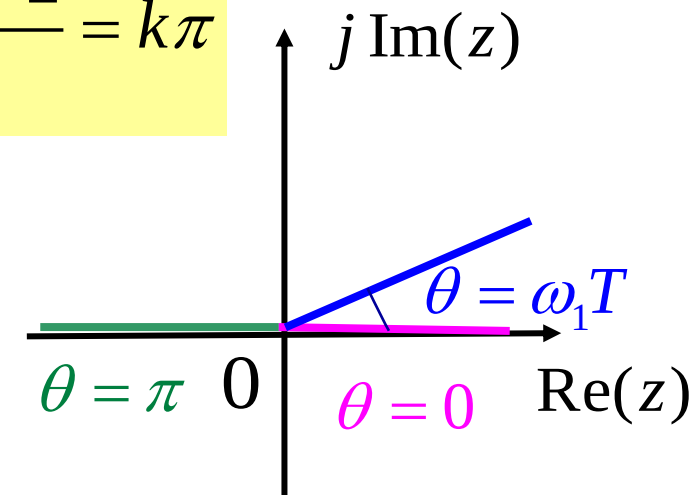
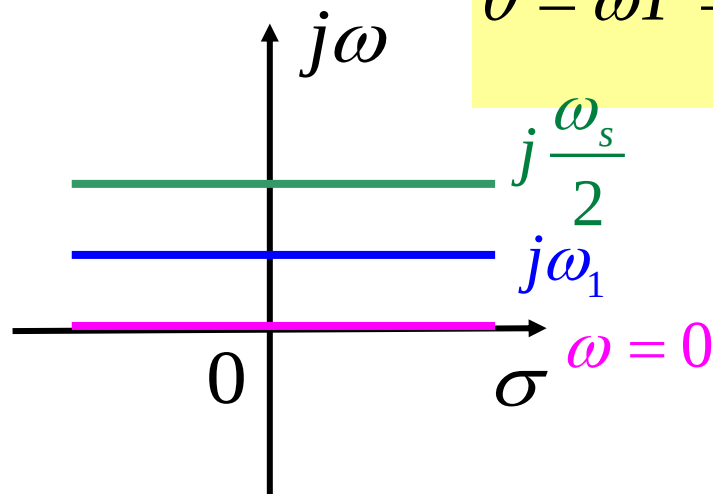
rays originating from the origin on the z-plane

Lines parallel to the real axis (passing through $j \frac{k\omega_s}{2}$ ($k=\pm 1, \pm 3, \dots$)) in the s-plane $\xrightarrow{\text{mapping}}$

the negative real axis in the z-plane $\left(\begin{matrix} \theta = \pi \\ r \text{ is arbitrary} \end{matrix} \right)$

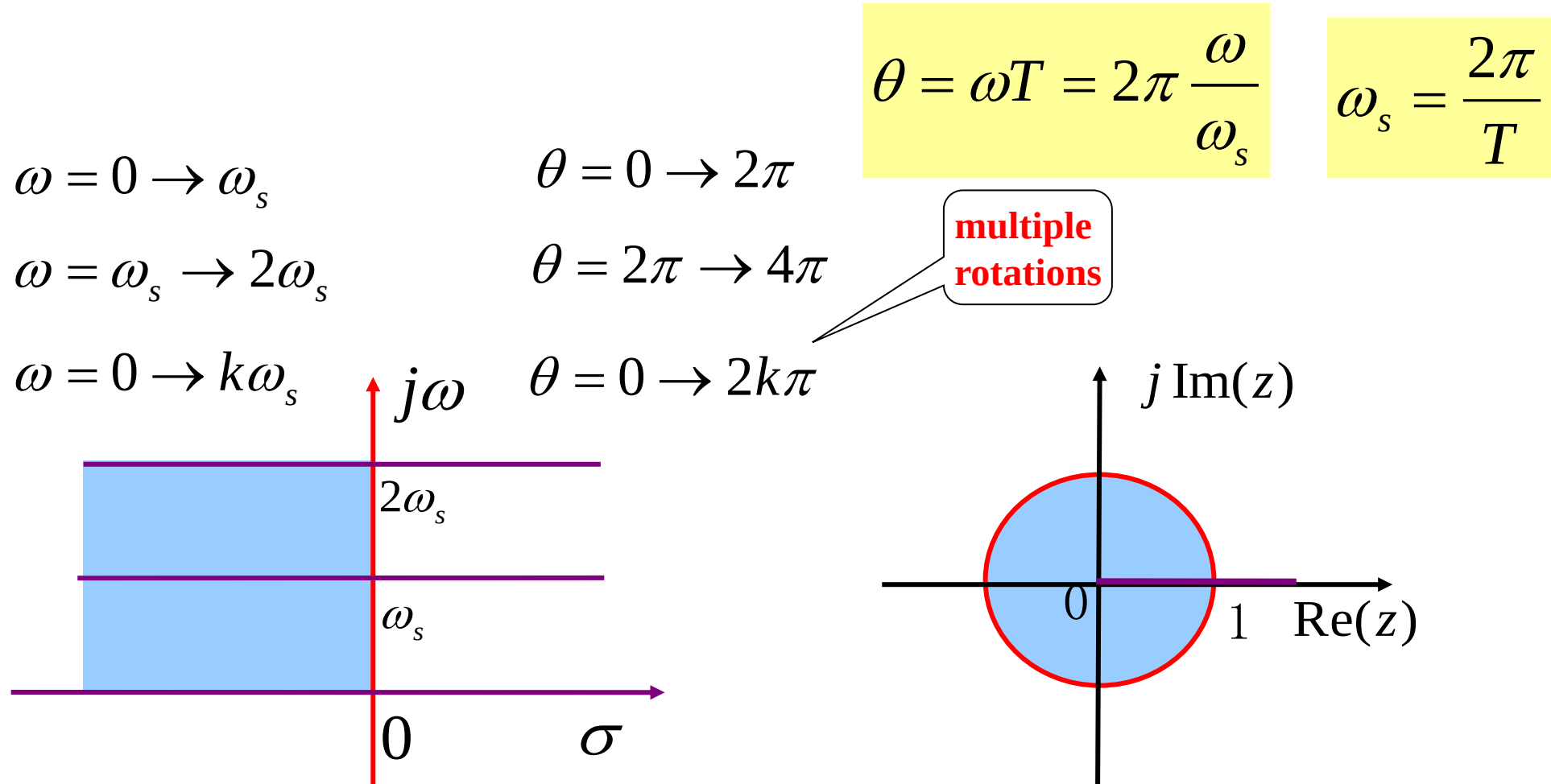
$$\omega_s = \frac{2\pi}{T}$$

$$\theta = \omega T = 2\pi \frac{\omega}{\omega_s} = 2\pi \frac{k\omega_s / 2}{\omega_s} = k\pi$$

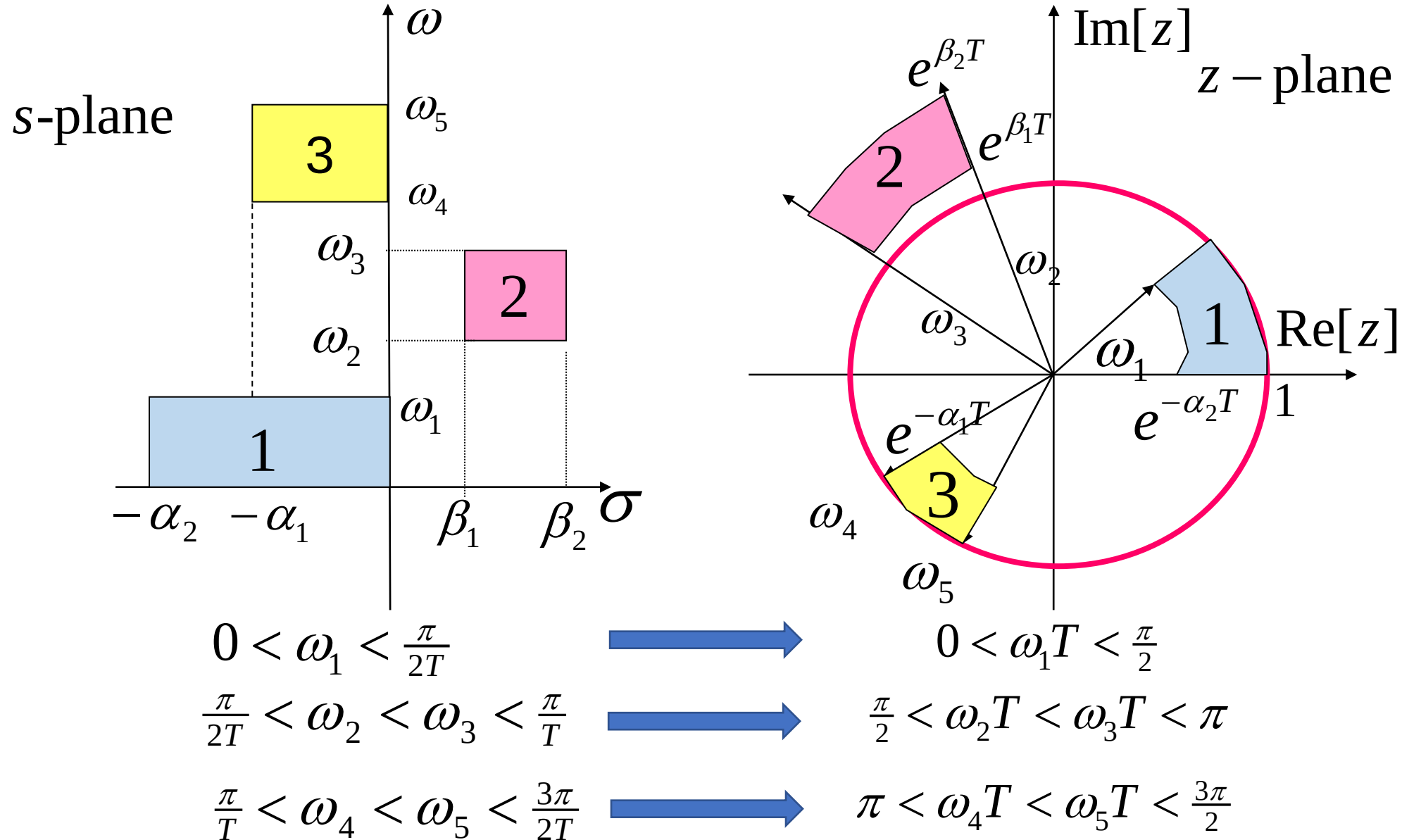


7.6 Relationship Between the z-Transform and the Laplace Transform

6) Moving along the imaginary axis in the s – plane $\xrightarrow{\text{mapping}}$ periodic rotation along the unit circle in the z – plane: each translation by ω_s corresponds to one full rotation around the unit circle, i.e., the mapping between the s-plane and z-plane is not single-valued.



7.6 Relationship Between the z-Transform and the Laplace Transform



1. Repeat all the examples in the lecture notes.
2. 7.21(a)(b)(f) in the textbook.

Review of Last Lecture

7.5 Basic Properties of the z-Transform

**7.6 Relationship between the z-Transform
and the Laplace Transform**

7.5 Basic Properties of the z-Transform

7.5.1 Linearity

if $ZT[x(n)] = X(z) \quad (R_{x1} < |z| < R_{x2})$, $ZT[y(n)] = Y(z) \quad (R_{y1} < |z| < R_{y2})$

then $ZT[ax(n) + by(n)] = aX(z) + bY(z)$

$$\max(R_{x1}, R_{y1}) < |z| < \min(R_{x2}, R_{y2})$$

the ROC is the overlapping part

Note: If **some zeros and poles cancel each other out** in the linear combination, **the ROC may expand**.

7.5.2 Shift Property

1. Bilateral z-transform of a Shifted Bilateral Sequence

$$ZT[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$$ZT[x(n-m)] = z^{-m}X(z)$$

$$ZT[x(n+m)] = z^mX(z) \quad \mathbf{m} \text{ is any positive integer.}$$

7.5 Basic Properties of the z-Transform

2. Unilateral z-transform of a Left-Shifted Bilateral Sequence

$$ZT[x(n+m)u(n)] = \sum_{n=0}^{\infty} x(n+m)z^{-n} = z^m \left[X(z) - \sum_{k=0}^{m-1} x(k)z^{-k} \right]$$

removing the contribution
of some points

3. Unilateral z-Transform of a Right-Shifted Bilateral Sequence

$$ZT[x(n-m)u(n)] = z^{-m} \left[X(z) + \sum_{k=-m}^{-1} x(k)z^{-k} \right]$$

adding the contribution
of some points

4. Unilateral z-Transform of a Shifted Causal Sequence $x(n)$

$$ZT[x(n-m)u(n)] = z^{-m} X(z)$$

$$ZT[x(n+m)u(n)] = z^m \left[X(z) - \sum_{k=0}^{m-1} x(k)z^{-k} \right]$$

Sequence shifting only changes the zero-pole situation of the z-transform at **$z=0$ and ∞** .

7.5 Basic Properties of the z-Transform

7.5.3 z-Domain Differentiation (Time-Domain Linear Weighting)

$$\text{if } ZT[x(n)] = X(z)$$

$$ZT[nx(n)] = -z \frac{d}{dz} X(z)$$

Multiplying the time-domain sequence by n is equivalent to differentiating in the z-domain and multiplying by $(-z)$.

$$ZT[n^2 x(n)] = -z \frac{d}{dz} \left[-z \frac{dX(z)}{dz} \right] = \left[-z \frac{d}{dz} \right]^2 X(z)$$

$$ZT[n^m x(n)] = \left[-z \frac{d}{dz} \right]^m X(z)$$

7.5 Basic Properties of the z-Transform

7.5.4 z-Domain Scaling (Time-Domain Exponential Weighting)

$$ZT[x(n)] = X(z) \quad (R_{x1} < |z| < R_{x2})$$

$$ZT[a^n x(n)] = X\left(\frac{z}{a}\right) \quad (R_{x1} < \left|\frac{z}{a}\right| < R_{x2} \Rightarrow |a|R_{x1} < |z| < |a|R_{x2}) \quad |a| > 1$$

$$ZT[a^{-n} x(n)] = X(az) \quad (R_{x1} < |az| < R_{x2} \Rightarrow \frac{R_{x1}}{|a|} < |z| < \frac{R_{x2}}{|a|}) \quad |a| > 1$$

$$ZT[(-1)^n x(n)] = X(-z) \quad (R_{x1} < |z| < R_{x2})$$

$$ZT[(-1)^n u(n)] = z/(z+1) \quad (|z| > 1)$$

Given $X(z) = \frac{z}{z^2 - \frac{1}{6}z - \frac{1}{6}} \left(|z| > \frac{1}{2} \right)$, $\lim_{n \rightarrow \infty} x(n) = (\quad)$

- ☐ A 1
- ☐ B -1
- ☒ C 0
- ☐ D 2

提交

7.5 Basic Properties of the z-Transform

7.5.5 Initial Value Theorem

If $\mathbf{x(n)}$ is a causal sequence, and $ZT[x(n)] = X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$

then $x(0) = \lim_{z \rightarrow \infty} X(z)$ (the signal jump at $\mathbf{n=0}$ corresponds to high-frequency components)

7.5.6 Final Value Theorem

If $\mathbf{x(n)}$ is a causal sequence, and $ZT[x(n)] = X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$

then $\lim_{n \rightarrow \infty} x(n) = \lim_{z \rightarrow 1} [(z-1)X(z)]$ ($\mathbf{n \rightarrow \infty}$ corresponds to the DC component, i.e., $\mathbf{s=0, z=e^s=1}$)

It requires that $\mathbf{x(n)}$ converges as $n \rightarrow \infty$, i.e., the poles of $X(z)$ must lie **inside the unit circle (if on the unit circle, it should be a first-order pole and can only be at $z=1$).**

Note the difference from the system stability condition: the condition **for the stability of a causal system** is that the poles of the system function must **lie inside the unit circle.**

7.5 Basic Properties of the z-Transform

7.5.7 Time-Domain Convolution Theorem

Given two sequences $x(n)$ and $h(n)$, their z-transforms are:

$$ZT[x(n)] = X(z) \quad (R_{x1} < |z| < R_{x2})$$

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Then the z-transform of the time-domain convolution of the two sequences is equivalent to the product of their z-transforms in the z-domain:

$$ZT[x(n) * h(n)] = X(z)H(z)$$
$$\max(R_{x1}, R_{h1}) < |z| < \min(R_{x2}, R_{h2})$$

The ROC is the overlapping part.

Note: If some zeros and poles cancel each other out, the ROC may expand.

Mapping Relationship Between the z-Plane and s-Plane Translation

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Express s in rectangular coordinate form and **z in polar coordinate form:**

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$$z = e^{(\sigma + j\omega)T} = e^{\sigma T} e^{j\omega T} = re^{j\theta}$$

$$\left\{ \begin{array}{l} r = e^{\sigma T} = e^{\frac{2\pi\sigma}{\omega_s}} \\ \theta = \omega T = 2\pi \frac{\omega}{\omega_s} \end{array} \right.$$

$$\omega_s = \frac{2\pi}{T} \text{ is the repetition frequency}$$

7.6 Relationship Between the z-Transform and the Laplace Transform



1) The imaginary axis on the s-plane $\xrightarrow{\text{mapping}}$ the unit circle ($r = 1$) in the z-plane

$$\sigma = 0, s = j\omega \quad r = |z| = e^{\sigma T} = 1$$

2) The left half-plane on the s-plane $\xrightarrow{\text{mapping}}$ the interior of the unit circle ($r < 1$) on the z-plane

$$\sigma < 0, s = \sigma + j\omega \quad r = |z| = e^{\sigma T} < 1$$

3) The right half-plane on the s-plane $\xrightarrow{\text{mapping}}$ the exterior of the unit circle ($r > 1$) on the z-plane

$$\sigma > 0, s = \sigma + j\omega \quad r = |z| = e^{\sigma T} > 1$$

4) Lines parallel to the imaginary axis on the s-plane $\xrightarrow{\text{mapping}}$ circles on the z-plane $\begin{pmatrix} \sigma > 0, r > 1 \\ \sigma < 0, r < 1 \end{pmatrix}$

7.6 Relationship Between the z-Transform and the Laplace Transform

5) The real axis ($\omega=0$) on the s-plane $\xrightarrow{\text{mapping}}$ the positive real axis ($\theta=0$) on the z-plane

Lines parallel to the real axis (ω is a constant) $\xrightarrow{\text{mapping}}$

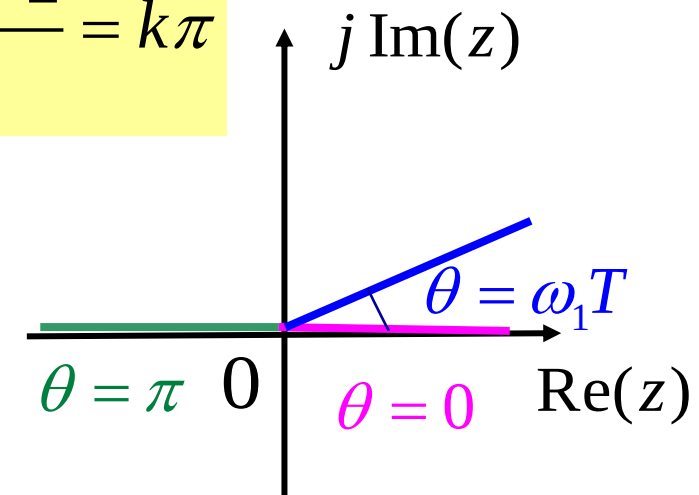
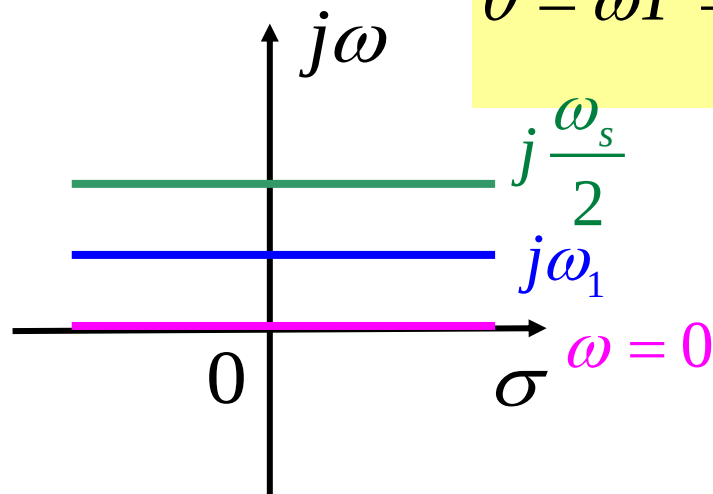
rays originating from the origin on the z-plane

Lines parallel to the real axis (passing through $j \frac{k\omega_s}{2}$ ($k=\pm 1, \pm 3, \dots$)) in the s-plane $\xrightarrow{\text{mapping}}$

the negative real axis in the z-plane $\left(\begin{matrix} \theta = \pi \\ r \text{ is arbitrary} \end{matrix} \right)$

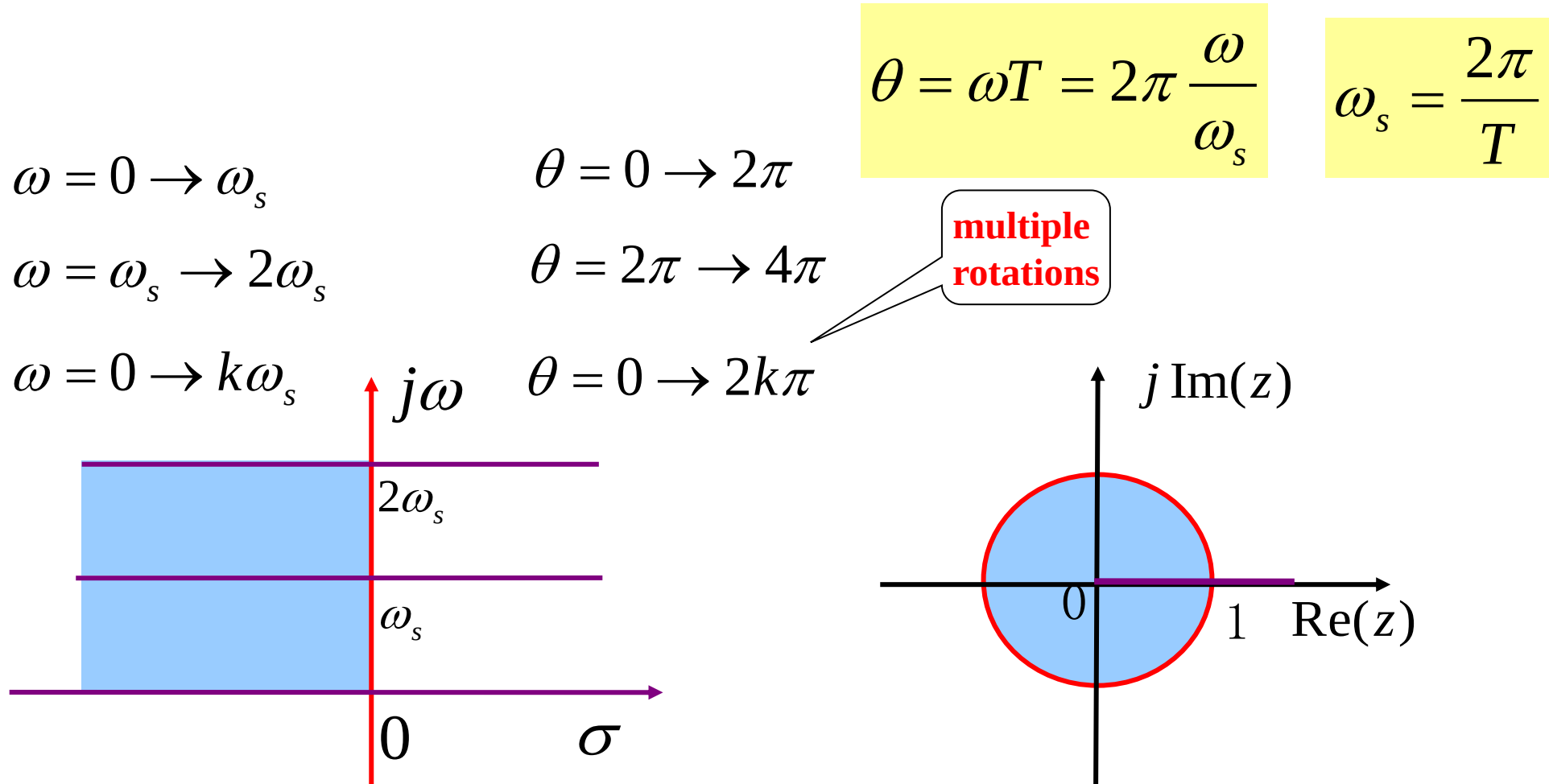
$$\omega_s = \frac{2\pi}{T}$$

$$\theta = \omega T = 2\pi \frac{\omega}{\omega_s} = 2\pi \frac{k\omega_s / 2}{\omega_s} = k\pi$$

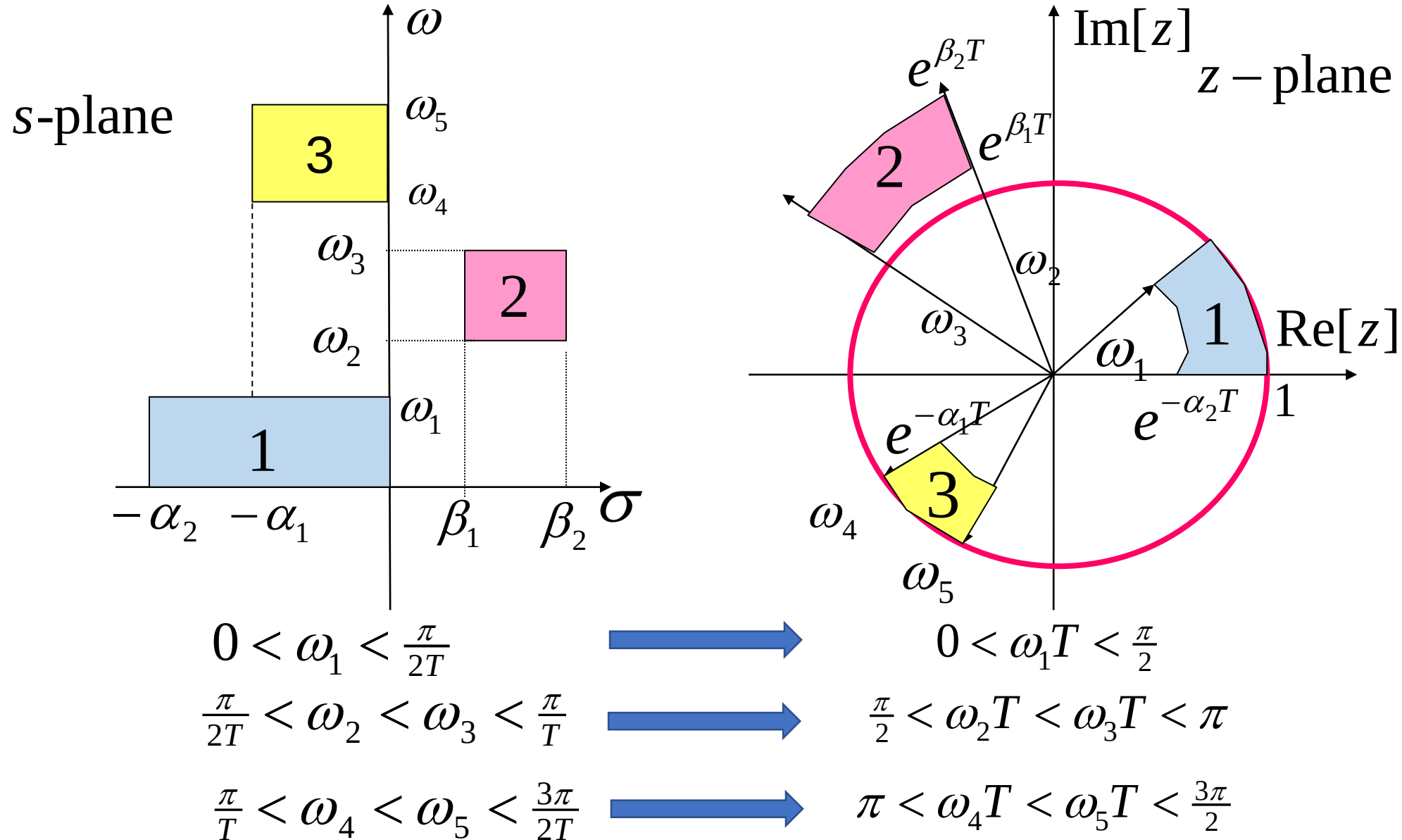


7.6 Relationship Between the z-Transform and the Laplace Transform

6) Moving along the imaginary axis in the s – plane $\xrightarrow{\text{mapping}}$ periodic rotation along the unit circle in the z – plane: each translation by ω_s corresponds to one full rotation around the unit circle, i.e., the mapping between the s-plane and z-plane is not single-valued.



7.6 Relationship Between the z-Transform and the Laplace Transform



Contents of This Lecture

7.7 Solving Difference Equations

Using Z-Transform

7.8 System Function of Discrete-Time Systems

7.9 Signal Flow Graph of Systems

Learning Objectives of This Lecture

1. Proficiency in calculating the **zero-state response** using z-transform;
2. Proficiency in the mapping relationships among **system functions, difference equations, and simulation block diagrams**;
3. Understand the relationship between **the pole distribution of the system function** and the characteristics of **the unit sample response**;
4. Familiarize with the methods for determining system **stability** and **causality**;
5. Given a **differential/difference equation/system function**, be able to proficiently draw the system's signal flow graph.

7.7 Solving Difference Equations using Z-Transform

Principle: Utilizing the **linearity** and **shift** property of the **unilateral z-transform**

1. Shift property of the unilateral z-transform

$$X(z) = \sum_{n=0}^{\infty} x(n)u(n)z^{-n}$$

$$ZT[x(n-m)u(n)] = z^{-m} \left[X(z) + \sum_{k=-m}^{-1} x(k)z^{-k} \right]$$

$$ZT[x(n-1)u(n)] = z^{-1}X(z) + x(-1)$$

$$ZT[x(n-2)u(n)] = z^{-2}X(z) + z^{-1}x(-1) + x(-2)$$

7.7 Solving Difference Equations using Z-Transform

2. Steps for solving difference equations using the unilateral z-transform

$$\sum_{k=0}^N a_k y(n-k) = \sum_{r=0}^M b_r x(n-r)$$

Both $x(n-r)$ and $y(n-k)$ are right-shifted sequences.

Take the unilateral z-transform of both sides:

$$\sum_{k=0}^N a_k z^{-k} [Y(z) + \sum_{l=-k}^{-1} y(l) z^{-l}] = \sum_{r=0}^M b_r z^{-r} [X(z) + \sum_{m=-r}^{-1} x(m) z^{-m}]$$

Initial state

If it is a causal signal,
this term is zero

7.7 Solving Difference Equations using Z-Transform

Zero-State Response

If the initial state of the system is $\mathbf{y}(l)=\mathbf{0}$ ($-N \leq l \leq -1$), the system is in the zero state, and the excitation $\mathbf{x}(n)$ is a causal sequence, then:

$$\sum_{k=0}^N a_k z^{-k} Y(z) = \sum_{r=0}^M b_r z^{-r} X(z)$$

The result obtained is the zero-state response:

$$Y_{zs}(z) = X(z) \cdot \frac{\sum_{r=0}^M b_r z^{-r}}{\sum_{k=0}^N a_k z^{-k}}$$

System Function:

$$H(z) = \frac{\sum_{r=0}^M b_r z^{-r}}{\sum_{k=0}^N a_k z^{-k}}$$

$$Y_{zs}(z) = X(z) H(z)$$

$$y_{zs}(n) = ZT^{-1}[X(z)H(z)]$$

7.7 Solving Difference Equations using Z-Transform

Example 7-10: Given the difference equation $y(n) - 0.9y(n-1) = 0.05x(n)$ of a causal system, with the excitation $x(n) = u(n)$ (unit step sequence), find the zero-state response $y_{zs}(n)$ of the system using the inverse z-transform method.

Solution: Take the z-transform of both sides of the equation:

$$Y(z) - 0.9z^{-1}Y(z) = \frac{0.05z}{z-1} \quad \text{Using linearity and shift property}$$

$$Y(z) = \frac{0.05z^2}{(z-0.9)(z-1)} \quad \frac{Y(z)}{z} = \frac{A_1}{z-0.9} + \frac{A_2}{z-1}$$

$$A_1 = \left. \frac{0.05z}{z-1} \right|_{z=0.9} = -0.45 \quad A_2 = \left. \frac{0.05z}{z-0.9} \right|_{z=1} = 0.5$$

$$Y_{zs}(z) = -\frac{0.45z}{z-0.9} + \frac{0.5z}{z-1} \quad (|z| > 1)$$

$$y_{zs}(n) = [-0.45(0.9)^n + 0.5]u(n)$$

7.8 System Function of Discrete-Time Systems

7.8.1 Definition of the System Function

1. Ratio of the z-transform of the system's zero-state response to the z-transform of the excitation

If $x(n)$ is a **causal sequence** and the system is in the **zero state**:

$$\sum_{k=0}^N a_k y(n-k) = \sum_{r=0}^M b_r x(n-r)$$

Zero-state

$$Y(z) \sum_{k=0}^N a_k z^{-k} = X(z) \sum_{r=0}^M b_r z^{-r}$$

Causal

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{r=0}^M b_r z^{-r}}{\sum_{k=0}^N a_k z^{-k}}$$

2. The z-transform of the unit sample response $h(n)$

The convolution sum of the excitation and the unit sample response is the zero-state response of the system:

$$y_{zs}(n) = x(n) * h(n)$$

By the time-domain convolution theorem:

$$Y_{zs}(z) = X(z)H(z)$$

$$H(z) = \frac{Y_{zs}(z)}{X(z)}$$

$$H(z) = \sum_{n=0}^{\infty} h(n)z^{-n}$$

What is the unit sample response of the discrete-time system described by the difference equation $y(n) - ay(n-1) = bx(n)$? ()

- ☐ A $h(n) = ab^n u(n)$
- ☐ B $h(n) = \left(\frac{a}{b}\right)^n u(n)$
- ☒ C $h(n) = ba^n u(n)$
- ☐ D $h(n) = \left(\frac{b}{a}\right)^n u(n)$

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7.8 System Function of Discrete-Time Systems

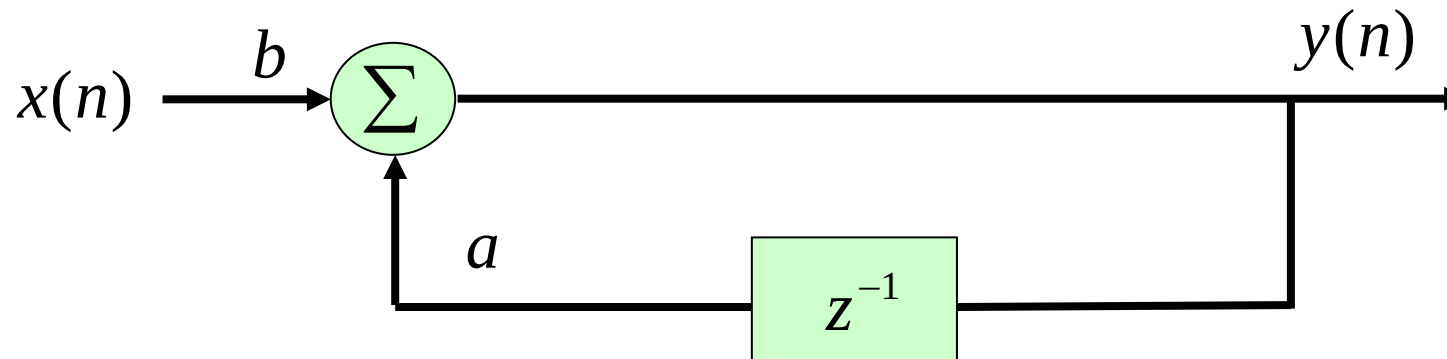
Example 7-11: Find the system function and unit sample response of the discrete-time system described by the following difference equation, and draw the system block diagram.

$$y(n] - ay(n-1) = bx(n]$$

Solution: If the system is in the zero state, then $y(-1)=0$. Taking the z-transform of both sides of the equation gives:

$$Y(z)(1 - az^{-1}) = bX(z)$$

$$H(z) = \frac{b}{1 - az^{-1}} = \frac{bz}{z - a} \quad (|z| > |a|) \quad h(n) = ba^n u(n)$$



7.8 System Function of Discrete-Time Systems

Example 7-12: Suppose the system function of a second-order discrete linear time-invariant system is:

$$H(z) = \frac{(1+z^{-1})^2}{(1-\frac{1}{2}z^{-1})(1+\frac{3}{4}z^{-1})}$$

Find the difference equation corresponding to this system, and draw the system simulation block diagram.

Solution:

$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}} = \frac{Y(z)}{X(z)}$$

$$\left(1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}\right)Y(z) = (1 + 2z^{-1} + z^{-2})X(z)$$

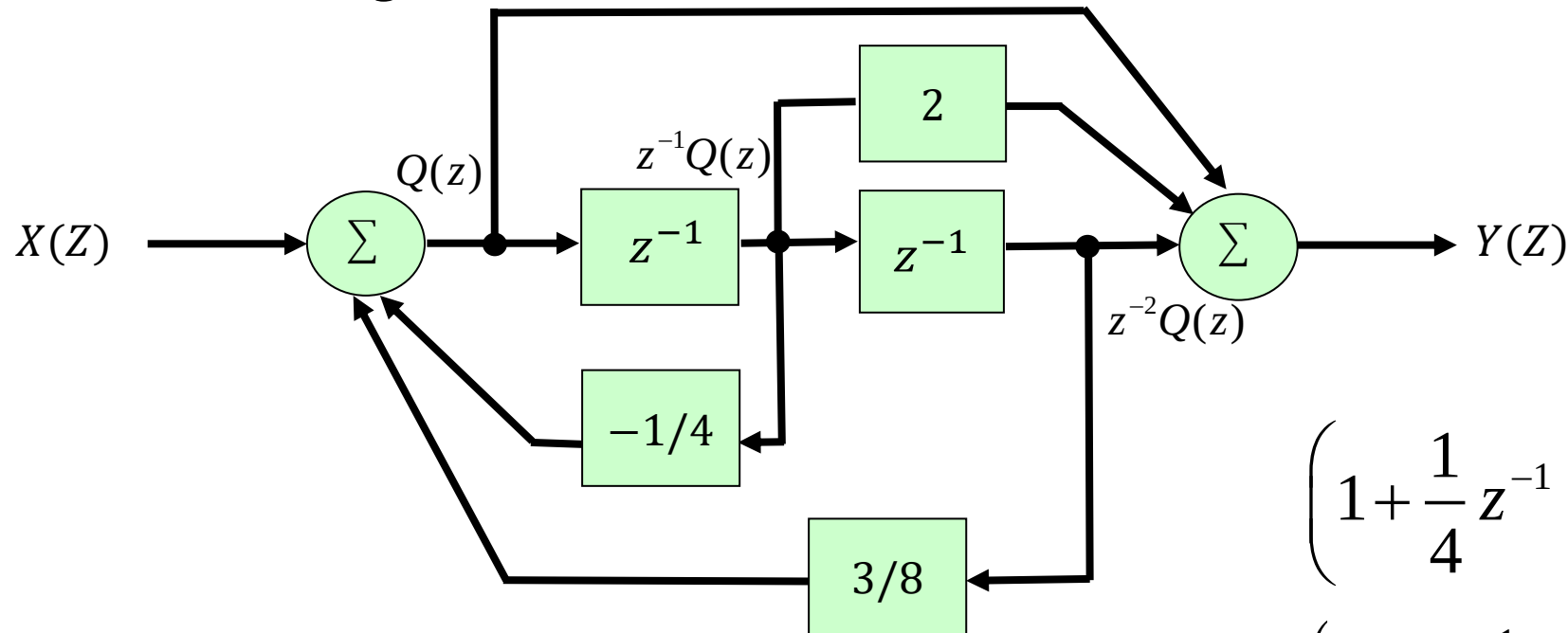
The system difference equation is

$$y(n) + \frac{1}{4}y(n-1) - \frac{3}{8}y(n-2) = x(n) + 2x(n-1) + x(n-2)$$

7.8 System Function of Discrete-Time Systems

$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}} = \frac{Y(z)}{X(z)}$$

system block diagram:



$$\left(1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}\right)Q(z) = X(z)$$

$$\left(1 + 2z^{-1} + z^{-2}\right)Q(z) = Y(z)$$

7.8 System Function of Discrete-Time Systems

7.8.2 Analyzing the Unit Sample Response by the Zero-Pole Distribution of the System Function

$$H(z) = \left[G \frac{\prod_{r=1}^M (1 - z_r z^{-1})}{\prod_{k=1}^N (1 - p_k z^{-1})} \right] = \sum_{k=0}^N \frac{A_k z}{z - p_k} = A_0 + \sum_{k=1}^N \frac{A_k z}{z - p_k}$$

where z_r and p_k are the zeros and poles of $H(z)$ respectively; $p_0 = 0$

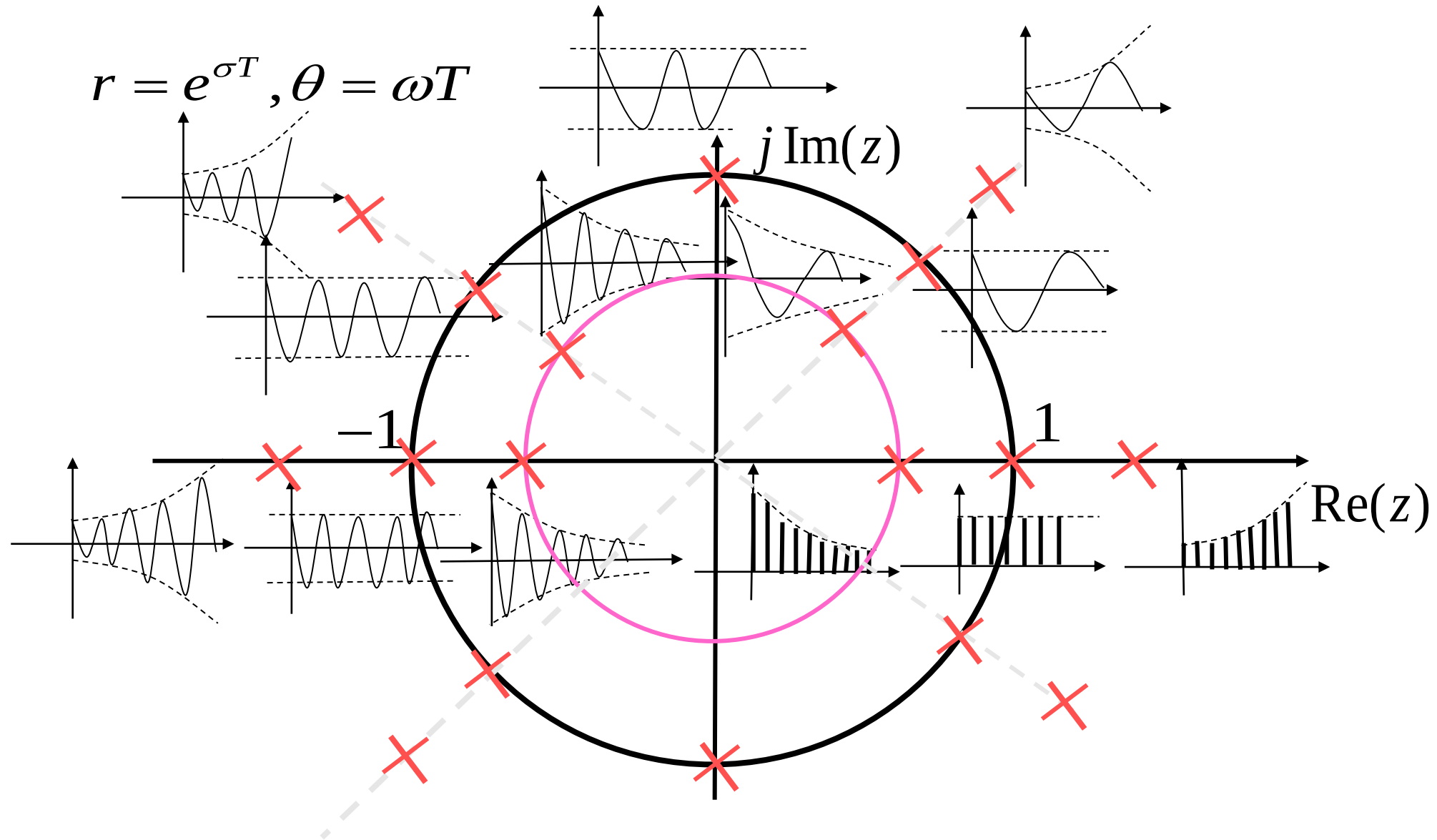
$$h(n) = ZT^{-1}[H(z)] = A_0 \delta(n) + \sum_{k=1}^N A_k (p_k)^n u(n)$$

$\therefore$ The characteristics of $h(n)$ depend on the poles p_k of $H(z)$

The amplitude A_k of $h(n)$ depends on the zeros z_r of $H(z)$

7.8 System Function of Discrete-Time Systems

Effects of the Pole Distribution on $h(n)$



7.8 System Function of Discrete-Time Systems

Example 7-13: Given the difference equation of the system below, find the unit sample response of the system using the z-domain analysis.

$$y(n) - 5y(n-1) + 6y(n-2) = x(n) - 3x(n-2)$$

Solution: (1) Take the z-transform of both sides of the difference equation, we get

$$(1 - 5z^{-1} + 6z^{-2})Y(z) = (1 - 3z^{-2})X(z)$$

(2) The system function is:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 - 3z^{-2}}{1 - 5z^{-1} + 6z^{-2}} = \frac{z^2 - 3}{z^2 - 5z + 6}$$

$$\frac{H(z)}{z} = \frac{z^2 - 3}{z(z-2)(z-3)} = -\frac{1}{2z} - \frac{\frac{1}{2}}{z-2} + \frac{2}{z-3}$$

$$H(z) = -\frac{1}{2} - \frac{1}{2} \frac{z}{z-2} + \frac{2z}{z-3} \quad (|z| > 3)$$

(3) Take the inverse z-transform of the system function, the unit sample response is:

$$h(n) = -\frac{1}{2}\delta(n) + (-2^{n-1} + 2 \times 3^n)u(n) = \delta(n) + 5\delta(n-1) + (2 \times 3^n - 2^{n-1})u(n-2)$$

7.8.3 Stability and Causality of Discrete-Time Systems

1. Conditions for Causal Stability of the System in the Time Domain

The **necessary and sufficient condition** for the stability of a discrete-time system is that the **unit sample response is absolutely summable**, i.e.:

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

The **necessary and sufficient condition** for a **causal stable system** is that **$h(n)$** is a **causal sequence and absolutely summable**, i.e.:

$$\left\{ \begin{array}{ll} h(n) = h(n)u(n) & \text{Causal} \\ \sum_{n=-\infty}^{\infty} |h(n)| < \infty & \text{Stable} \end{array} \right.$$

2. Conditions for Causal Stability in the z-Domain

$$H(z) = \sum_{n=-\infty}^{\infty} h(n)z^{-n}, \text{ its existence requires: } \sum_{n=-\infty}^{\infty} |h(n)z^{-n}| < \infty$$

when $|z|=1$, $\sum_{n=-\infty}^{\infty} |h(n)| < \infty$, just satisfies the condition for system stability.

Therefore, for a **stable system**, the ROC of $H(z)$ should **include the unit circle** (regardless of whether the system is causal).

For a **causal stable system**, the ROC is $|z| > a$ ($a < 1$), which means **all poles lie inside the unit circle**.

7.8 System Function of Discrete-Time Systems

Example 7-14: Given the system function below, explain the stability of the system under three ROC conditions, and find the unit sample response.

$$H(z) = \frac{-9.5z}{(z - 0.5)(z - 10)} \quad \begin{array}{ll} (1) 10 < |z| \leq \infty & (2) 0.5 < |z| < 10 \\ (3) |z| < 0.5 \end{array}$$

Solution: (1) $10 < |z| \leq \infty$

Method 1: From the ROC, this is a causal system. The ROC does not include the unit circle, so the system is unstable.

Method 2: Judge from the ROC that this system is causal.

$$z_1 = 0.5 \quad z_2 = 10 \quad |z_2| > 1$$

The poles of a causal stable system lie inside the unit circle. Here, one pole is outside the unit circle, so the system is unstable.

$$h(n) = [(0.5)^n - (10)^n]u(n)$$

7.8 System Function of Discrete-Time Systems



$$(2) \quad 0.5 < |z| < 10$$

Right-sided
sequence

Left-sided
sequence

The ROC includes the unit circle, so the system is stable.

Judge from the ROC that this system is non-causal.

$$H(z) = \frac{z}{z - 0.5} - \frac{z}{z - 10}$$

$$h(n) = (0.5)^n u(n) + (10)^n u(-n-1)$$

$$(3) \quad |z| < 0.5 \quad \text{Left-sided sequence}$$

The ROC does not include the unit circle, so the system is unstable.

Judge from the ROC that this system is non-causal.

$$h(n) = [(10)^n - (0.5)^n] u(-n-1)$$

From the above, **the stability condition for a non-causal system** is also that **the ROC of the system function includes the unit circle.**

Given the difference equation of a causal system:

$$y(n) + 0.2y(n - 1) - 0.24y(n - 2) = x(n) + x(n - 1)$$

Is this system stable?

- ☒ A Stable
- ☐ B Unstable

提交

7.8 System Function of Discrete-Time Systems

Example 7-15:

Given $y(n) + 0.2y(n-1) - 0.24y(n-2) = x(n) + x(n-1)$, find $H(z)$ and $h(n)$, discuss the ROC and stability of this causal system. If the excitation signal is $x(n) = u(n)$, find the zero – state response.

Solution : (1) Find the system function

Take the z-transform of both sides of the difference equation:

$$Y(z) + 0.2z^{-1}Y(z) - 0.24z^{-2}Y(z) = X(z) + z^{-1}X(z)$$

$$\text{Rearrange to get: } \left(1 + \frac{0.2}{z} - \frac{0.24}{z^2}\right)Y(z) = \left(1 + \frac{1}{z}\right)X(z)$$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{z(z+1)}{z^2 + 0.2z - 0.24} = \frac{z(z+1)}{(z-0.4)(z+0.6)}$$

7.8 System Function of Discrete-Time Systems



(2) Find the ROC and determine stability

$$H(z) = \frac{z(z+1)}{(z-0.4)(z+0.6)} \quad \because z_1 = 0.4, \quad z_2 = -0.6$$

The system is causal, $\therefore |z| > 0.6$

The ROC includes the unit circle, so the system is stable.

(3) Find the unit sample response $h(n)$

$$\frac{H(z)}{z} = \frac{1.4}{z-0.4} - \frac{0.4}{z+0.6} \quad H(z) = \frac{1.4z}{z-0.4} - \frac{0.4z}{z+0.6}$$

$$h(n) = [1.4(0.4)^n - 0.4(-0.6)^n]u(n)$$

7.8 System Function of Discrete-Time Systems



(4) Find the zero-state response

$$x(n) = u(n) \qquad X(z) = \frac{z}{z-1} \quad (|z| > 1)$$

$$Y_{zs}(z) = H(z) X(z) = \frac{z^2(z+1)}{(z-1)(z-0.4)(z+0.6)}$$

$$Y_{zs}(z) = \frac{2.08z}{z-1} - \frac{0.93z}{z-0.4} - \frac{0.15z}{z+0.6} \quad (|z| > 1)$$

zero-state response

$$y_{zs}(n) = [2.08 - 0.93(0.4)^n - 0.15(-0.6)^n]u(n)$$

7.9 Signal Flow Graph of Systems

1. Overview

Systems (continuous or discrete) can be described using block diagrams, which are more intuitive than differential equations or difference equations.

Simulation of linear systems:

- Continuous systems — addition, multiplication, integration
- Discrete systems — addition, multiplication, delay

System block diagram $\xrightarrow{\text{simplify}}$ **Signal flow graph**

The **Signal Flow Graph** was proposed by Mason (from MIT) in the 1950s. It is applied in feedback system analysis, linear equation system solving, linear system simulation, digital filter design, etc.

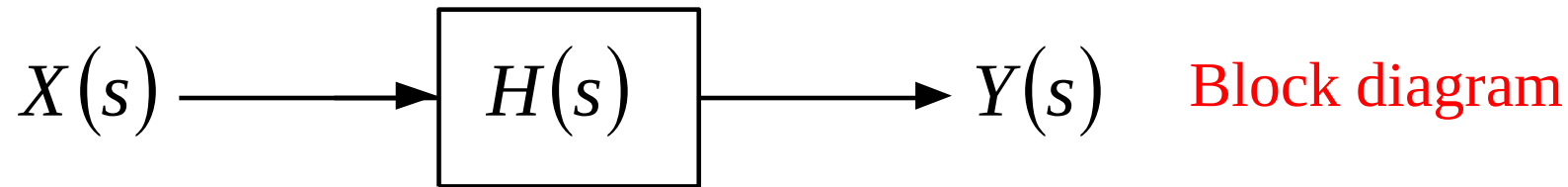
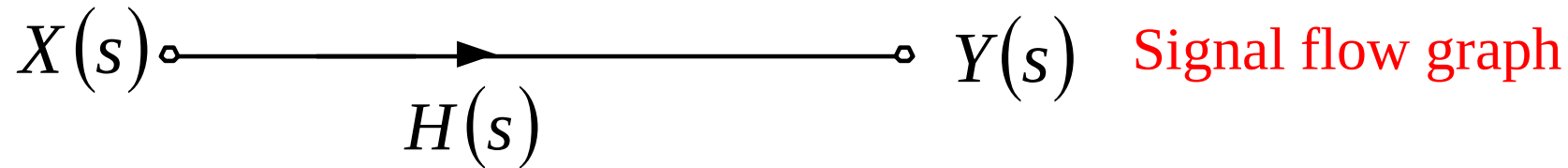
Key advantages of the signal flow graph method:

- The system model is represented concisely and clearly;
- Simplifies the calculation equations of the system function.

7.9 Signal Flow Graph of Systems

2. Representation of Systems Using Signal Flow Graphs

In fact, systems are described using **nodes** and **branches**:



$X(s)$ 、 $Y(s)$ are called **nodes**.

The line segment representing the signal transmission path is called a **branch**.

The signal transmission direction is indicated by an arrow; the **system function (transfer function) is marked near the arrow, which acts as a multiplier.**

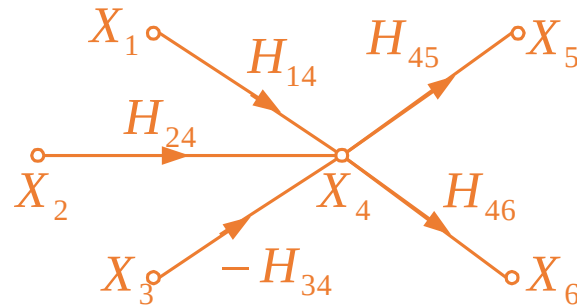
7.9 Signal Flow Graph of Systems

3. Properties of Signal Flow Graphs

(1) A branch represents the functional relationship between two signals; signals can only pass along the direction of the arrow on the branch:

$$X(s) \xrightarrow{H(s)} Y(s) \quad Y(s) = H(s)X(s)$$

(2) A node can superimpose the signals of all input branches, and transmit the summed signal to all output branches:



For example, node X_4

$$X_4 = H_{14}X_1 + H_{24}X_2 - H_{34}X_3$$

(3) The **transfer function** remains unchanged after the signal flow graph is transposed. Transposition means reversing the signal transmission direction of all branches in the graph, while swapping the input and output nodes.

(4) For a given system, **the signal flow graph is not unique in form.**

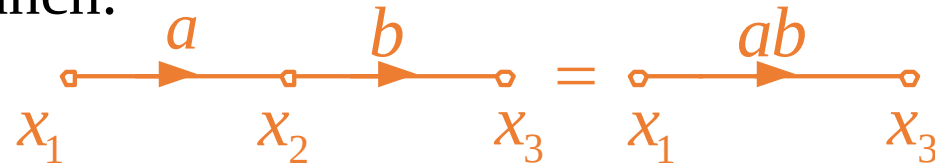
7.9 Signal Flow Graph of Systems

4. Algebraic Operations of Signal Flow Graphs

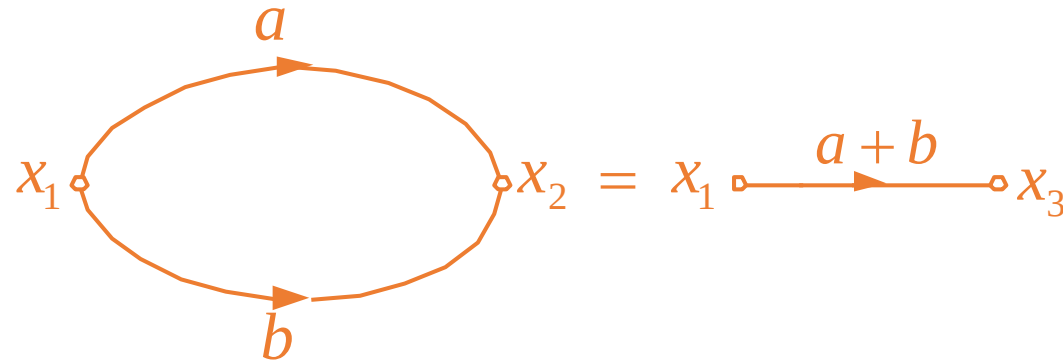
(1) The value of a node with one input branch equals the input signal multiplied by the branch gain:



(2) **Merging of series branches:** The total gain equals the product of the gains of each branch:

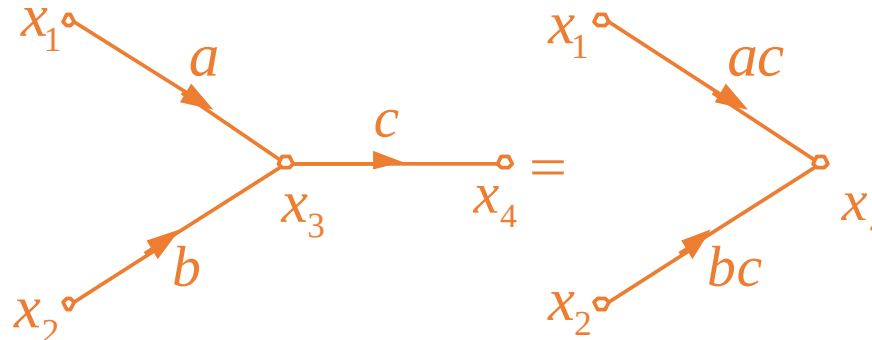


(3) **Merging of parallel branches:** Sum the gains of parallel branches

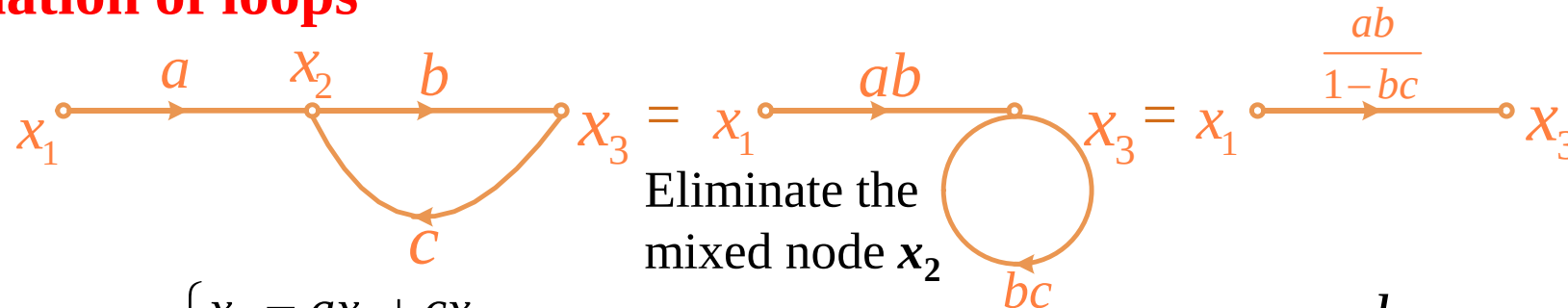


7.9 Signal Flow Graph of Systems

(4) Elimination of mixed nodes



(5) Elimination of loops



$$\text{Thus } \begin{cases} x_2 = ax_1 + cx_3 \\ x_3 = bx_2 \end{cases} \Rightarrow x_3 = abx_1 + bcx_3 \Rightarrow x_3 = \frac{ab}{1-bc} x_1$$

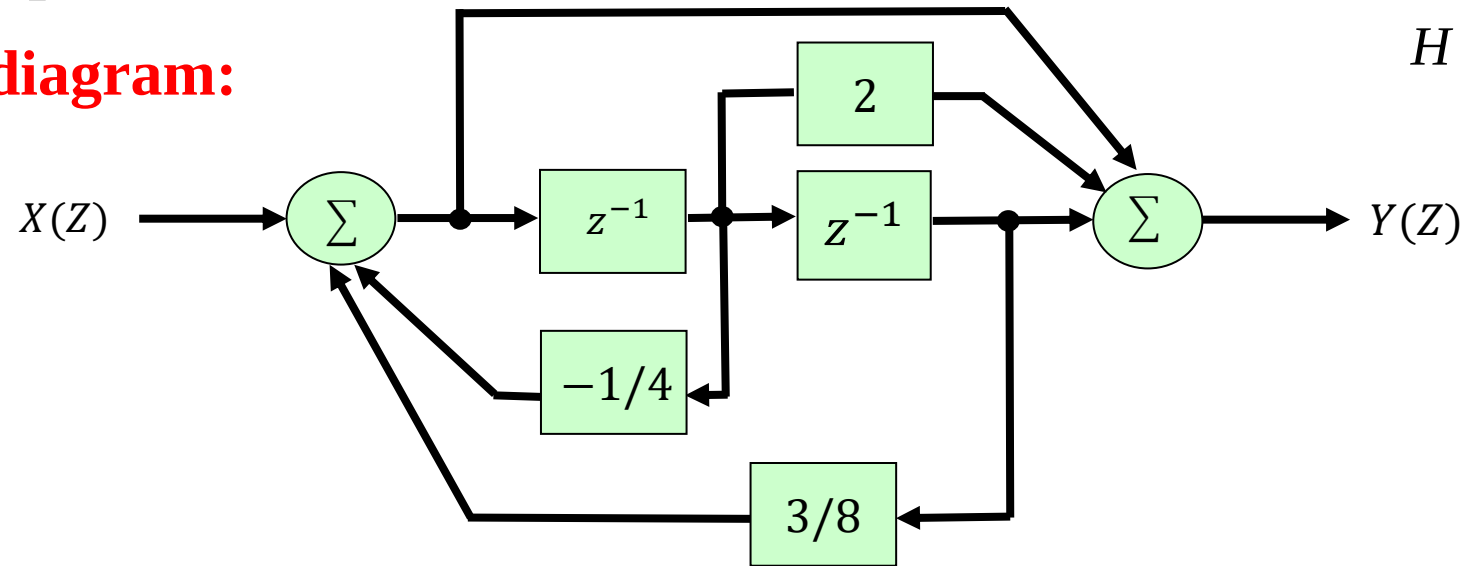
Summary: The signal flow graph can be simplified through the following steps to obtain the system function:

- ① **Merge series branches to reduce nodes;**
- ② **Merge parallel branches to reduce branches;**
- ③ **Eliminate loops**

7.9 Signal Flow Graph of Systems

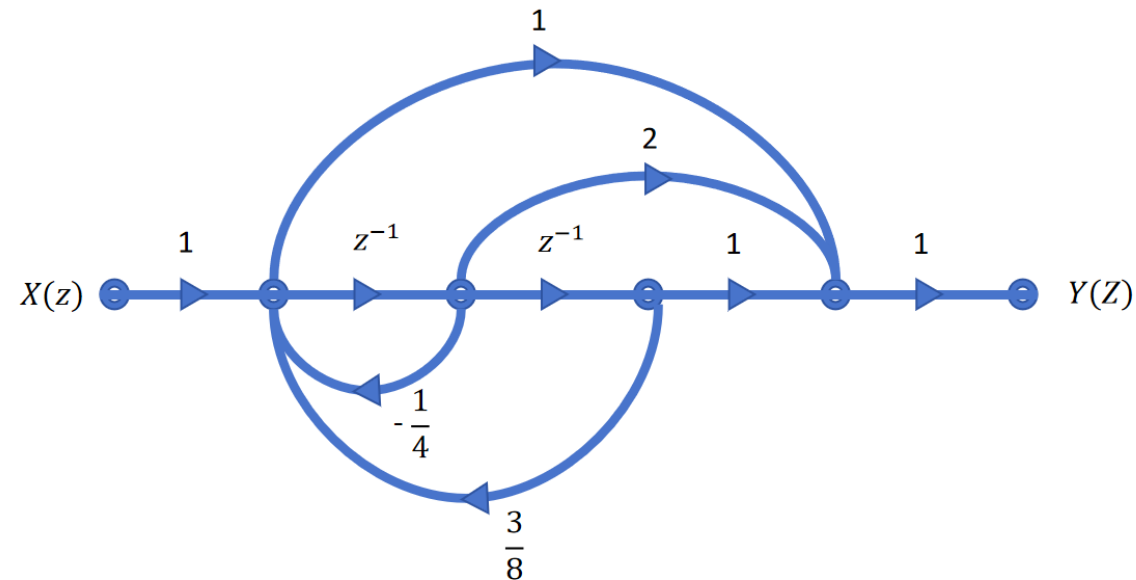
Example 7-12

Block diagram:



$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}} = \frac{Y(z)}{X(z)}$$

Signal flow graph:



1. Repeat all the examples in the lecture notes.
2. 7.34(a) (judge only causality and stability) in the textbook.
3. Given the following system difference equation, find the system function $H(z)$ and the unit sample response $h(n)$. If the excitation signal is $x(n) = u(n)$, find the zero-state response.

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = 2x(n)$$

Review of Last Lecture

7.7 Solving Difference Equations Using z-Transform

7.8 System Function of Discrete-Time Systems

7.9 Signal Flow Graph of Systems

7.7 Solving Difference Equations using Z-Transform

Zero-State Response

$$\sum_{k=0}^N a_k y(n-k) = \sum_{r=0}^M b_r x(n-r)$$

If the initial state of the system is $\mathbf{y(l)=0}$ ($-N \leq l \leq -1$), the system is in the zero state, and the excitation $\mathbf{x(n)}$ is a causal sequence, then:

$$\sum_{k=0}^N a_k z^{-k} Y(z) = \sum_{r=0}^M b_r z^{-r} X(z)$$

System Function:

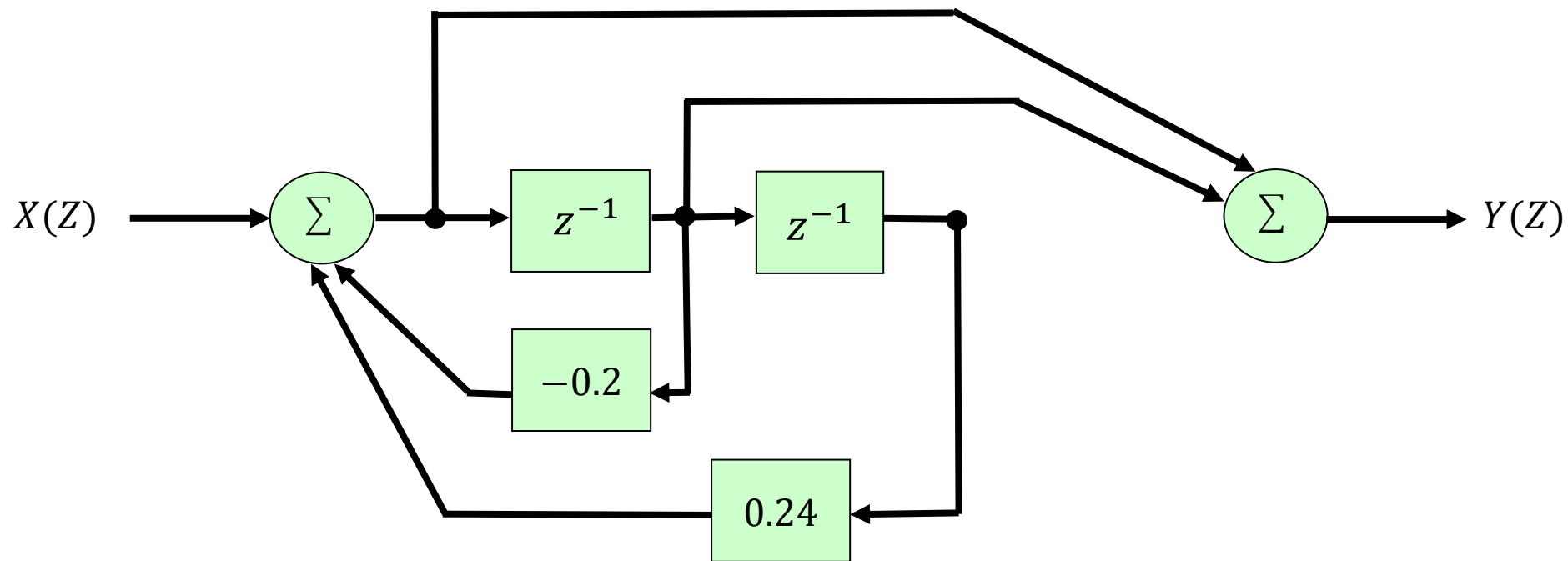
$$H(z) = \frac{\sum_{r=0}^M b_r z^{-r}}{\sum_{k=0}^N a_k z^{-k}}$$

$$Y_{zs}(z) = X(z) H(z)$$

$$y_{zs}(n) = ZT^{-1}[X(z) H(z)]$$

Fill the blanks in the system difference equation with the block diagram below:

$$y(n) + [\text{填空1}] y(n-1) + [\text{填空2}] y(n-2) = x(n) + [\text{填空3}] x(n-1) + [\text{填空4}] x(n-2)$$



作答

$$H(z) = \sum_{n=0}^{\infty} h(n)z^{-n}$$


7.8.3 Stability and Causality of Discrete-Time Systems

1. Conditions for Causal Stability of the System in the Time Domain

The **necessary and sufficient condition** for the stability of a discrete-time system is that the **unit sample response is absolutely summable**, i.e.:

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

The **necessary and sufficient condition** for a **causal stable system** is that **$h(n)$** is a **causal sequence and absolutely summable**, i.e.:

$$\left\{ \begin{array}{ll} h(n) = h(n)u(n) & \text{Causal} \\ \sum_{n=-\infty}^{\infty} |h(n)| < \infty & \text{Stable} \end{array} \right.$$

2. Conditions for Causal Stability in the z-Domain

$$H(z) = \sum_{n=-\infty}^{\infty} h(n)z^{-n}, \text{ its existence requires: } \sum_{n=-\infty}^{\infty} |h(n)z^{-n}| < \infty$$

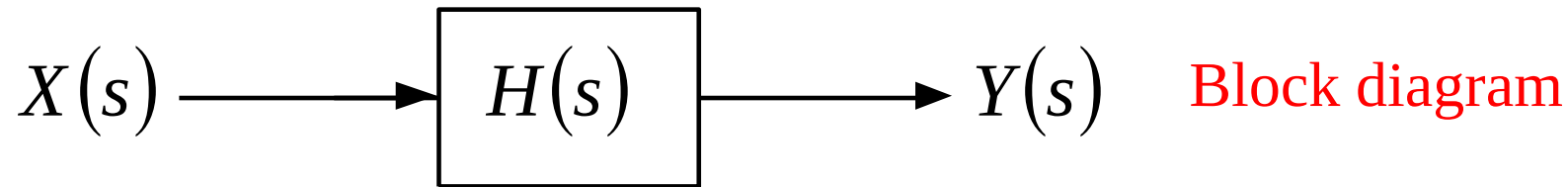
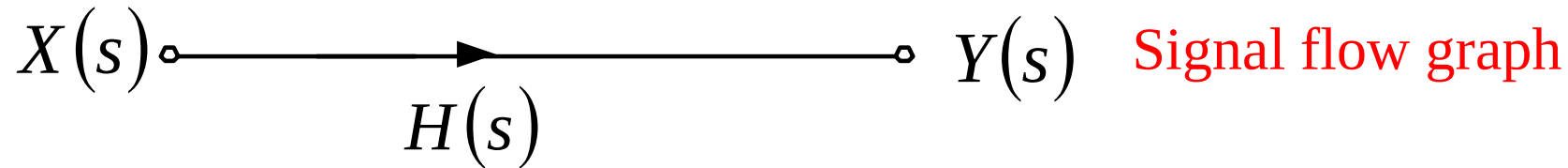
when $|z|=1$, $\sum_{n=-\infty}^{\infty} |h(n)| < \infty$, just satisfies the condition for system stability.

Therefore, for a **stable system**, the ROC of $H(z)$ should **include the unit circle** (regardless of whether the system is causal).

For a **causal stable system**, the ROC is $|z| > a$ ($a < 1$), which means **all poles lie inside the unit circle**.

Representation of Systems Using Signal Flow Graphs

Systems are described using **nodes** and **branches**:



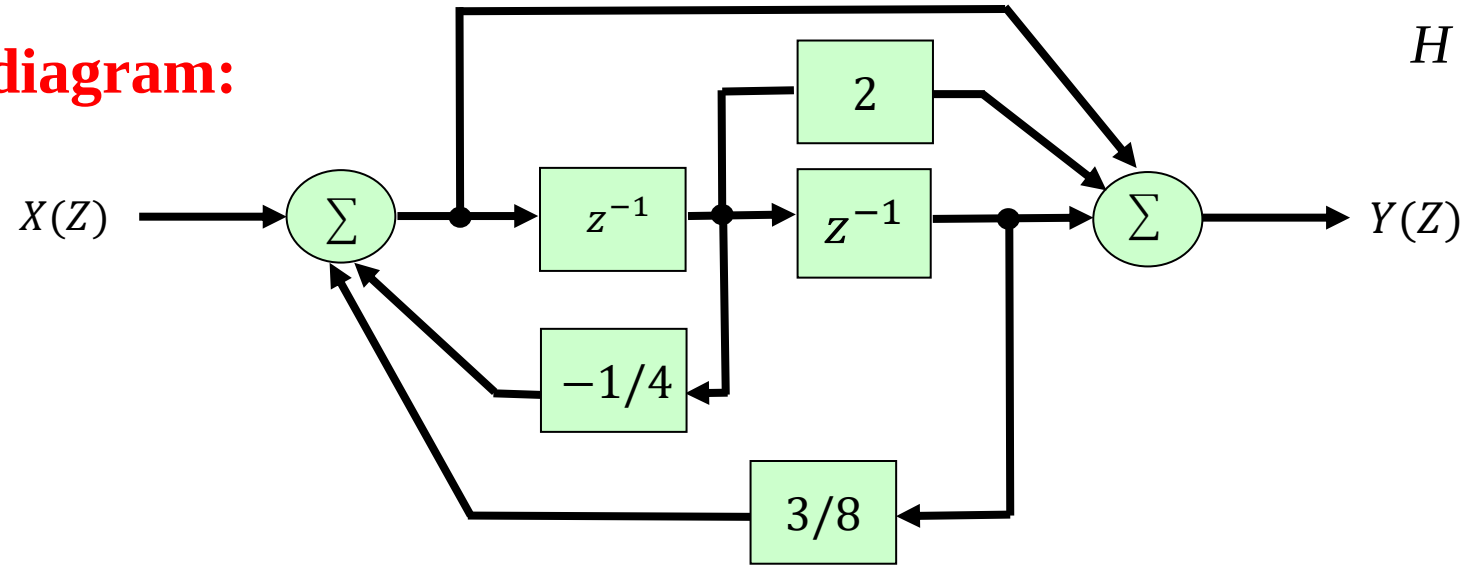
$X(s)$ 、 $Y(s)$ are called **nodes**.

The line segment representing the signal transmission path is called a **branch**.

The signal transmission direction is indicated by an arrow; the **system function (transfer function) is marked near the arrow, which acts as a multiplier**.

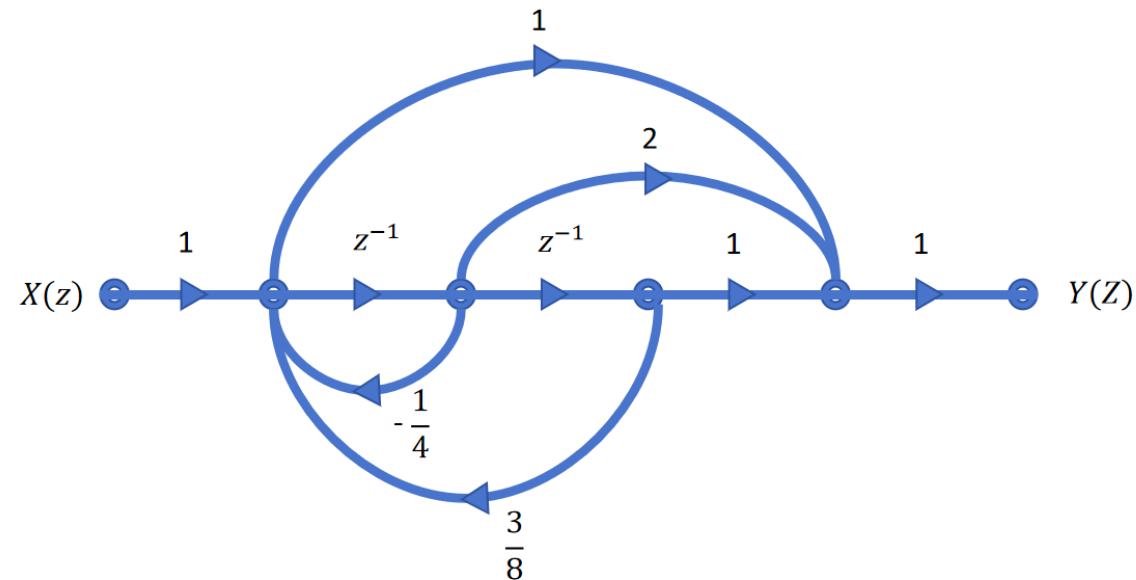
7.9 Signal Flow Graph of Systems

Block diagram:



$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}} = \frac{Y(z)}{X(z)}$$

Signal flow graph:



Contents of This Lecture

7.10 Fourier Transform of Sequences

7.11 Frequency Response of
Discrete-Time Systems

Learning Objectives of This Lecture

1. Understand the difference between **analog angular frequency** and **digital angular frequency**, and grasp the **periodicity** of the Fourier transform of sequences;
2. Skillfully use the frequency response of discrete-time systems to calculate the **steady-state response**;
3. Skillfully use the **pole-zero plot** of the system function to roughly sketch the **frequency response characteristic curve**.

7.10 Fourier Transform of Sequences (DTFT)

7.10.1 Definition

The Fourier transform of a sequence is also called the **Discrete-time Fourier Transform (DTFT)**, which serves as a foundation for studying the frequency response of discrete-time systems.

Fourier Transform of Continuous Signals

$$X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega$$

Fourier Transform of Discrete Sequences

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

Comparison of Frequencies

Analog angular frequency Ω , dimension: radians/second ;

Digital angular frequency ω , dimension: **radians**;

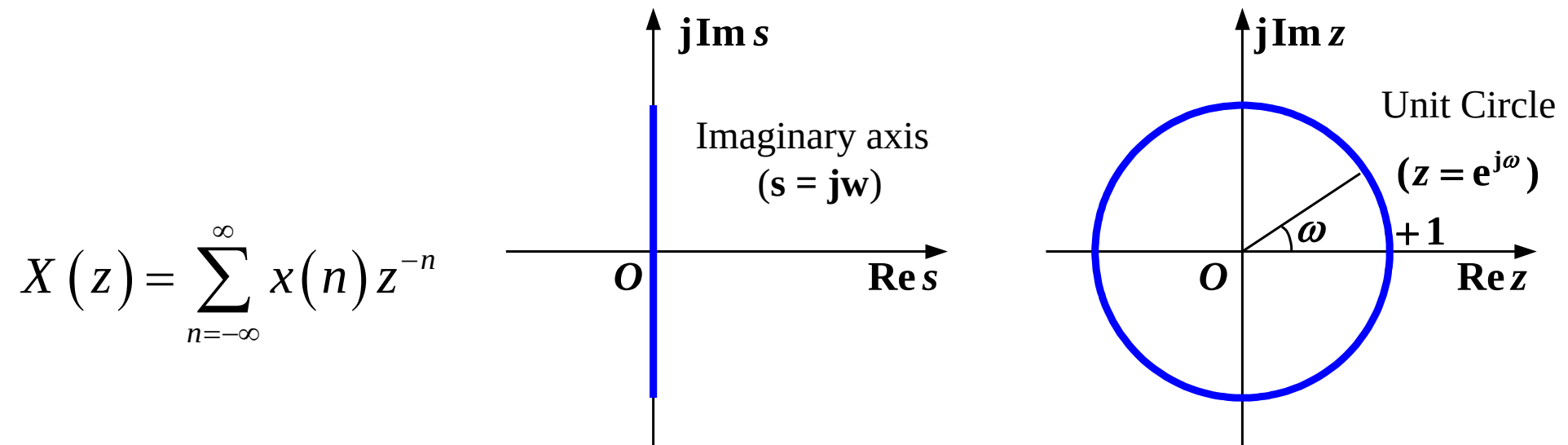
$e^{j\omega}$ is a periodic function with a period of 2π .

$$z = e^{sT} = e^{j\Omega T} = e^{j\omega}$$

$$\omega = \Omega T$$

7.10 Fourier Transform of Sequences (DTFT)

Relationship between DTFT and z-Transform



Let $z = e^{j\omega}$, $|z| = 1$, i.e., the z – transform on the unit circle is the DTFT :

Period is 2π

$$X(e^{j\omega}) = X(z) \Big|_{z=e^{j\omega}} = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

7.10 Fourier Transform of Sequences (DTFT)

7.10.2 Spectrum of Sequences

$$X(e^{j\omega}) = |X(e^{j\omega})| e^{j\varphi(\omega)} \quad X(e^{j\omega}) \text{ is the spectrum of } x(n)$$

$|X(e^{j\omega})|$: Magnitude spectrum of $x(n)$, $\varphi(\omega)$: Phase spectrum.

Both are continuous functions of ω .

Since $X(e^{j\omega}) = X(e^{j(\omega+2\pi)})$, $X(e^{j\omega})$ is a periodic function with period 2π .

For the Fourier transform $X(e^{j\omega})$ of a **real sequence** $x(n)$:

Its real part is **even-symmetric**, and its imaginary part is **odd-symmetric**;

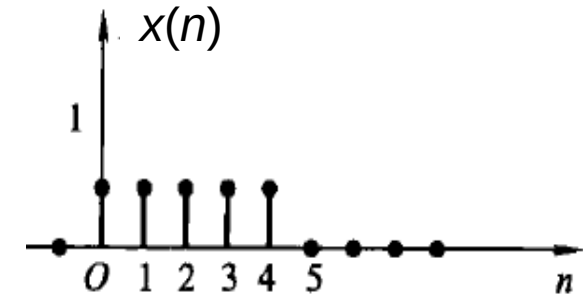
Its magnitude spectrum is **even-symmetric**, and its phase spectrum is **odd-symmetric**.

7.10 Fourier Transform of Sequences (DTFT)

Example 7-16: Given the sequence $x(n]=u(n)-u(n-5)$, find its Fourier transform $X(e^{j\omega})$.

Solution:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} = \sum_{n=0}^4 e^{-j\omega n} = \frac{1-e^{-j5\omega}}{1-e^{-j\omega}} \\ &= \frac{e^{-j5\omega/2}}{e^{-j\omega/2}} \cdot \frac{e^{j5\omega/2}-e^{-j5\omega/2}}{e^{j\omega/2}-e^{-j\omega/2}} = e^{-j2\omega} \cdot \frac{\sin(5\omega/2)}{\sin(\omega/2)} \\ &= |X(e^{j\omega})| e^{j\varphi(\omega)} \end{aligned}$$

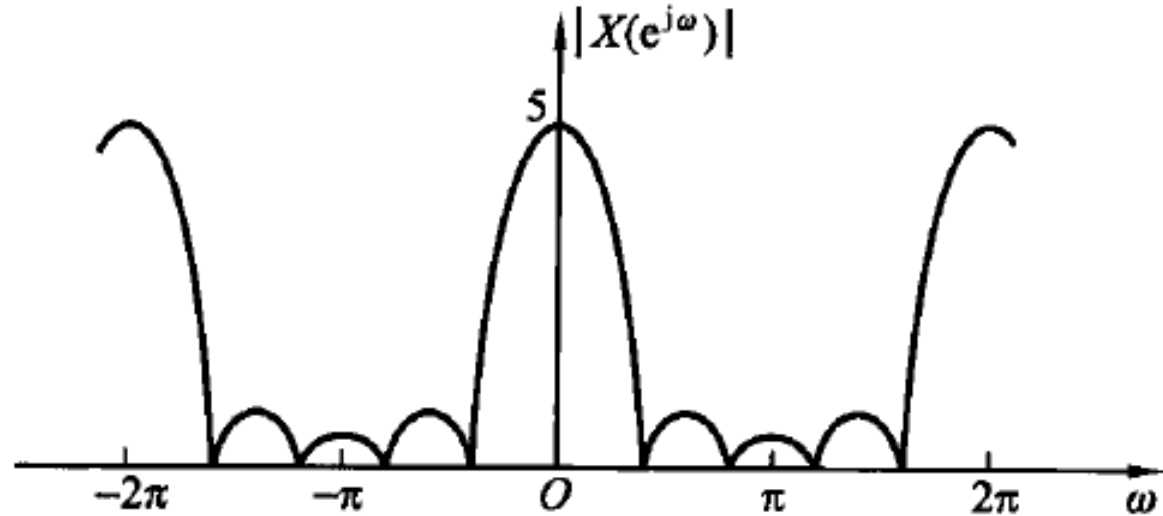


$$|X(e^{j\omega})| = \left| \frac{\sin(5\omega/2)}{\sin(\omega/2)} \right| \quad \varphi(\omega) = -2\omega + \arg \left[\frac{\sin(5\omega/2)}{\sin(\omega/2)} \right]$$

↑
arg[]--Phase shift of the expression
in the bracket

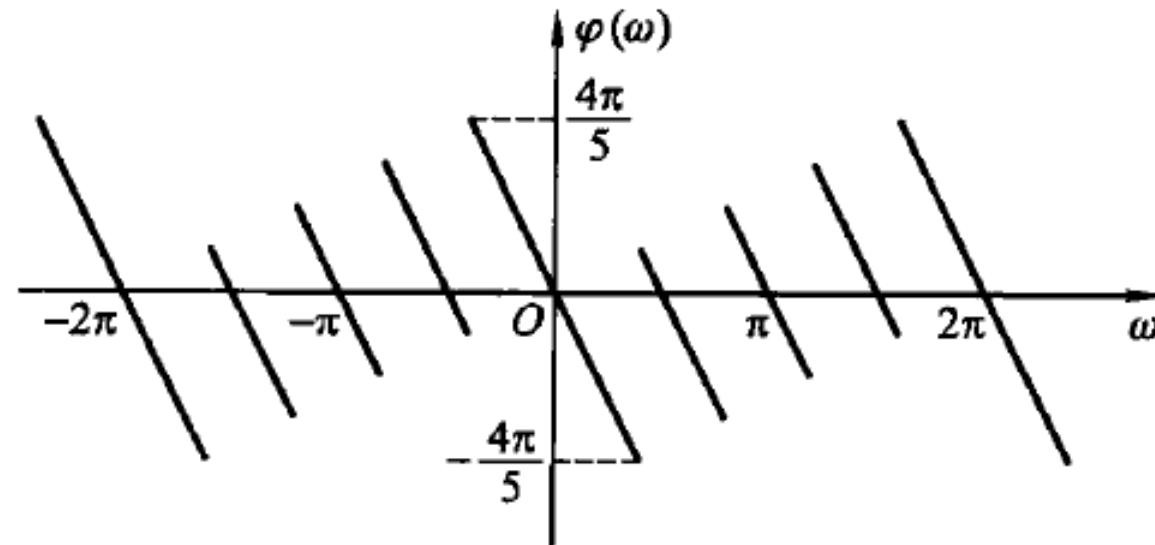
7.10 Fourier Transform of Sequences (DTFT)

$$|X(e^{j\omega})| = \left| \frac{\sin(5\omega/2)}{\sin(\omega/2)} \right|$$



$$\varphi(\omega) = -2\omega + \arg \left[\frac{\sin(5\omega/2)}{\sin(\omega/2)} \right]$$

$$= \begin{cases} -2\omega & \omega \in [0, 2\pi/5) \\ -2\omega + \pi & \omega \in [2\pi/5, 4\pi/5) \\ -2\omega + 2\pi & \omega \in [4\pi/5, 6\pi/5) \\ -2\omega + 3\pi & \omega \in [6\pi/5, 8\pi/5) \\ -2\omega + 4\pi & \omega \in [8\pi/5, 2\pi) \end{cases}$$



7.10 Fourier Transform of Sequences (DTFT)

Fourier Transform of Sampled Signals

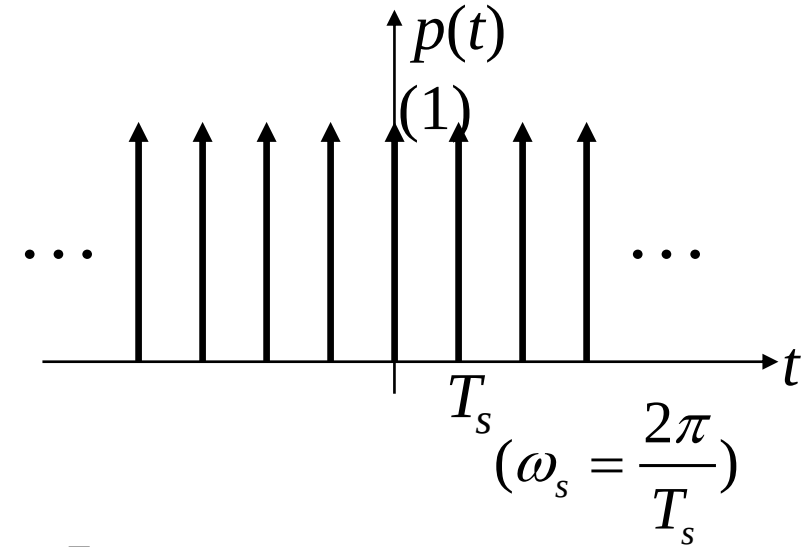
Let

Fourier transform of continuous signal $f(t)$: $F(\Omega)$

Fourier transform of sampling pulse $p(t)$: $P(\Omega)$

Fourier transform of sampled signal $f_s(t)$: $F_s(\Omega)$

Sampled signal: $f_s(t) = f(t)p(t) = f(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$



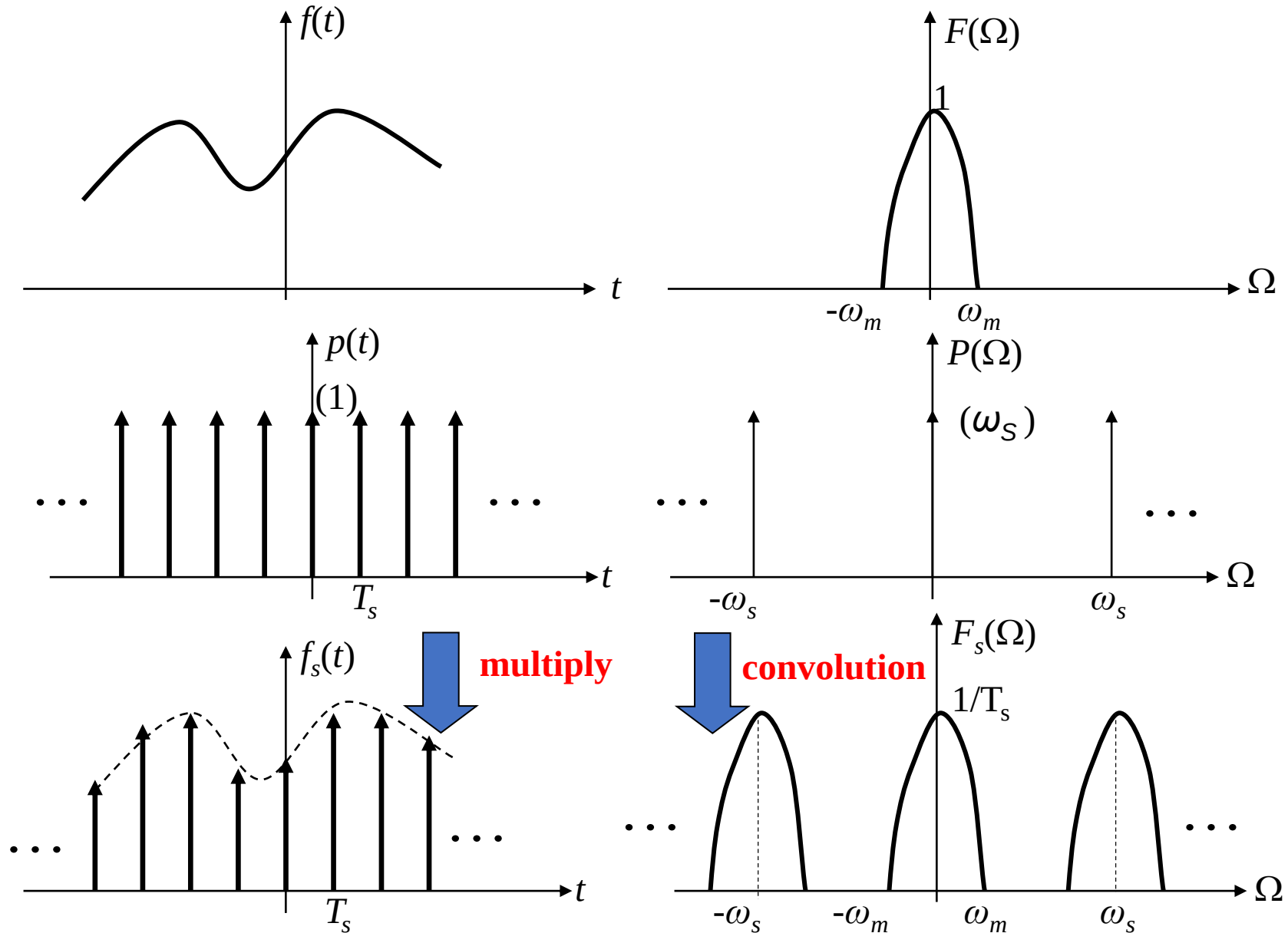
$$P(\Omega) = 2\pi \sum_{n=-\infty}^{\infty} P_n \delta(\Omega - n\omega_s) \quad \text{where} \quad P_n = \frac{1}{T_s} \int_{-\frac{T_s}{2}}^{\frac{T_s}{2}} \delta(t) e^{-jn\omega_s t} dt = \frac{1}{T_s}$$

$$F_s(\Omega) = \frac{1}{2\pi} F(\Omega) * P(\Omega) = \frac{1}{2\pi} F(\Omega) * \frac{2\pi}{T_s} \sum_{n=-\infty}^{\infty} \delta(\Omega - n\omega_s)$$

$$= \frac{1}{T_s} \sum_{n=-\infty}^{\infty} F(\Omega - n\omega_s) \quad \leftarrow F_s(\Omega) \text{ is a periodic function with period } \omega_s$$

$$\omega = \Omega T_s, \quad \omega_s T_s = 2\pi \quad F_s(e^{j\omega}) = F_s(e^{j\Omega T_s}) \text{ is a periodic function with period } 2\pi$$

7.10 Fourier Transform of Sequences (DTFT)



7.10 Fourier Transform of Sequences (DTFT)

7.10.3 Conditions for the Existence of DTFT

(1) For the DTFT to exist, $|X(e^{j\omega})| < \infty$ must hold for all ω .

$$|X(e^{j\omega})| = \left| \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \right| \leq \sum_{n=-\infty}^{\infty} |x(n)| |e^{-j\omega n}| \leq \sum_{n=-\infty}^{\infty} |x(n)| < \infty$$

This shows that if a sequence is **absolutely summable**, its Fourier transform exists. However, due to the inequality scaling above, **absolute summability** is only a **sufficient condition** for the existence of the sequence's Fourier transform.

(2) If a sequence is **square-summable**, $\sum_{n=-\infty}^{\infty} |x(n)|^2 < \infty$, the sequence has **finite energy**. In this case, $\sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$ converges in mean square to $X(e^{j\omega})$, so the Fourier transform of the sequence also exists.

The two conditions above are only **sufficient conditions** for the existence of a sequence's Fourier transform. For some sequences that do not satisfy these conditions (e.g., periodic sequences, unit step sequences), their Fourier transforms can still be obtained by introducing the unit sample signal $\delta(n)$.

To date, **the necessary and sufficient condition for the existence of a sequence's Fourier transform has not been found.**

Which of the following statements about the Fourier transform of sequences is **incorrect** ()?

- ☐ A The Fourier transform of a real sequence satisfies conjugate symmetry: the real part of $X(e^{j\omega})$ is even-symmetric, the imaginary part is odd-symmetric; the magnitude is even-symmetric, and the phase angle is odd-symmetric.
- ☐ B If a sequence satisfies the absolute summability condition, it must be energy-limited; however, energy limitation does not guarantee absolute summability.
- ☒ C $X(e^{j\omega})$ is a complex function of ω with a period of 2π , but its magnitude and phase angle do not necessarily have a period of 2π .
- ☐ D Both the absolute summability and the square summability conditions are sufficient but not necessary condition of the existence of DTFT.

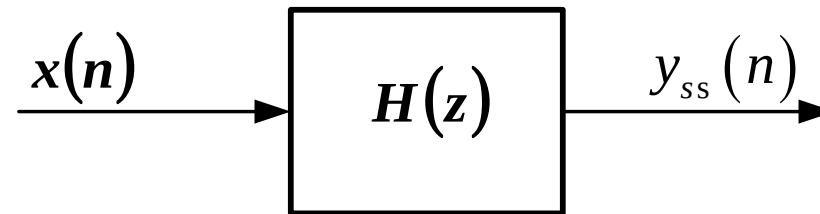
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7.11 Frequency Response of Discrete-Time Systems

7.11.1 Significance of Discrete System's Frequency Response Characteristic

1. Definition of Frequency Response Characteristic

Frequency response characteristic: The variation of the **steady-state response (SS)** of a discrete system with frequency, when the system is excited by a sinusoidal sequence.



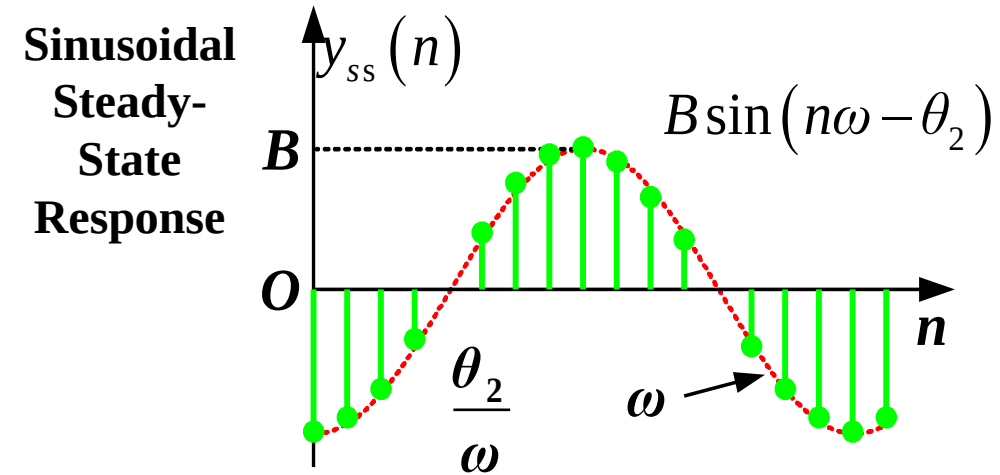
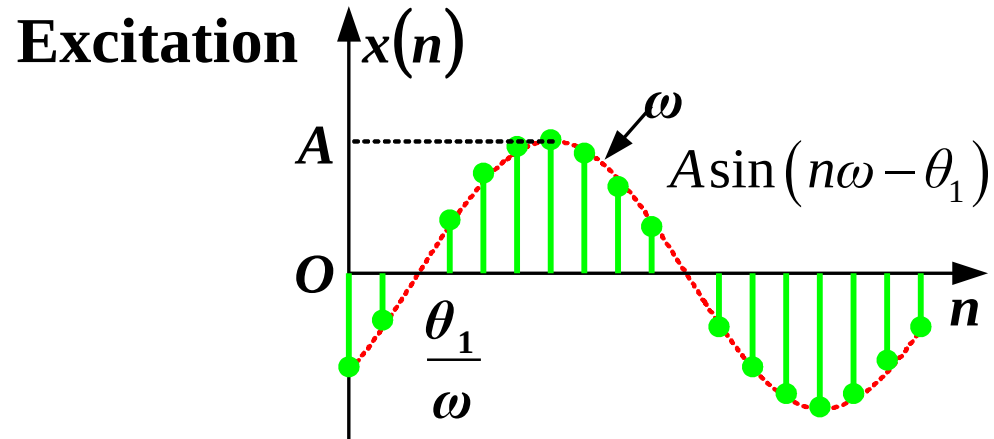
Stable discrete causal system

$$x(n) = A \sin(n\omega_c) u(n)$$

$$H(e^{j\omega}) = H(z) \Big|_{z=e^{j\omega}} = |H(e^{j\omega})| e^{j\varphi(\omega)}$$

$$y_{ss}(n) = A |H(e^{j\omega_c})| \sin[n\omega_c + \varphi(\omega_c)] u(n)$$

7.11 Frequency Response of Discrete-Time Systems



$$H(e^{j\omega}) = |H(e^{j\omega})| e^{j\varphi(\omega)} = \frac{B}{A} e^{j[-(\theta_2 - \theta_1)]}$$

$$|H(e^{j\omega})| = \frac{B}{A} \quad \varphi(\omega) = -(\theta_2 - \theta_1)$$

Significance of the frequency response characteristic: It represents the changes in amplitude and phase of the output sinusoidal sequence relative to the input sinusoidal sequence.

7.11 Frequency Response of Discrete-Time Systems

2. Deriving Frequency Response from System Function

The z-transform of the system function **on the unit circle** is the frequency response characteristic of the system:

$$H(e^{j\omega}) = H(z) \Big|_{z = e^{j\omega}} = |H(e^{j\omega})| e^{j\varphi(\omega)}$$

$|H(e^{j\omega})| \sim \omega$: Magnitude response or magnitude-frequency characteristic
Ratio of the amplitude of the output sinusoidal sequence
to that of the input sinusoidal sequence

$\varphi(\omega) \sim \omega$: Phase response (or phase-frequency characteristic)
Phase shift of the output sinusoidal sequence relative to the
input sinusoidal sequence

Since $e^{j\omega}$ is a periodic function, $H(e^{j\omega})$ is also a periodic function with a period of 2π .
This is a prominent feature that distinguishes discrete systems from continuous systems.

7.11 Frequency Response of Discrete-Time Systems

Example 7-17: Given the system function of a discrete-time system below:

$H(z) = \frac{z-1}{(z-1/3)(z-1/2)} \quad \left(|z| > \frac{1}{2} \right)$, find the sinusoidal steady-state response $y_{ss}(n)$ when the excitation is $x(n) = \cos(\pi n / 2)u(n)$.

Solution: The frequency response of the system is $H(e^{j\omega}) = \frac{e^{j\omega} - 1}{(e^{j\omega} - 1/3)(e^{j\omega} - 1/2)}$

$$H(e^{j\pi/2}) = \frac{e^{j\pi/2} - 1}{(e^{j\pi/2} - 1/3)(e^{j\pi/2} - 1/2)} = \frac{j-1}{(j-1/3)(j-1/2)} = \frac{6-1+j}{5-1-j} = \frac{6}{5}e^{-j\pi/2}$$

$$|H(e^{j\pi/2})| = \frac{6}{5} \quad \varphi(\pi/2) = -\frac{\pi}{2}$$

$$y_{ss}(n) = |H(e^{j\pi/2})| \cos[\pi n / 2 + \varphi(\pi/2)]u(n) = \frac{6}{5} \sin(\pi n / 2)u(n)$$

7.11 Frequency Response of Discrete-Time Systems

3. Relationship Between Frequency Response & Unit Sample Response

The frequency response of a discrete-time system is the **Fourier transform of the system's unit sample response**:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n}$$

Physical significance: $H(e^{j\omega})$ describes how each harmonic component is modified, with $h(n)$ serving as the weighting coefficient.

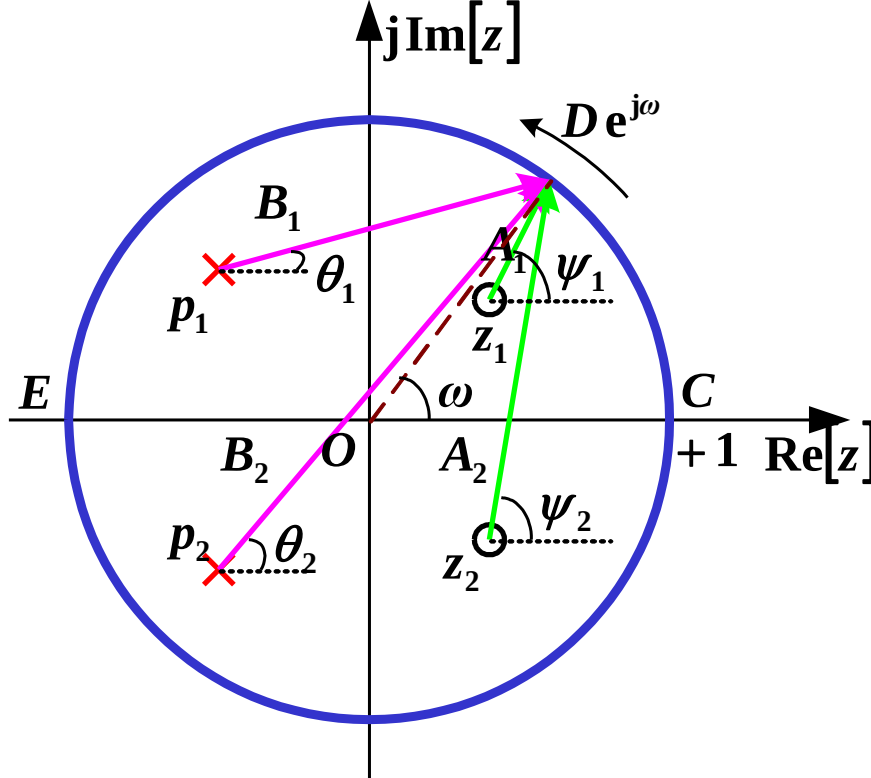
Since $h(n)$ is generally a real sequence, $|H(e^{j\omega})|$ is an even function, and $\varphi(\omega)$ is an odd function.

7.11 Frequency Response of Discrete-Time Systems

7.11.2 Geometric Determination of the Frequency Response Characteristic

$$H(z) = \frac{\prod_{r=1}^M (z - z_r)}{\prod_{k=1}^N (z - p_k)}$$

$$H(e^{j\omega}) = \frac{\prod_{r=1}^M (e^{j\omega} - z_r)}{\prod_{k=1}^N (e^{j\omega} - p_k)} = |H(e^{j\omega})| e^{j\varphi(\omega)}$$

$$\begin{aligned} \angle (e^{j\omega} - z_r) &= \psi_r \\ e^{j\omega} - p_k &= B_k e^{j\theta_k} \end{aligned}$$


$$\text{Magnitude - frequency response } |H(e^{j\omega})| = \frac{\prod_{r=1}^M A_r}{\prod_{k=1}^N B_k} \quad \text{Phase - frequency response } \varphi(\omega) = \sum_{r=1}^M \psi_r - \sum_{k=1}^N \theta_k$$

Notes:

(1) Zeros or poles at $z = 0$ have no effect on the magnitude-frequency response. Thus, adding or removing zeros/poles at $z = 0$ does not change the magnitude-frequency response, but it will affect the phase-frequency response.

(2) When the $e^{j\omega}$ point rotates near a pole p_i the length B_i of the pole vector becomes the shortest, so the frequency response may have a peak at this point:

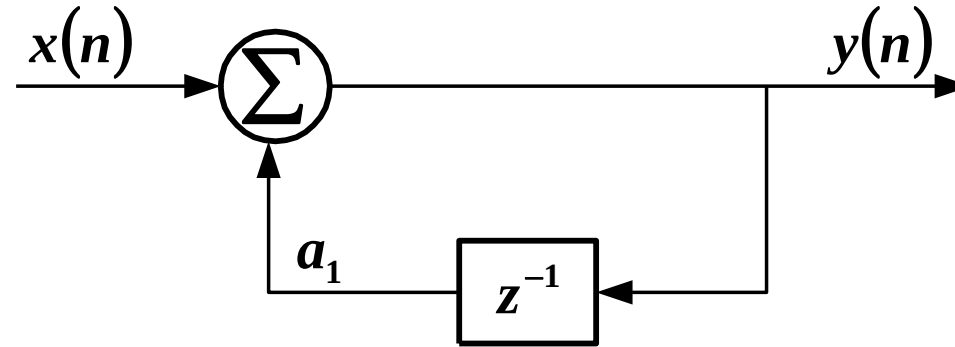
The closer the pole is to the unit circle, the shorter is, and the sharper the peak of the frequency response becomes;

If the pole lies **on the unit circle**, and the peak of the frequency response tends to infinity.

(3) The effect of zeros is the opposite of that of poles.

7.11 Frequency Response of Discrete-Time Systems

Example 7-18: Find the frequency response of the first-order discrete causal stable system shown below, and analyze its characteristics.



Solution: Difference equation $y(n] - a_1 y[n-1] = x(n]$

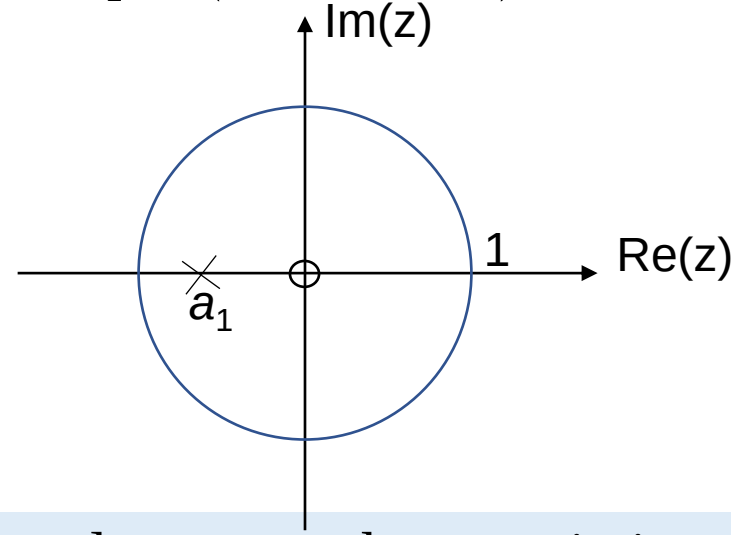
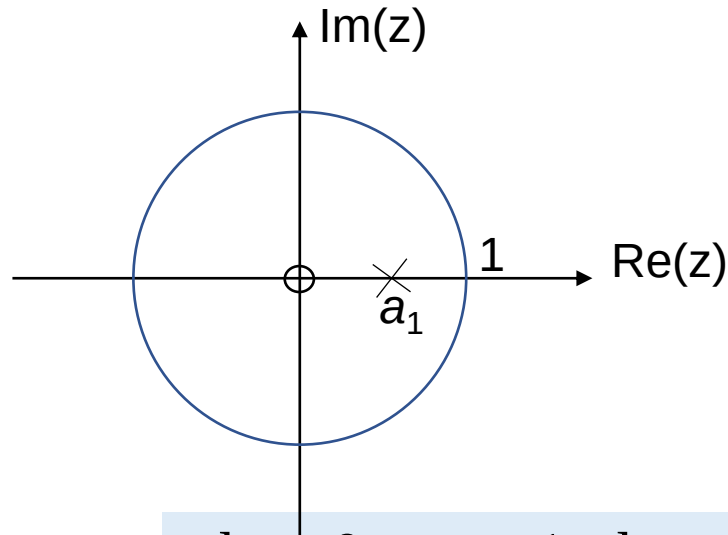
For a causal system, the system function is $H(z) = \frac{z}{z - a_1} \quad |z| > |a_1|$

To ensure the system is stable, $|a_1| < 1$ is required.

Unit sample response $h(n] = a_1^n u(n]$

7.11 Frequency Response of Discrete-Time Systems

$$\text{Frequency Response } H(e^{j\omega}) = \frac{e^{j\omega}}{e^{j\omega} - a_1} = \frac{1}{(1 - a_1 \cos \omega) + j a_1 \sin \omega}$$



when $0 < a_1 < 1$, the system has a low-pass characteristic
 when $-1 < a_1 < 0$, the system has a high-pass characteristic.

$$\text{Magnitude-frequency response } |H(e^{j\omega})| = \frac{1}{\sqrt{1 + a_1^2 - 2a_1 \cos \omega}}$$

$$\text{Phase-frequency response } \varphi(\omega) = -\arctan\left(\frac{a_1 \sin \omega}{1 - a_1 \cos \omega}\right)$$

A discrete system has the difference equation below:

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + \frac{1}{3}x(n-1).$$

The unit sample response of this system is ()

☐ A $h(n) = \left[\frac{1}{2} \left(\frac{10}{3} \right)^n - \frac{1}{4} \left(\frac{7}{3} \right)^n \right] u(n)$

☐ B $h(n) = \left[\frac{10}{3} \left(\frac{1}{8} \right)^n - \frac{7}{3} \left(\frac{3}{4} \right)^n \right] u(n)$

☒ C $h(n) = \left[\frac{10}{3} \left(\frac{1}{2} \right)^n - \frac{7}{3} \left(\frac{1}{4} \right)^n \right] u(n)$

☐ D $h(n) = \left[\frac{1}{8} \left(\frac{10}{3} \right)^n - \frac{3}{4} \left(\frac{7}{3} \right)^n \right] u(n)$

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7.11 Frequency Response of Discrete-Time Systems

Example 7-19: Given the difference equation of a discrete system below:

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + \frac{1}{3}x(n-1)$$

- 1) Find the system function and unit sample response;
- 2) Draw the pole-zero plot of the system function;
- 3) Roughly sketch the magnitude-frequency response characteristic curve;
- 4) Draw the simulation block diagram of the system.

Solution: 1) system function $H(z) = \frac{Y(z)}{X(z)} = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}} = \frac{z(z + \frac{1}{3})}{z^2 - \frac{3}{4}z + \frac{1}{8}}$

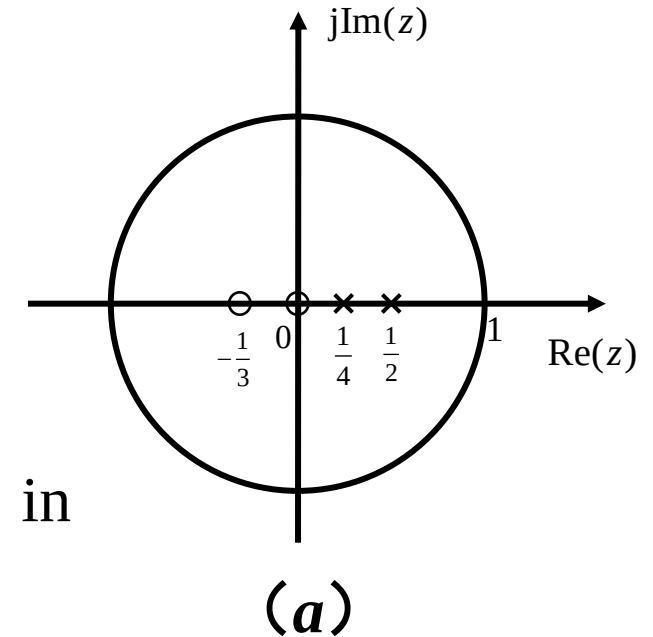
$$= \frac{10}{3} \left(\frac{z}{z - \frac{1}{2}} \right) - \frac{7}{3} \left(\frac{z}{z - \frac{1}{4}} \right) \quad (|z| > \frac{1}{2})$$

Unit Sample Response $h(n) = z^{-1}[H(z)] = \left[\frac{10}{3} \left(\frac{1}{2} \right)^n - \frac{7}{3} \left(\frac{1}{4} \right)^n \right] u(n)$

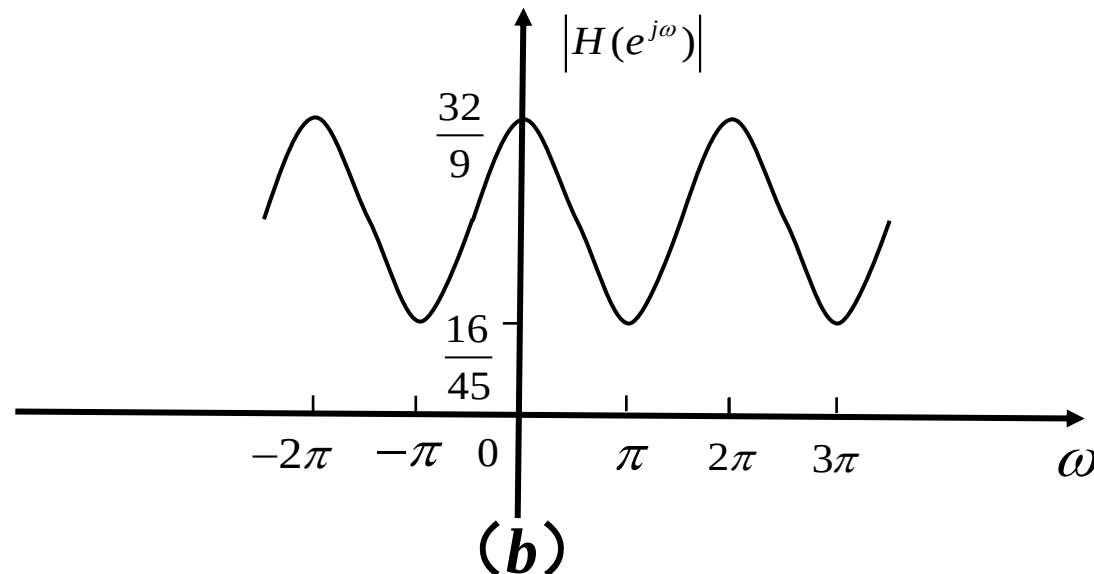
7.11 Frequency Response of Discrete-Time Systems

$$2) \quad H(z) = \frac{z(z + \frac{1}{3})}{z^2 - \frac{3}{4}z + \frac{1}{8}} = \frac{z(z + \frac{1}{3})}{(z - \frac{1}{2})(z - \frac{1}{4})}$$

The pole-zero distribution is shown in Figure (a).

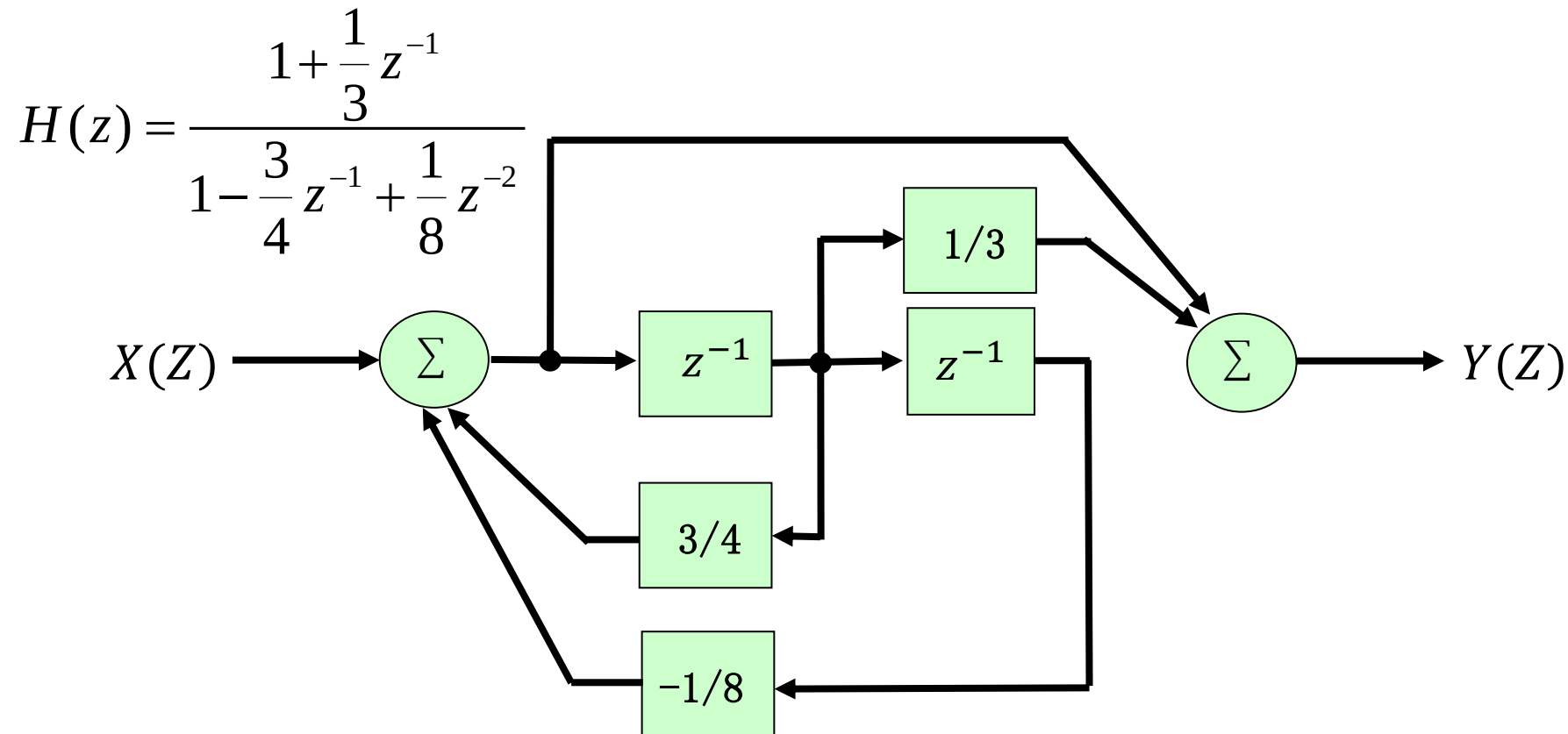


3) The magnitude-frequency response of the system (shown in Figure (b)) is derived from the pole-zero distribution.



7.11 Frequency Response of Discrete-Time Systems

4) The system structure block diagram (shown in Figure (c)) is obtained from the difference equation or the system function.



(c)

7.11.3 Introduction to Filters

Functions of Filters:

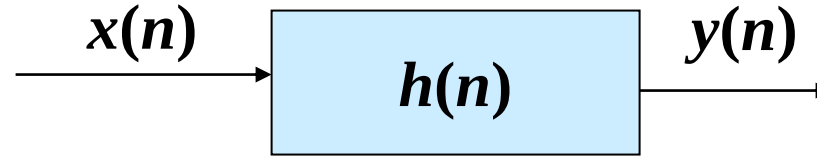
- 1) Remove unwanted parts of a signal (signal separation), e.g., random noise;
- 2) Extract useful parts of a signal (signal recovery), e.g., extract components within a certain frequency band.

From the system perspective, filters are divided into two main categories: analog filters (AF) and digital filters (DF). Both are realizable **LTI systems**, but they differ greatly in physical composition and working methods:

➤ **Analog Filters:** Use analog circuits to filter analog signals. They can only be implemented with hardware, using components such as resistors (R), inductors (L), capacitors (C), operational amplifiers, or switched capacitors.

➤ **Digital Filters:** Use discrete-time systems to filter digital signals. They can be implemented with **hardware** (digital signal processors) or **computer software**. They are significantly superior to analog filters in terms of volume, weight, precision, stability, reliability, storage function, flexibility, and cost-performance ratio

7.11.4 Working Principle of Digital Filters



The output of an **LTI system** is:

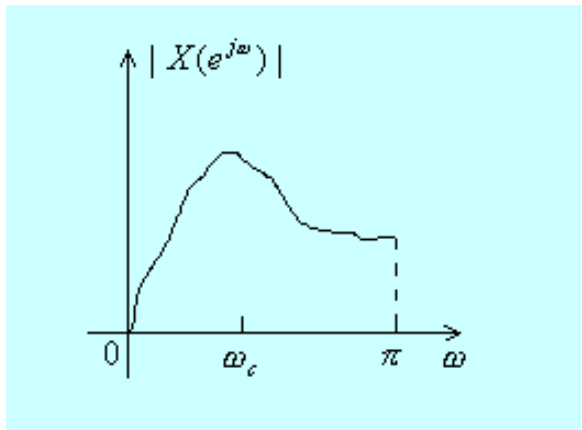
$$y(n) = x(n) * h(n) = \text{IDTFT}[X(e^{j\omega})H(e^{j\omega})]$$

The spectrum $X(e^{j\omega})$ of the input sequence is transformed into $X(e^{j\omega})H(e^{j\omega})$ after passing through the filter $H(e^{j\omega})$

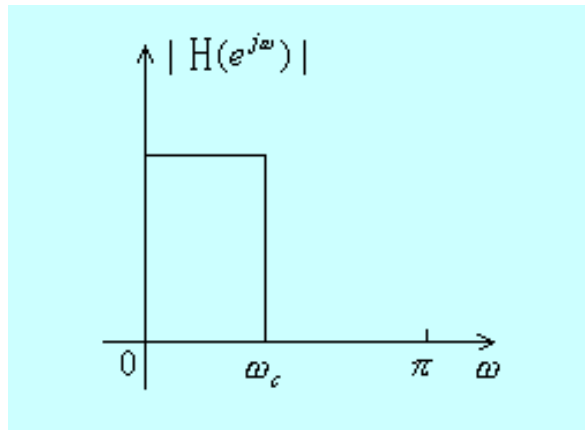
By selecting $H(e^{j\omega})$, we make the filter output $X(e^{j\omega})H(e^{j\omega})$ meet our requirements-----this is the working principle of digital filters.

7.11 Frequency Response of Discrete-Time Systems

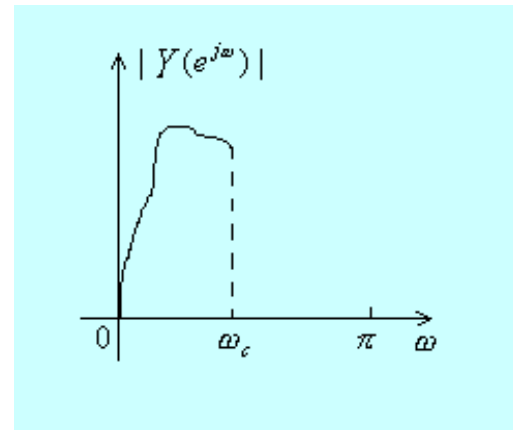
Filtering Principle



(a) $|X(e^{j\omega})|$



(b) $|H(e^{j\omega})|$



(c) $|Y(e^{j\omega})|$

when $x(n)$ passes through the system $h(n)$, the output $y(n)$ no longer contains frequency components with $|\omega| > \omega_c$,
while components with $|\omega| < \omega_c$ "pass without distortion".

7.11.5 Classical Filters and Modern Filters

1. Classical Filters

$$x(n) = s(n) + u(n) \xrightarrow{\text{Additive noise}}$$

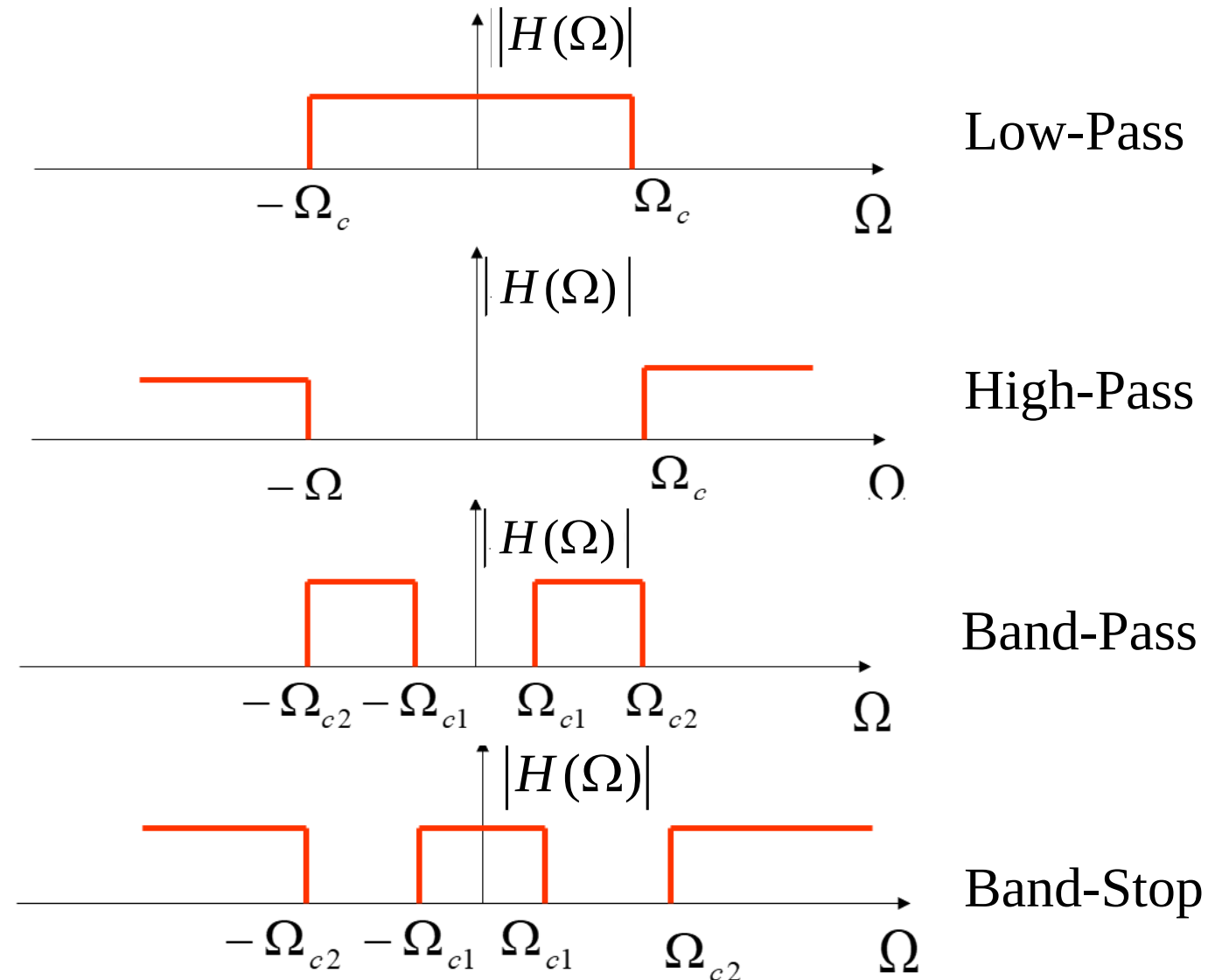
If the useful component $u(n)$ and the component $s(n)$ to be removed in $x(n)$ occupy different frequency bands, $u(n)$ can be effectively removed via a linear system.

Classified by function: Low-Pass (LP), High-Pass (HP), Band-Pass (BP), Band-Stop (BS), All-Pass

Each type has two versions: Analog Filter (AF) and Digital Filter (DF).

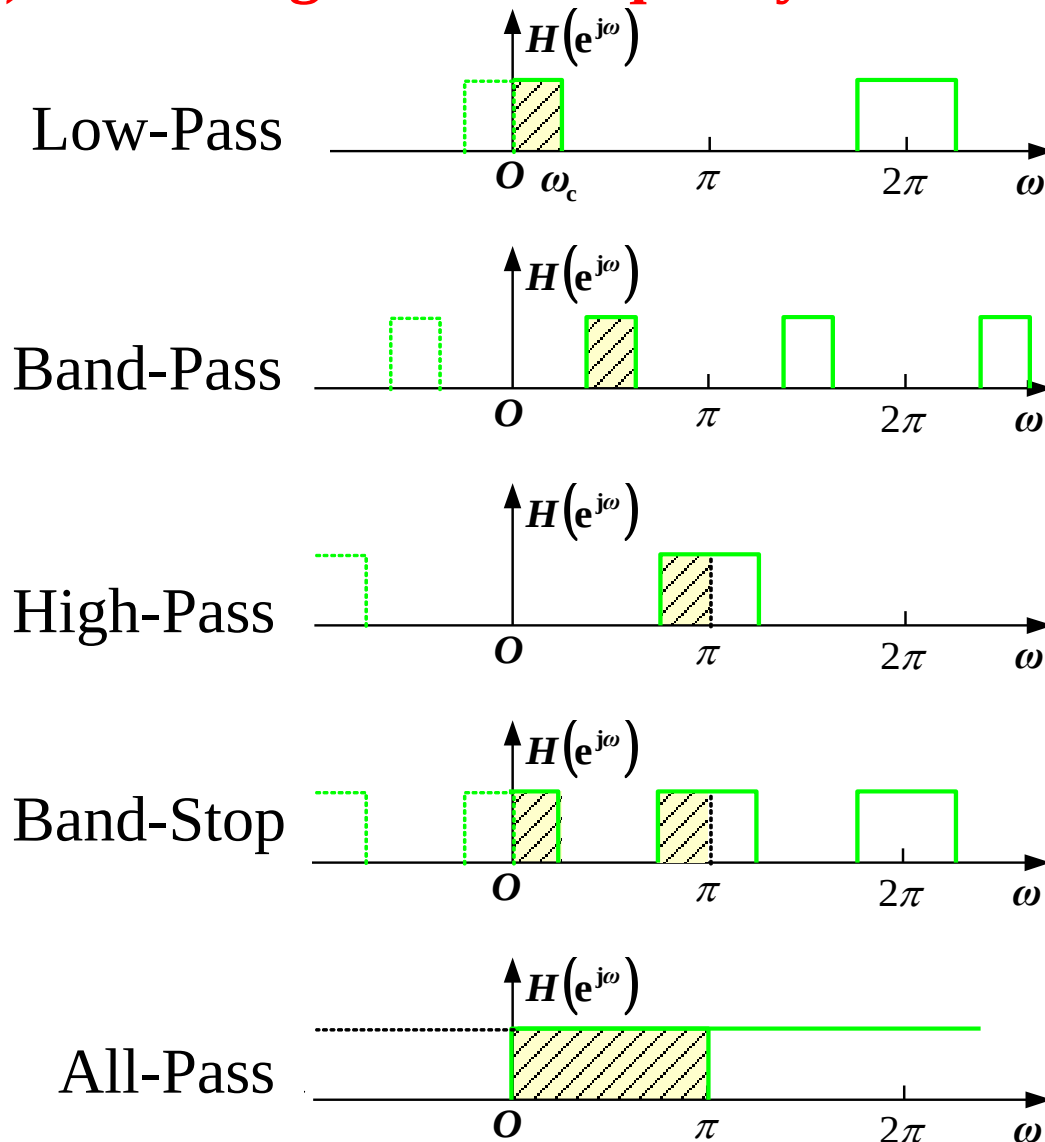
7.11 Frequency Response of Discrete-Time Systems

1) Ideal Magnitude-Frequency Characteristics of Analog Filters



7.11 Frequency Response of Discrete-Time Systems

2) Ideal Magnitude-Frequency Characteristics of Digital Filters



Due to periodicity and symmetry, it is sufficient to study the range $0 \leq \omega \leq \pi$

The phase-frequency response is assumed to be 0 in the left figure.

Which of the following statements about filters is **incorrect** ?

- ☐ A A filter "filters out" a frequency by reducing the gain at that frequency.
- ☒ B The region near $\omega = 2\pi$ in a digital filter corresponds to the "high frequency" of the actual frequency.
- ☐ C Analog filters can process analog signals directly.
- ☐ D Only signals converted via analog-to-digital (A/D) conversion can be processed by digital filters.

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7.11 Frequency Response of Discrete-Time Systems

Summary

1. The frequency response characteristic of a system: $H(e^{j\omega}) = H(z) \Big|_{z=e^{j\omega}} = |H(e^{j\omega})| \cdot e^{j\varphi(\omega)}$
 $|H(e^{j\omega})|$: Magnitude-frequency characteristic (ratio of the amplitude of the output sequence to the input sequence)
 $\varphi(\omega)$: Phase-frequency characteristic (phase shift of the output sequence relative to the input sequence)
2. The frequency response of a system is the dynamic behavior of the system function **on the unit circle**; it changes with ω and affects the amplitude and phase of the output.
3. Since $e^{j\omega}$ is a periodic function with period 2π , the frequency response characteristic $H(e^{j\omega})$ of the system is also a periodic function with period 2π .
4. $|H(e^{j\omega})|$ is an **even function** of ω and $\varphi(\omega)$ is an **odd function** of ω .
5. The region near $\omega = \pi$ in a digital filter corresponds to the **high frequency** of the actual frequency.

1. Repeat all the examples in the lecture notes.
2. 7.37(a) in the textbook.